



Atag Celsius induction cooker

Phase 2 | Initial Redesign Proposal

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Introduction

We aim to redesign the Atag Induction Celsius Cooking Hob and equip **regular cooks** with **the confidence** to let the ATAG cooking plate **empower** them through their culinary experimentation in **an intuitive manner**. Our goal is to make them **feel in control** and **excited** during the onboarding and about the cooking process, with the integration of new technology that Atag unlocks in their daily lives.

In this report, we aim to propose an improved redesign that fits our scope. The proposed design reflects the interaction qualities that emerged from our interaction vision of how cooking with the hob is like a jamming session with friends. Our concept is positioned between routine cooking and exploratory interaction, taking into account guests who might come over to cook. As a result, we have identified two user personas in this report: **the simple routine chef** and **the exploratory chef**.

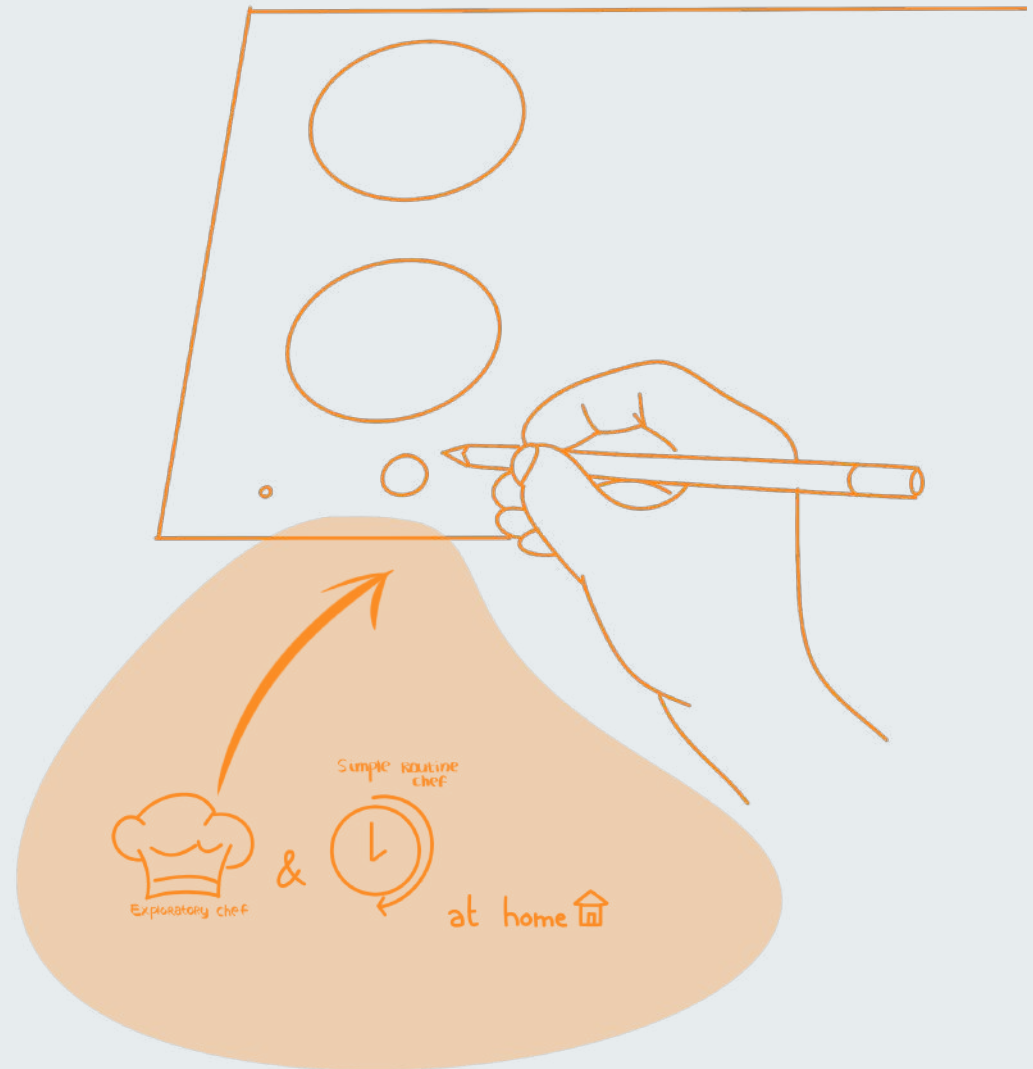


Figure 1: Illustration of redesigning the hob.

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1 ■ Recap of Phase 1

This section presents an outline of the previous research phase of our project, primarily focusing on the conclusions derived from it. It commences by capturing the design brief, following with the client's feedback and building our user personas, which will be used as a guide to redesign the interaction for the remainder of the report.

Quick Chapter glance:

Design Brief

Design scope

Design goal

Interaction Vision

Pain points in user flow

Client feedback

User Personas

Conclusion



Design brief

On this page we present a summary of the key elements the redesign of the ATAG Celsius hob is based on. These elements are defined in report one and represent the features the final concept aims to implement. The visualisations on this page show the **design scope** (Figure 2), **design goal** (Figure 3) and **interaction vision and qualities** (Figure 4).



Figure 2: Design goal

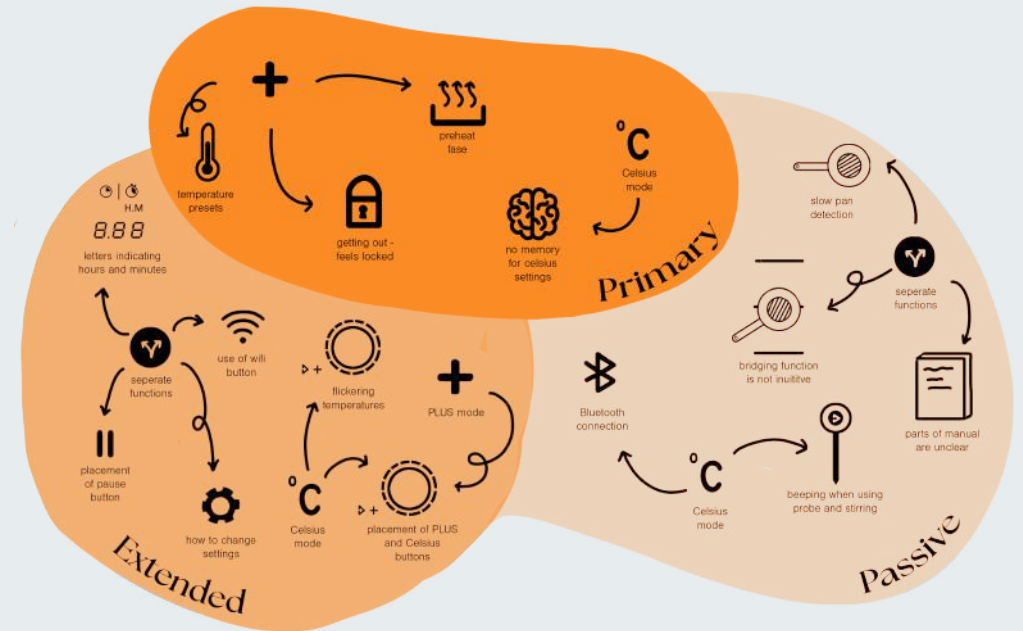


Figure 3: Design scope



Figure 4: Interaction vision and qualities

Pain points in user flow

Phase 1 involved creating detailed flow charts to better understand the current interaction and user flow. These charts helped us identify pain points in the induction hob's design, which we aim to address in the redesign process. **By analyzing the flow chart and combining it with our knowledge of the hob's problems, we were able to pinpoint**

areas for improvement. Flow charts are a useful tool for visualizing and understanding key parts of the interaction that require redesign. Figure 5 shows the main pain points we identified with the black ! icon, while the complete flow chart can be found in Appendix A.

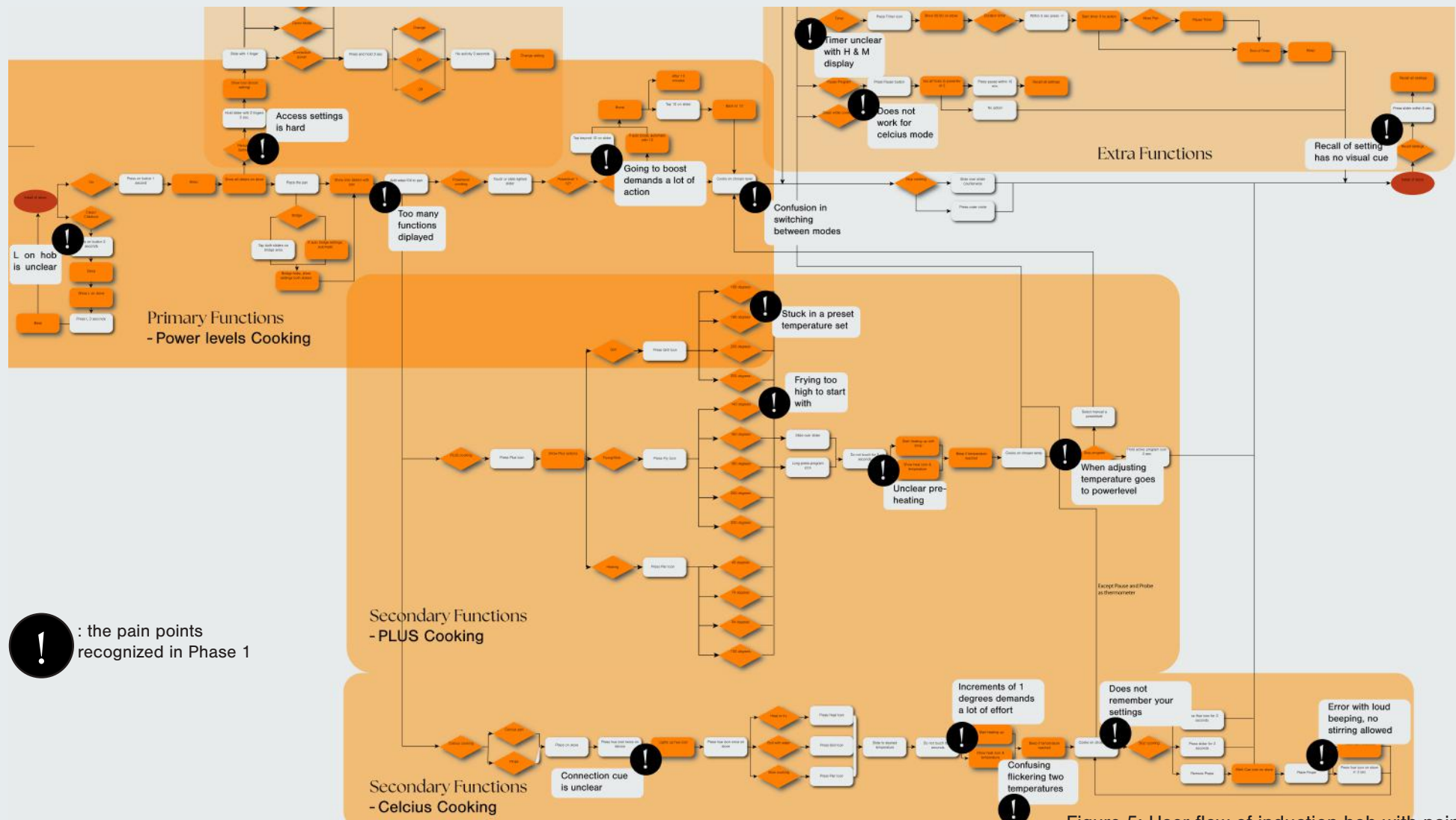


Figure 5: User flow of induction hob with pain points.

Client feedback

After sending the 'Design Brief' from Phase 1's report to our client, we held a meeting with them to gain valuable feedback. The meeting with Björn Flinterhoff had an overall positive response. There were 5 main discussion points which are presented in Figure 6. These provided us with valuable insights on:

- Why and **how** certain **functions** were **designed**?
- What were the **technical limitations** of the hob we currently have?
- What is our **focus** for the **future development** of the hob?

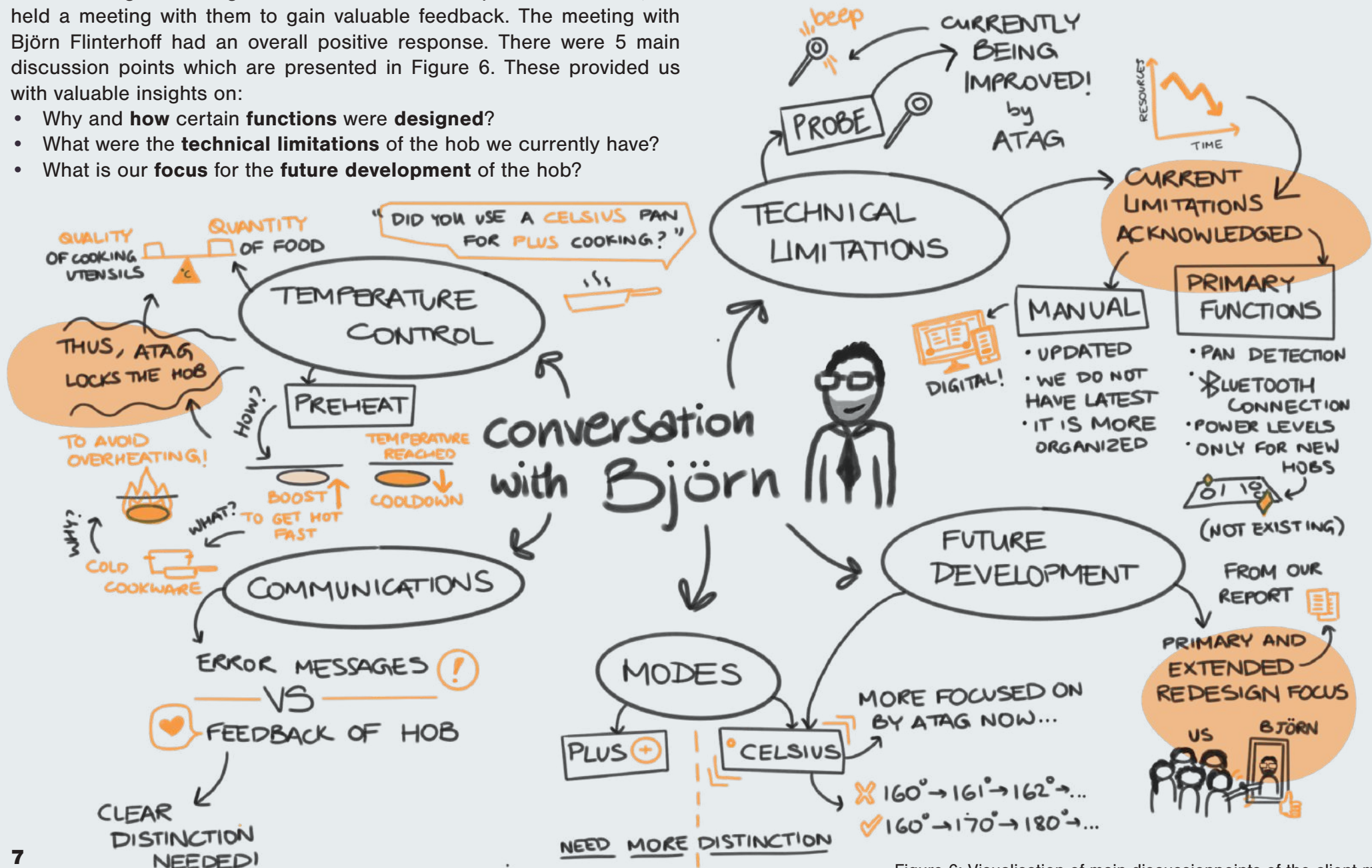


Figure 6: Visualisation of main discussion points of the client meeting,

User personas

Exploratory chef

Elena is an ambitious chef who loves exploring new recipes and cooking techniques. She is especially interested in the function of the Atag Celsius to cook with degrees. To create refined dishes and explore culinary possibilities.

Simple routine chef

Els is a mom that wants to have quick meals, so uses mostly the powerlevels on the induction hob. She also wants friends that are not familiar with the hob. She likes using the hob's basic functions to make simple and fast meals, skipping the complex settings.



Figure 7: Visualisation exploratory chef.



Figure 8: Visualisation simple routine chef.

Conclusion

In conclusion, this chapter has summarized the **beginning of our redesign journey** for ATAG's Celsius hob. By highlighting the design scope, goal, and interaction vision from the previous report, we start Phase 2 with a clear baseline. Additionally, the analysis of **pain points in the interaction flowchart** provided crucial insights into **areas that require improvement**, guiding our redesign focus. Thirdly, the fruitful **discussion with our client**, Björn Flinterhoff, allowed us to **better understand the hob's technical limitations** and gain valuable feedback for **future development**. And finally, **defining the user personas** (the exploratory and routine chef) for the purpose of **scoping** as well as to keep us in line with **our vision when we redesign**.



2.

Conceptualization

In this chapter, we move from the phase of establishing design intentions to the phase of conceptualization. Here, we provide an overview of the ideation process, which involved utilizing various creative facilitation techniques to address the pain-points associated with the proposed interaction qualities. This chapter concludes by presenting five interaction concept prototypes, which will be evaluated in the subsequent chapter.

Quick Chapter glance:

Ideation

- HMWs
- Figure storming
- Eidetic image
- What if?

Interaction Concepts

- Concept 1: Less buttons
- Concept 2: Big family
- Concept 3: Voice control
- Concept 4: Touch screen
- Concept 5: Learning hob

Conclusion



Ideation

After revisiting the pain points, design goals and intended interaction, we proceeded to generate as many ideas as possible to arrive at the most comprehensive and effective interaction designs. To achieve this, we employed various brainstorming techniques, each with its unique approach to addressing the problem. This allowed us to use our minds in distinct creative ways, which we then converged using the lens of our scope. In Figure 9, you can see an overview of the methods we used to arrive at the final concept.

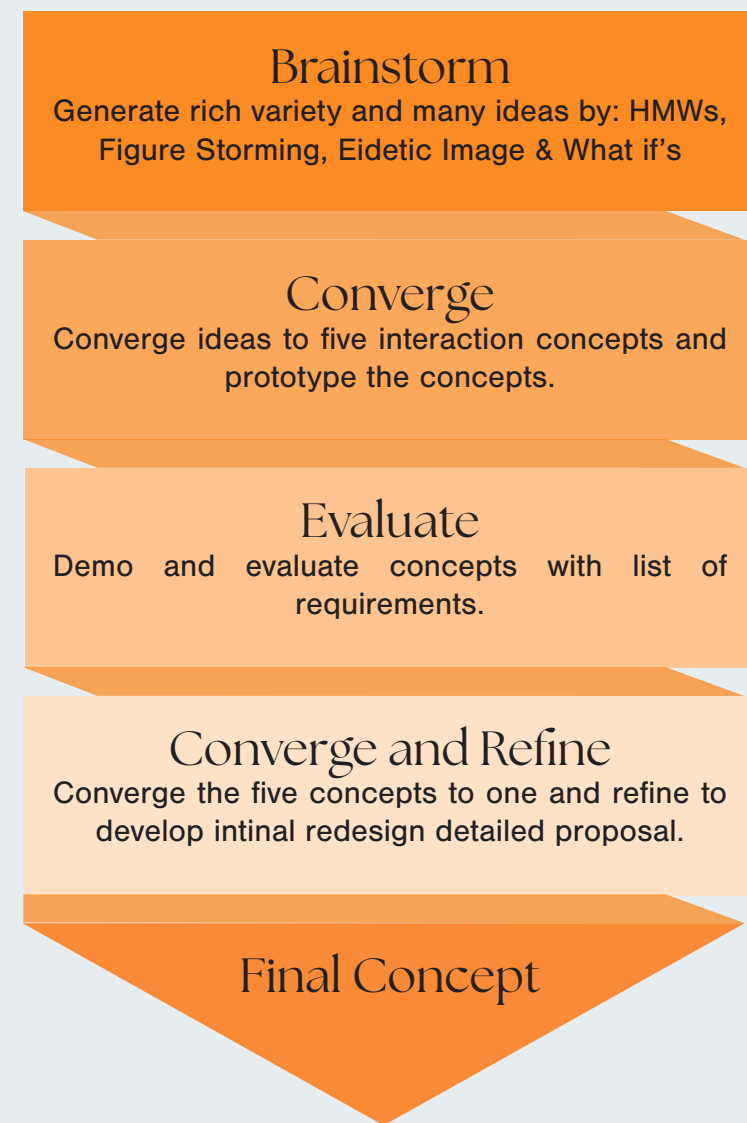


Figure 9: Overview of activities Phase 2.

How Might We's (HMWs)

What

'HMWs' breaks down complex issues into sub-questions, each member brainstorms on them for two minutes and passes it along. This technique quick and freely generates ideas and insights on specific subtopics.

Why

HMWs helped us approach brainstorming sessions with a fresh perspective by clearing our minds of preconceived notions and initial ideas from phase one.

We asked ourselves the following questions, based on our design scope:

- How might we **onboard users first hand?**
- How might we **give control during PLUS cooking?**
- How might we **switch between cooking settings?**
- How might we **indicate pre-heat?**
- How might we **improve user flow of Celsius cooking?**
- How might we **empower the user with the hob?**

This resulted in a wide range of ideas that all in some way improve the usability of the current product. The complete process of the HMW's can be found in Appendix B. We converged all ideas that came out of this method into the categories as shown in Figure 10.

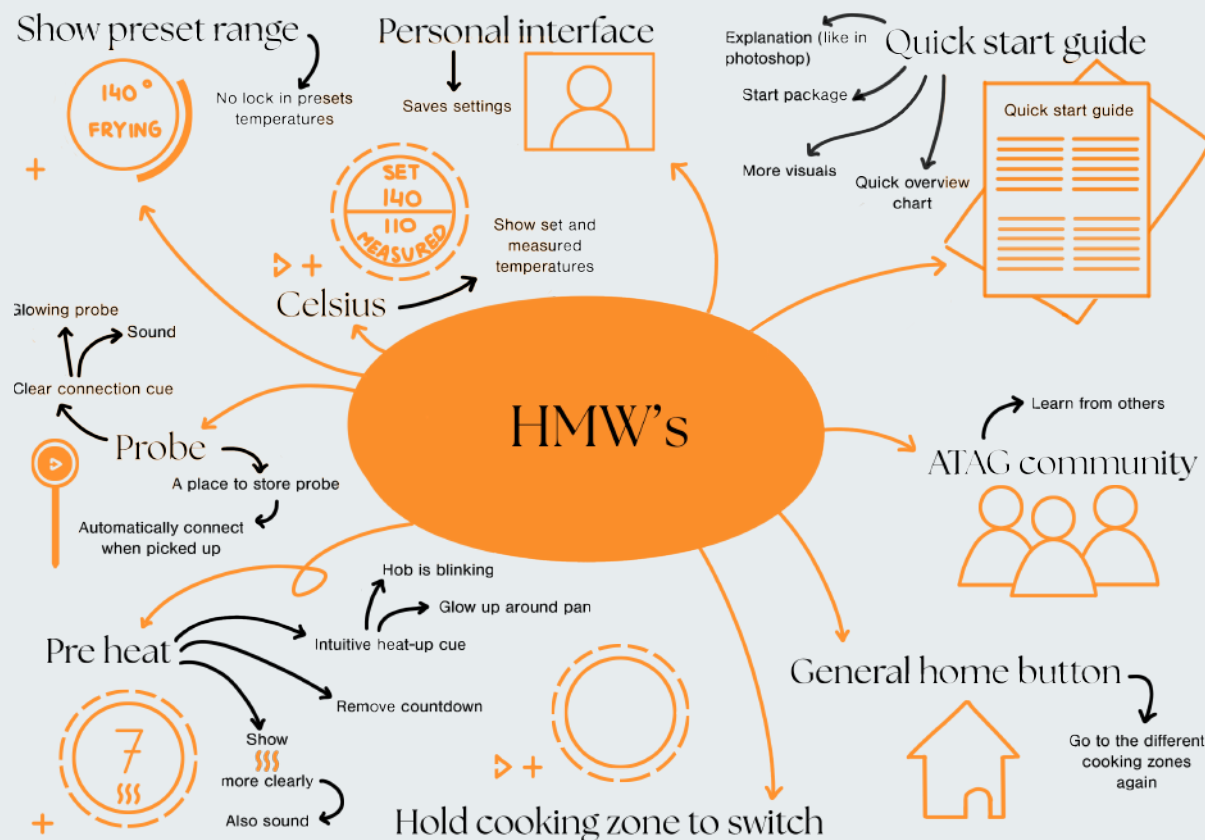


Figure 10: Visualisation of the HMW's

Figure Storming

What

In Figure Storming, one takes on the persona of a well-known Figure and tries to imagine how they would approach the world. In this case, how the selected person would want to use a hob. The used Figures were: Donald Trump, Oprah Winfrey, Cristiano Ronaldo, Kim Kardashian and Gordon Ramsey.

Why

It is helpful to think beyond the conventional ways of using a hob. This approach has helped us overcome the limitations of our minds and generate innovative solutions for various real-life scenarios. To achieve this, we considered different Figures who might have distinct requirements and preferences when it comes to using a hob.

Outcomes

After obtaining the Figures, we conducted five rounds of brainstorming sessions. In each round, every member was allotted five minutes to write down their ideas on how a specific Figure would utilize the hob. After which we concluded the session how we could translate the personality points into functionality points for the hob. The main valuable resulting ideas from these brainstorming sessions and conclusions are presented below. For the complete brainstorm sheets, please refer to Appendix C.



Donald Trump

- Quick recall settings with clear visual cues, such as glowing plates.
- A home button
- A straightforward boost button to automatically activate the hob's boost mode in the primary power level modes.

Figure 11: Visualisation of the Donald Trump



Figure 12: Visualisation of Oprah Winfrey

Oprah Winfrey

- The hob inspires.
- Motivate users to share recipes, along with presets used and cooking tips.
- The hob has a pre-heat breath like animation that creates a relaxing atmosphere.
- All sounds are subtle.



Figure 13: Visualisation of Cristiano Ronaldo

Cristiano Ronaldo

- Quick, healthy meals for someone that can be clumsy in the kitchen.
- The buttons should be bigger.
- It should connect to his food plan in a one button experience.
- PLUS cooking allows for changeable temperatures.



Figure 14: Visualisation of Kim Kardashian

Kim Kardashian

- The settings should be easy icons.
- Living in a big family, so clear safety locks.
- Recognizable health settings, and simple icons allowing promotion.
- A monitoring system for slow cooking while showing the health benefits.



Figure 15: Visualisation of Gordon Ramsay

Gordon Ramsay

- The hob must keep up with him, rapid connection within Celsius Cooking.
- He glance quickly and discern what is happening, so displaying the current mode, the current and target temperature.
- The surface of the hob should be easy to clean.

Eidetic Image

What

This visualization brainstorm technique involves visualizing the current interaction and proposing a new intention based on it while altering it to one of the interaction vision qualities. Each person then visualizes what they will create for the product to facilitate that interaction quality.

Why

By setting our desired interaction qualities as intentions, this method helped us to focus our ideas within the scope of our intended goals. We were challenged to think more strategically, allowing us to approach the brainstorming in a new and creative way.

Outcomes

We drew a lot and came up with many different ideas from this brainstorm, the images of the brainstorm are presented in Appendix D. One example card for the quality 'In control' is presented in Figure 16.

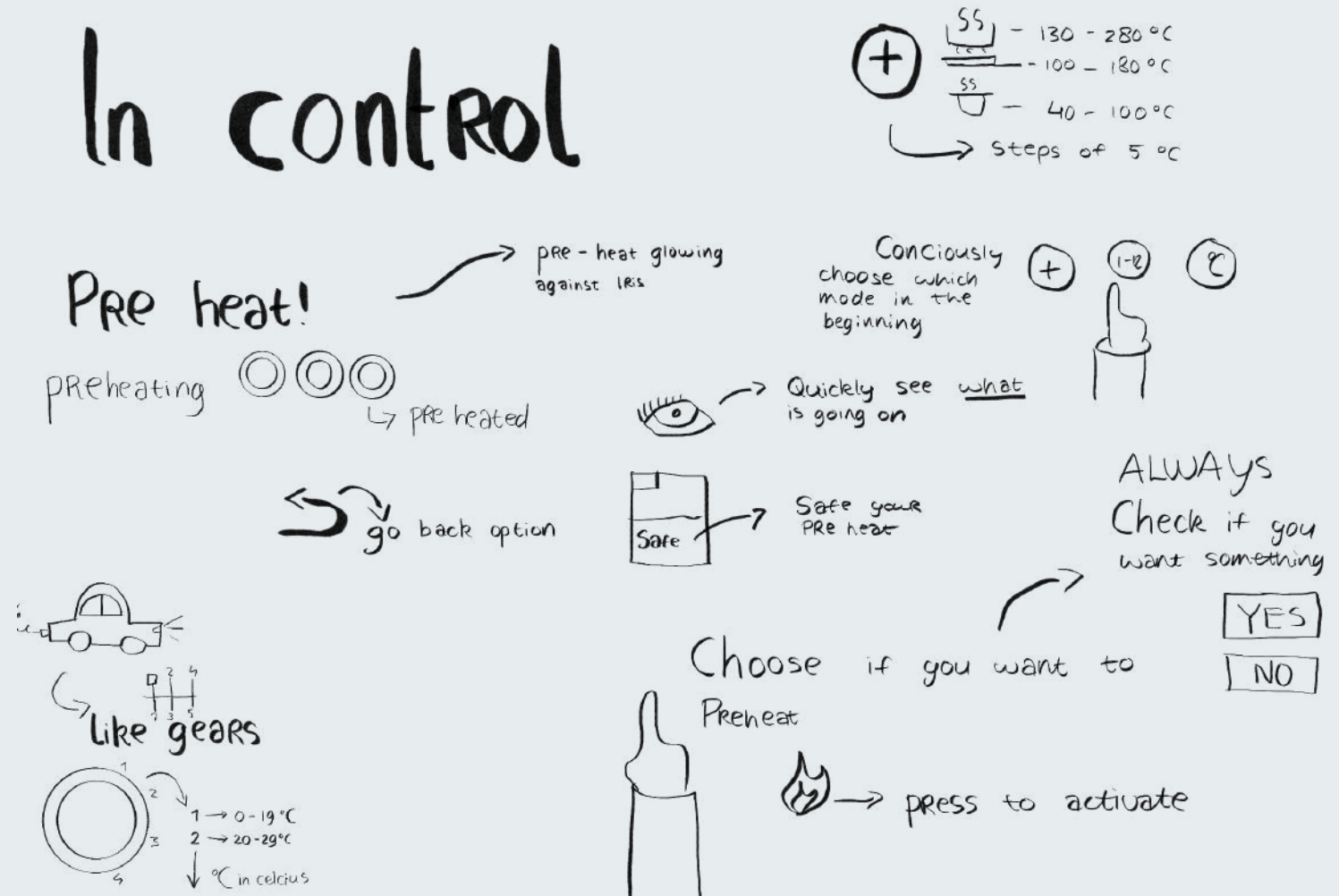


Figure 16: Visualisation of the Eidetic Image brainstorm around the theme 'In control'.

What if?

What

The “what if” brainstorming technique asks “what if” questions that explore different possibilities and potential outcomes. It encourages to think creatively and consider scenarios outside of the usual constraints.

Why

This method helped streamlining our ideas into clusters, which served as a starting point for narrowing down to five interaction concepts. The use of what-if questions ensured that we generated diverse concepts that would work well in diverse settings.

Outcomes

The main valuable resulting ideas from these brainstorming sessions and conclusions are presented below. For the complete brainstorm sheets, please refer to Appendix E. The main what if questions that captured the diversity of our previous brainstroms yet also provoked different creative design directions are presented within the different interaction concept below in Figure 17.

Interaction concepts

The big variety of ideas generated within the different brainstorm rounds formed the foundation for our journey towards an end concept. After the analysis of all the ideas, and the different clusters of the What if’s we created five interaction concepts that each team member is developed

in tandem. This allowed us to discover ideas within different parts of the product, and also with different mindset. Each member created a low-fi prototype that allowed for a quick demo enabling evaluation kickstarting for the converging to the final concept. See Figure 17 for overview interaction concepts.

What if....

Less buttons

The control of the hob would be central around less buttons?

Big family

The hob had personalising, making it suitable for a big family?

Voice control

The control of the would only be with voice control?

Touch screen

The control of the hob would only be with touch screen?

Learning hob

The experience with the hob would be gamified?

Figure 17: Overview of the interaction concepts.

Concept 1: Less Buttons

How

This idea arose from the cluster in the what if question with the core message of: **‘What if the hob had less buttons’?** The idea integrated in the design also arose from the Figure brainstorm of combination of Donald Trump (fast and intuitive interaction) and Gordon Ramsey (a fast hob that picks up the pace of the cook and sees in one glance what the hob is doing).

Why

Our aim is to make the cooking process an exciting and empowering experience while giving the user complete control. To achieve this, we have introduced a central control concept that lets the user choose their preferred mode at the beginning. The home button is always available for them to return to if needed. We have also made some changes to provide unity across the different modes, making it more intuitive. We have improved the product’s communication of the preheat process to make it clearer and guide the user through the cooking process, e.g. show that it is in boost so why the user might be locked. Our use of colors helps to create excitement, making the process understandable and intuitive for the user. This allows regular cooks to experiment with their culinary skills with ease, thanks to the integration of Atag’s new technology.

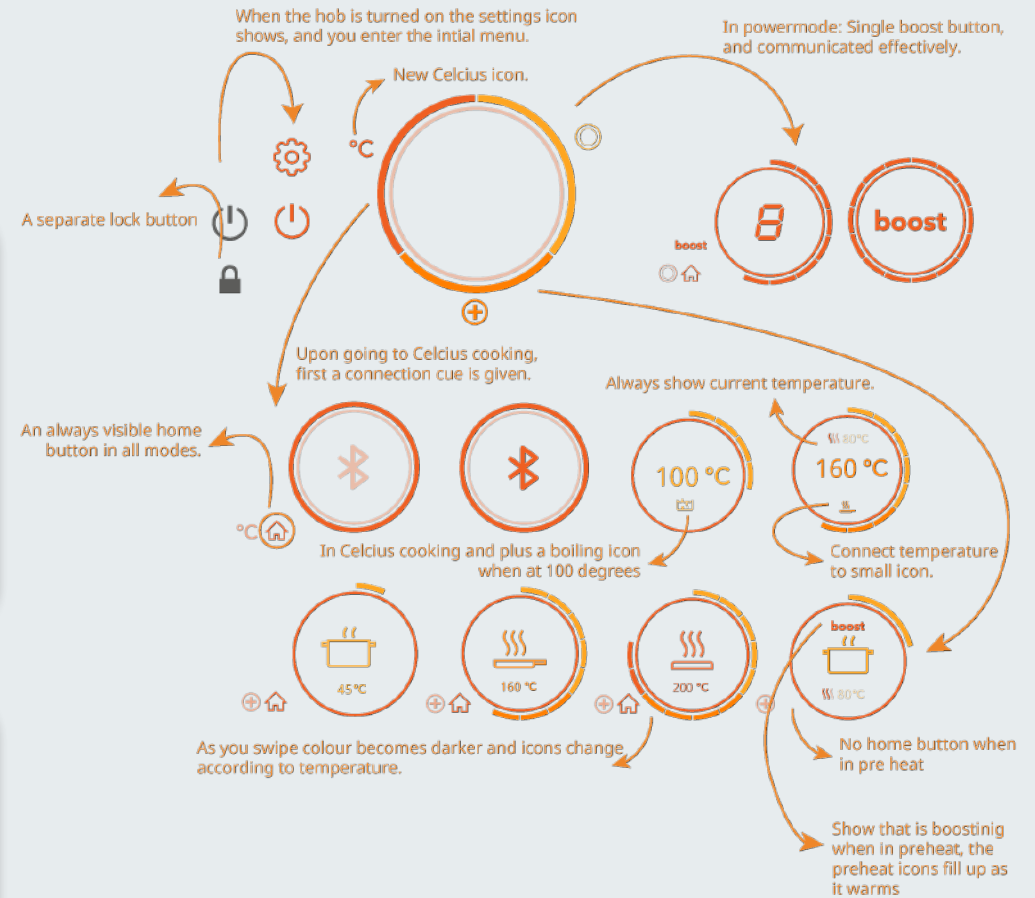


Figure 18: Overview of new features in the concept

What

To implement fewer buttons, the following changes have been made, as shown in Figure 18. The user flow is minimized and tried to be clearer by allowing more one-touch experiences, more visual cues (colours and symbols) and a conscious choice of which mode you are cooking accompanied with a Home button.

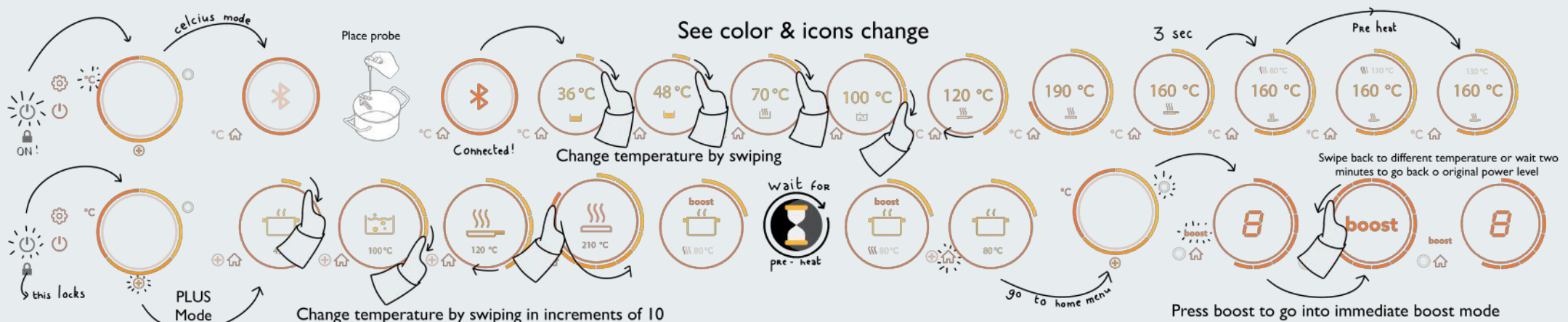


Figure 19: Flow of using the concept 'less buttons'.

Storyboard

This workflow demonstrates only the modified functions and their operations. For instance, the timer and pause functions remain unchanged in this concept and are thus omitted. To showcase the icons' usage flow, simplified pictures have been used instead of illustrating the entire interaction in Figure 19. The purpose of Figure 19 is to provide an overview of the flow of using the icons.

Conclusion & Addressed Pain points

The hob's overwhelming functions interfered with the cooking process during testing. To simplify control, the design incorporates intuitive controls revolving around a single button, color, and icons that correspond to user actions. This allows users to feel in control and select modes with ease, leveraging the power of simplicity.

Concept 2: Big family

How

This concept is based on the problems that were found during testing, mainly in the direction of understanding the userface as an user. During the different brainstorm sessions we found that personalisation was a good way to do that. This also makes it more useful for different kind of users and scenarios. A guest account was another solution that is a good way to make first time use less complex.

Why

In the primary redesign focus, it becomes clear that users are locked in the temperature settings of the PLUS cooking and there is no memory of settings for the Celsius cooking. This concept takes care of both of these problems. In the extended redesign focus, the placement of the PLUS and Celsius buttons is also confusing. By enabling users to determine how much is shown, the interface becomes less overwhelming and confusing. All the improvements together should improve the product on all predefined interaction qualities.

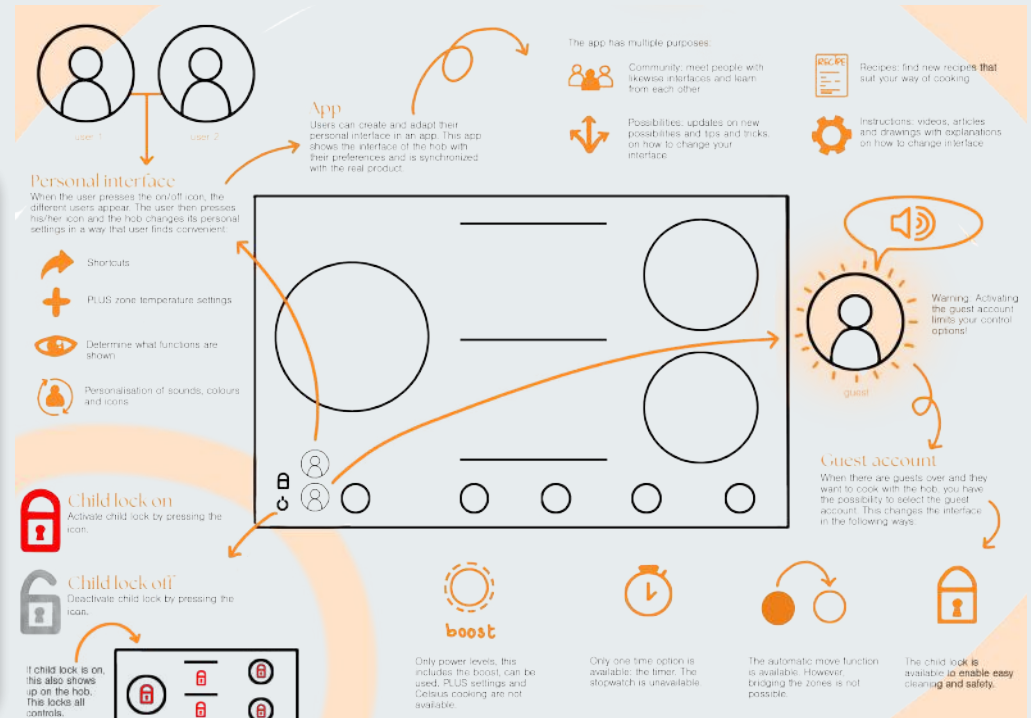


Figure 20: Visualisation of concept 2: personal hob.

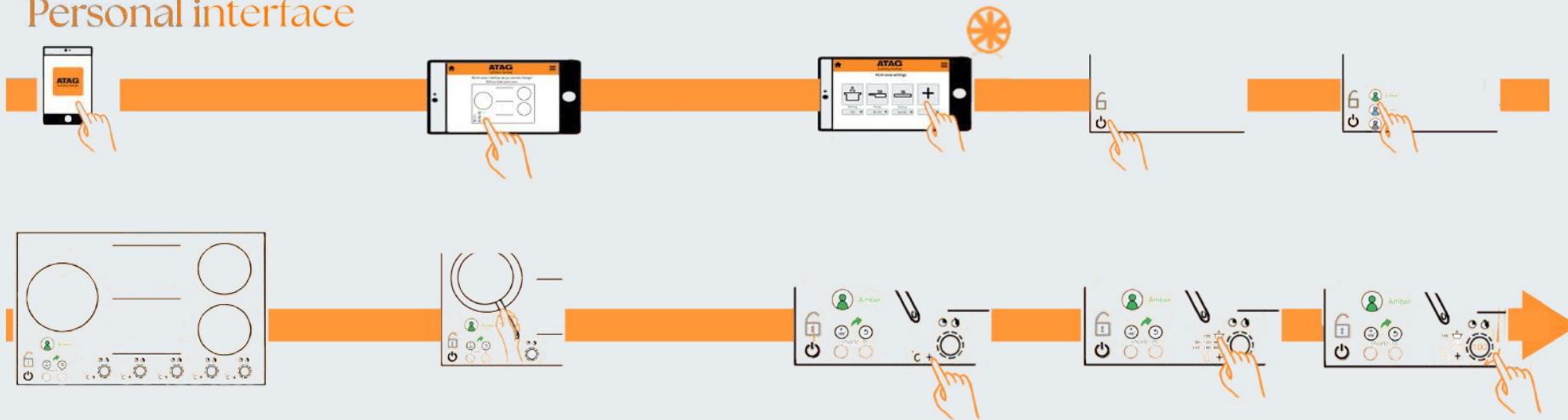
What

The concept consists of 3 main improvements: the child lock is moved outside of the on/off button, users can personalise the interface of the hob and there is a guest account with a simplified interface. Changing the personal interface can be done in the app, see Appendix F for a walkthrough of the app.

Storyboard

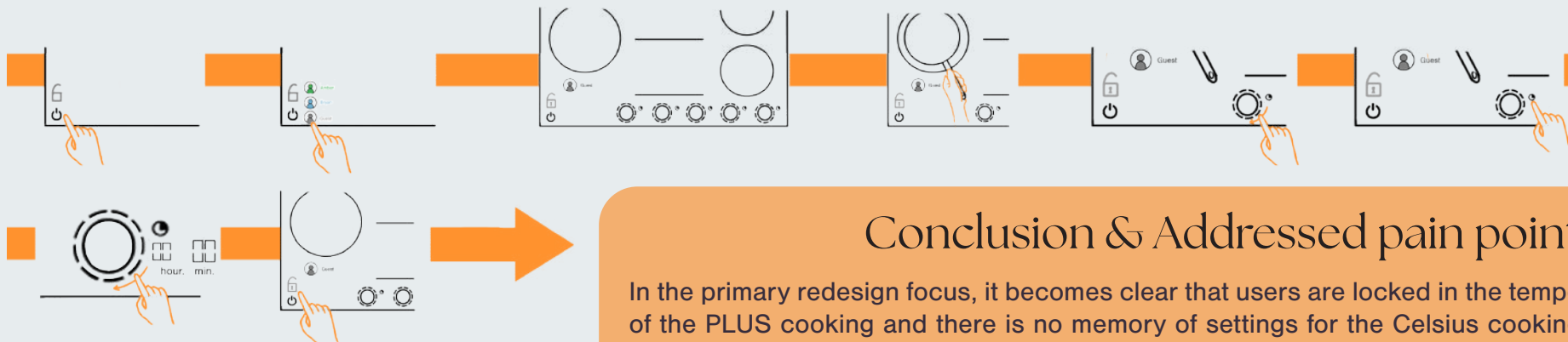
The Figure 21 depicts the interaction flow of the concept.

Personal interface



☀ In the app other changes are also possible: personalisation of icons and sounds, make new shortcuts, determine what functions are shown. This adds another PLUS setting preset. The app also has a community, shows recipes and has instructions.

Guest interface



Conclusion & Addressed pain points

In the primary redesign focus, it becomes clear that users are locked in the temperature settings of the PLUS cooking and there is no memory of settings for the Celsius cooking. This concept takes care of both of these problems. In the extended redesign focus, the placement of the PLUS and Celsius buttons is also confusing. By enabling users to determine how much is shown, the interface becomes less overwhelming and confusing. All the improvements together should improve the product on all predefined interaction qualities.

Figure 21: Storyboard of Concept 2: 'Big family'.

Concept 3: Voice Control

What

This concept's **core** is that the **hob is completely voice controlled**. As a result of our last brainstorming session, a concept emerged focusing on a cluster containing 'What If' questions related to voice control (Figure 22).

How

The main brainstorming activity that inspired this concept was 'Figure Storming', where **Oprah** was one of the figures. Despite the activity's intention to imagine the figures using the hob, Oprah's personality was perceived as the hob itself. As a result, we imagined the voice of her to be **a gentle and calm companion while cooking, as well as clear and motivating**.

Why

This concept was designed with a focus on the **guiding** and **in control** interaction qualities of our interaction vision. The addressed pain points cover the exact functionalities taken into account for these qualities.

Making it **intuitive** and **exciting** was where the...

- tone of voice
- structure of sentences
- clear/informative nature of the hob's response

...comes into play. Along with the fact that you get to talk with your kind and friendly hob, of course.

"WHAT IF THE HOB WAS COMPLETELY VOICE CONTROLLED?"

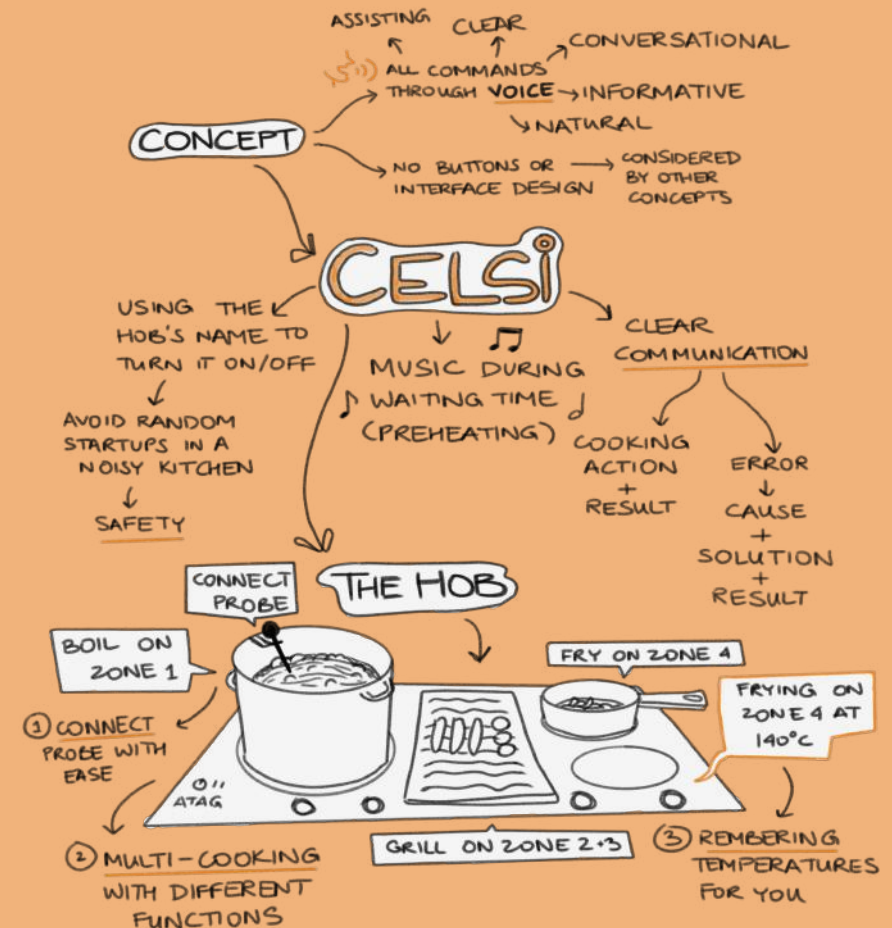


Figure 22: Visualisation of the concept voice control.

Conclusion & Addressed Pain Points

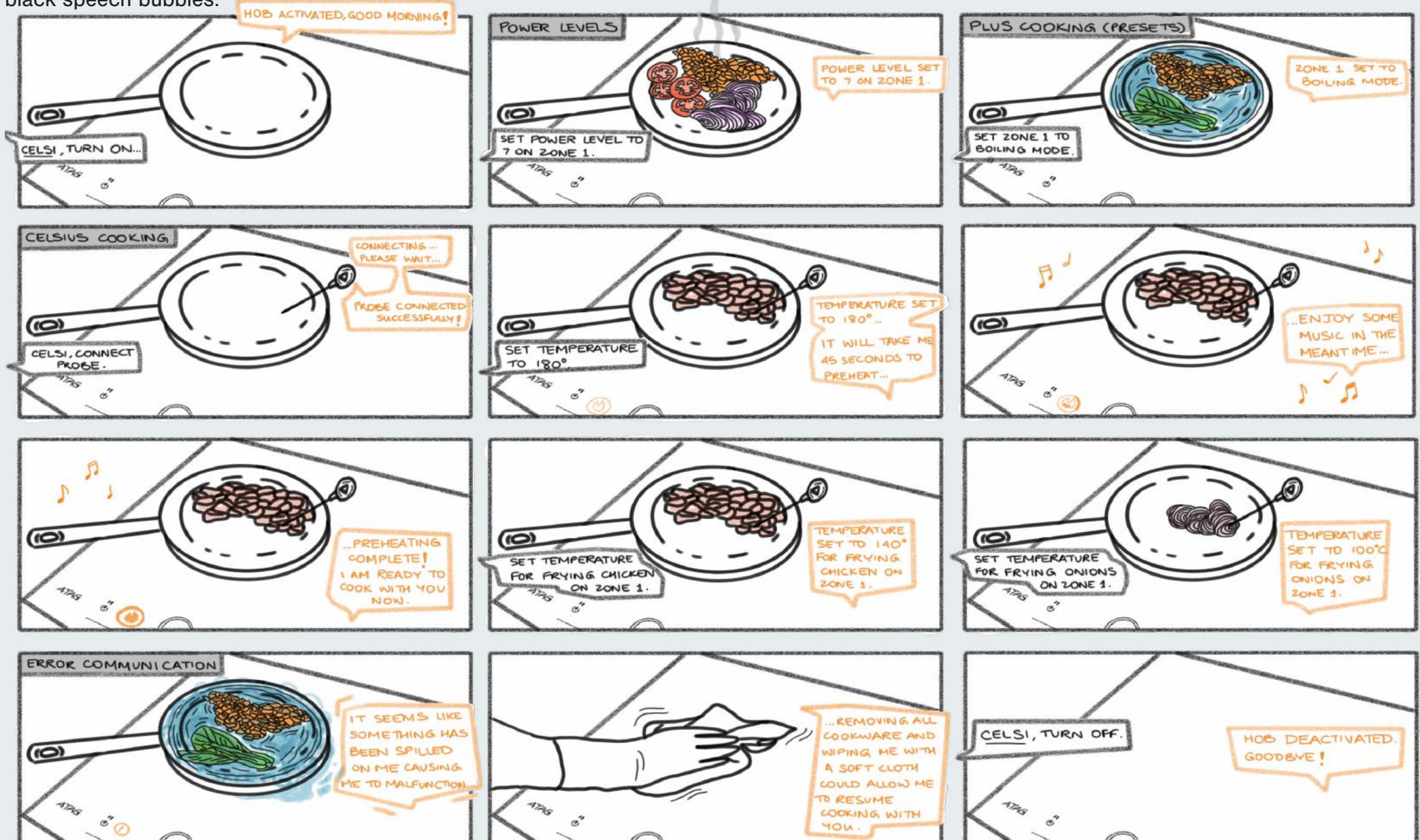
- **Controlling presets or power levels** through voice commands
- **Connecting the probe** through voice
- **Lower cognitive load** on user for remembering temperatures in Celsius precise cooking (like temperature required for frying onions or chicken)
- **Clear communication** of
 - cooking action and task completion by the hob
 - error and related solution (like spills, for example)
- **Multi-cooking** on different zones with different functions smoothly

In conclusion, this concept focused on voice activation and responsiveness.

Storyboard

Figure 23 shows the interaction flow with the concept's voice 'Celsi', whose responses are in orange, to the user whose voice is in the black speech bubbles.

cooking with CELSI



Concept 4: Touch screen

How

This idea is inspired on one of the What if questions explained previous. The question follows: **'What if the hob was a touch screen'**? In this question we refer to touch screens, such as in mobile phones. The concept is an integration of ideas from the Figure brainstorm and eidetic image.

Why

As said in our scope, we want the user to feel more in control and following their intuition while cooking, as the hob is guiding them. The biggest changes of the product are in making the user more conscious in sudden choices, by guiding them better. Not showing inactive functions of the hob and only buttons that can be used, leading to the hob being more intuitive. Giving the user the ability to change the temperature of the presets in the plus mode, so the user is more in control of the hob. These Celsius cooking is now an addition to the plus cooking, using the same presets as guide.



Figure 24: Visualisation Concept touchscreen.

What

In Figure 24, the lay out of the whole redesign is presented, in which the touch screen with and without a pan detected and its basic functions are shown. The settings can only be used when there is no pan detected. The timer, boost and return buttons can be used in all three the cooking modes. Refer to Appendix G for a more elaborate explanation of the concept.

Storyboard

Figure 25 shows the user interaction with the concept.

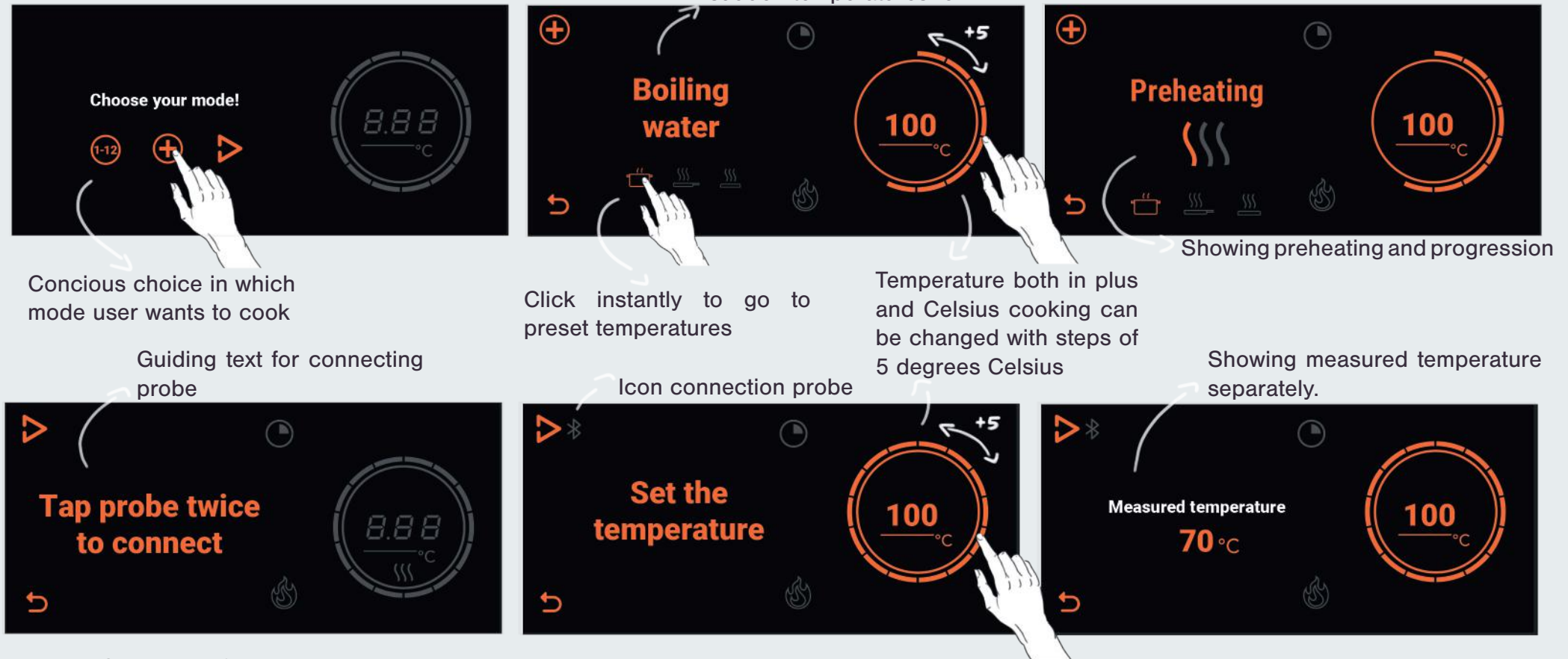


Figure 25: Story board of concept 'touch screen.'

Conclusion & Addressed Pain points

In the primary redesign focus, the main pain points were about people feeling stuck in the preset temperatures in the plus cooking. The preheating countdown was quite stressful. This redesign solves these problems by giving users the freedom to explore the plus cooking and to go a step back. The user will feel more relaxed because of the removed countdowns. In the extended redesign focus, the wifi, pause button and setting were confusing. As well as the placement of the plus and Celsius cooking. The hob is now more intuitive by only showing which options can be used. Removing the wifi and pause button, but adding a clear setting button. The placement of the plus and Celsius cooking is simplified and users are guided with clear text.

Concept 5: Learning the hob

How

In the “what if” stage, we explored ways to boost user engagement. We realized that a quick-start guide could swiftly introduce users to the basics. To further engage the community, we introduced gamified elements, like turning cooking into a game to enhance enjoyment. We plan to develop a **comprehensive quick-start kit**, which would include a manual integrating tutorials, usage logs, and community interactions through an app or interface, encouraging experimentation and enhancing user satisfaction.

Why

This concept enhances user comfort through intuitive guides and gamified elements, allowing users to quickly learn how to use the system. This approach gradually boosts their confidence and encourages the use of more advanced features.

How it fits the design goal and interaction vision:

Guiding: Clear instructions help users easily understand and use the ATAG cooking plate, from basic to advanced functionalities.

Excitement: Successfully completing cooking tasks and leveling up brings a sense of satisfaction and pride.

Intuitive: The user interface is intuitive, enabling users to rely on their instincts for operation.

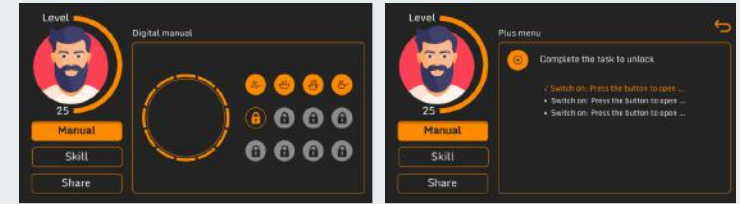
Confidence: A clear guide and user-friendly interface boost users' confidence in their cooking abilities.

Digital manual

The manual of the hob is digitalized, and explained below.

Guide system

Users can freely use the hob or follow systematic learning through levels provided in the guide



Record of use

Records the number of times users utilize these functions.



The usage count is converted into an experience bar recorded in the profile, enhancing the user's experience level, to motivate users to use advanced features.

Atag community

Allows users to share their hob profiles with each other



The goal is to create a community where everyone can share new uses for the hob or help each other solve problems.

Interaction flow

 From hob interface to the digital manual

Figure 26 shows how the interface on the mobile app interacts with the buttons on the hob when the user is using it. When a button on the hob is pressed, the experience points are briefly displayed around the button, and then the data of the experience points are recorded on the mobile app. For the full interaction flow see Appendix H.



Figure 26: The user flow of the concept 'learning the hob'.

What

This concept primarily focuses on creating a **digital user guide**, which displays three main features in its simple interface: tutorial for using features, usage record, and sharing ATAG profiles.

Storyboard

Figure 27 shows the user scenarios of the digital manual

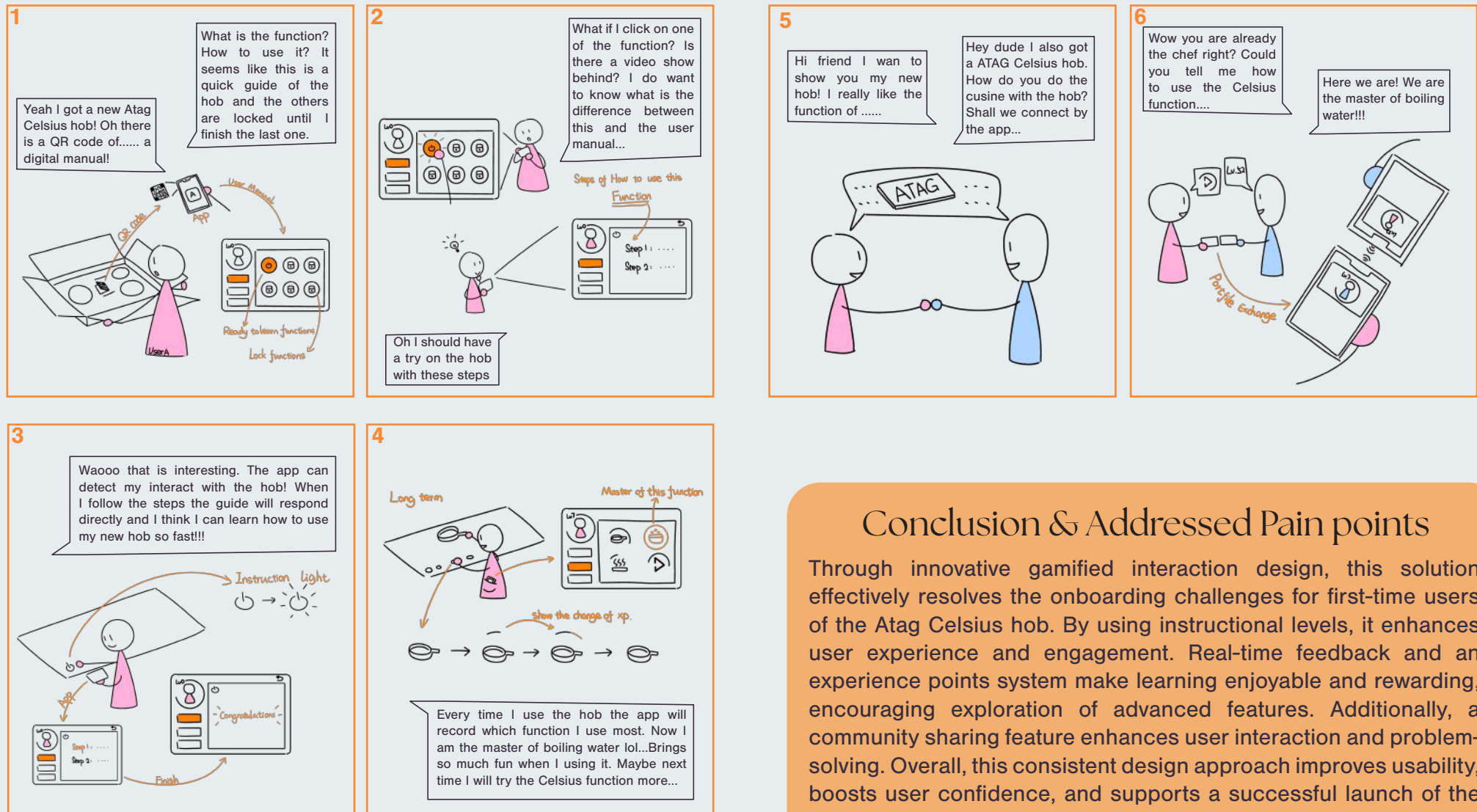


Figure 27: The storyboard of the concept 'learning the hob'.

Conclusion & Addressed Pain points

Through innovative gamified interaction design, this solution effectively resolves the onboarding challenges for first-time users of the Atag Celsius hob. By using instructional levels, it enhances user experience and engagement. Real-time feedback and an experience points system make learning enjoyable and rewarding, encouraging exploration of advanced features. Additionally, a community sharing feature enhances user interaction and problem-solving. Overall, this consistent design approach improves usability, boosts user confidence, and supports a successful launch of the Atag Celsius hob.

Conclusion

To come up with ideas that would improve the ATAG Celsius hob within the predefined design scope, several ideation methods were used. From these different ideation sessions, 5 concepts were formed that address the identified pain points. The 5 concepts will be evaluated by the other team members and by using a converging method the strong points of the 5 concepts will be converged into a redesign concept.



3 ■ Evaluation of Concepts

After gathering many ideas and converging them into concepts, in this chapter we evaluate the proposed interaction concepts, to allow us to move to one central concept in the next chapter. We will evaluate the concepts based on previous pain points of the products (see report Phase 1) and a list of requirements. After which we conclude on the strong points of all concepts to move towards one central concept.

Quick Chapter glance:

Evaluation

- List of requirements
- Interactive Demos
- Evaluation Overview
- Harris profiles

Redesign Scope



List of Requirements

To be able to make better guided decisions on the concepts we came up with, a list of requirements is made, see Figure 28. These requirements are based on the interaction qualities we have defined in report 1.

We combined the interaction quality exciting and confident, because to us they go hand in hand. The way the interaction qualities are placed in Figure 28 shows their importance to us (left = most important, right = least important). Per interaction quality there is a list of requirements and their importance is shown in the order they are listed (top = most important, bottom = least important).

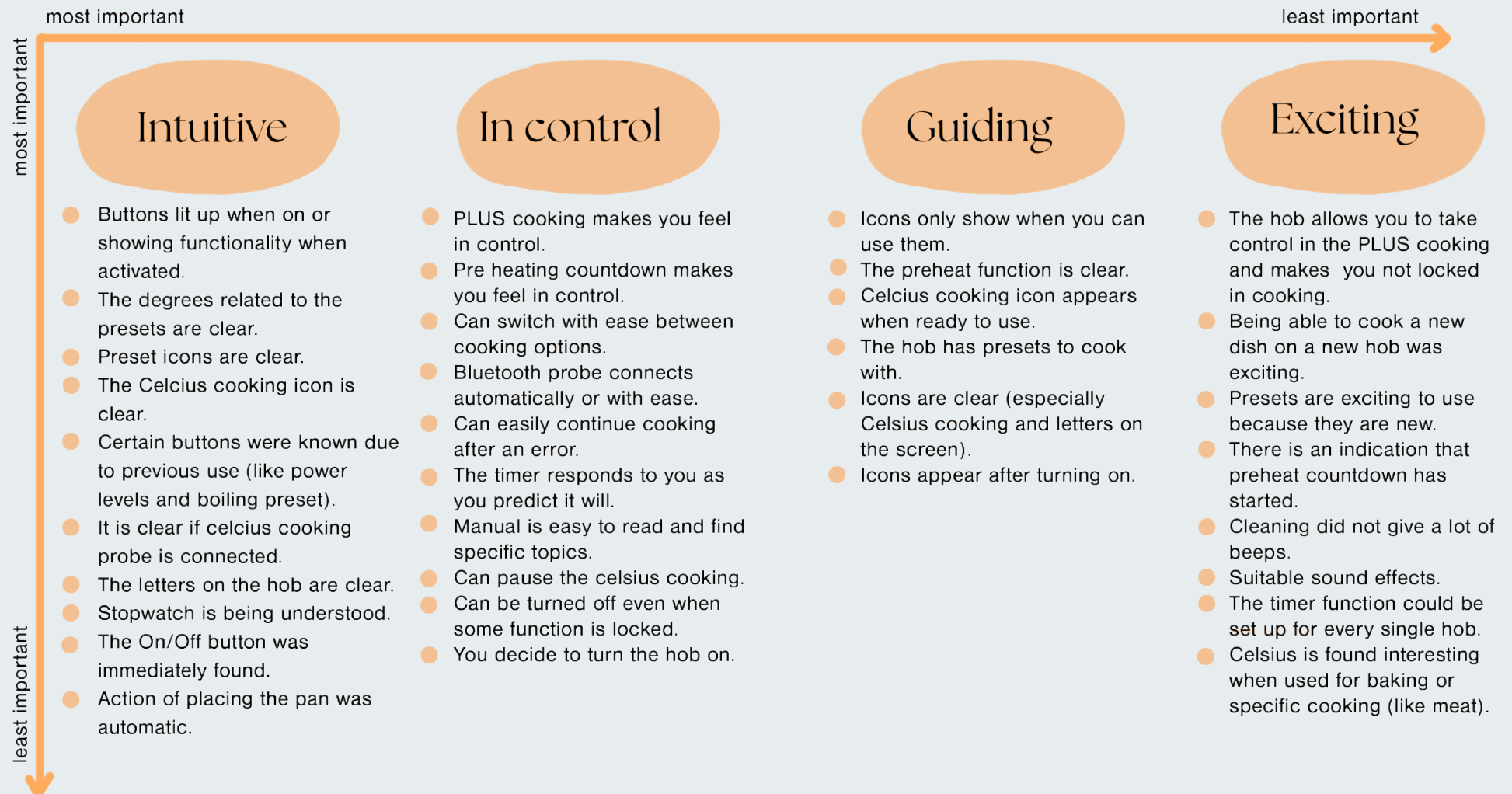


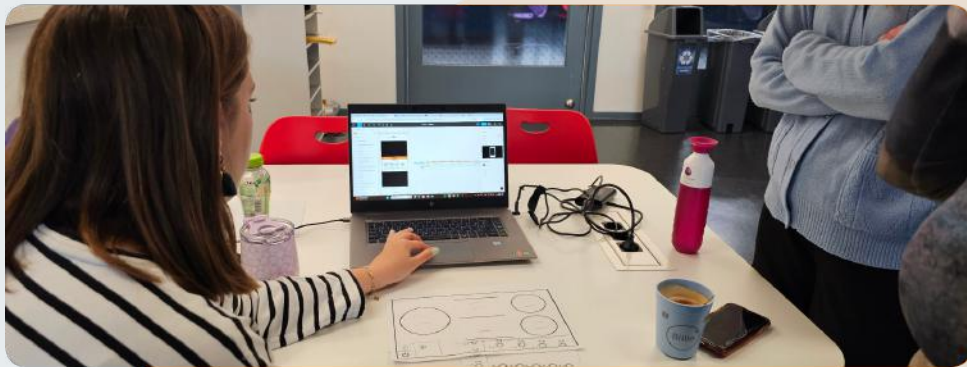
Figure 28: The list of requirements.

Interactive Demos

Low-fidelity version of each concept (Chapter 2) was prototyped and presented to the remaining four teammates, who evaluated it. These demos were explanatory and playable tours of various redesigned elements from each prototype, see Figure 29.



C1: Less Buttons



C2: Big Family



C3: Voice Control



C4: Touch Screen



C5: Learning the Hob

Evaluation Overview

To delve into the evaluation process, we identified four key steps and two additional ones to reach the redesigned proposal. The process overview is depicted in Figure 30.

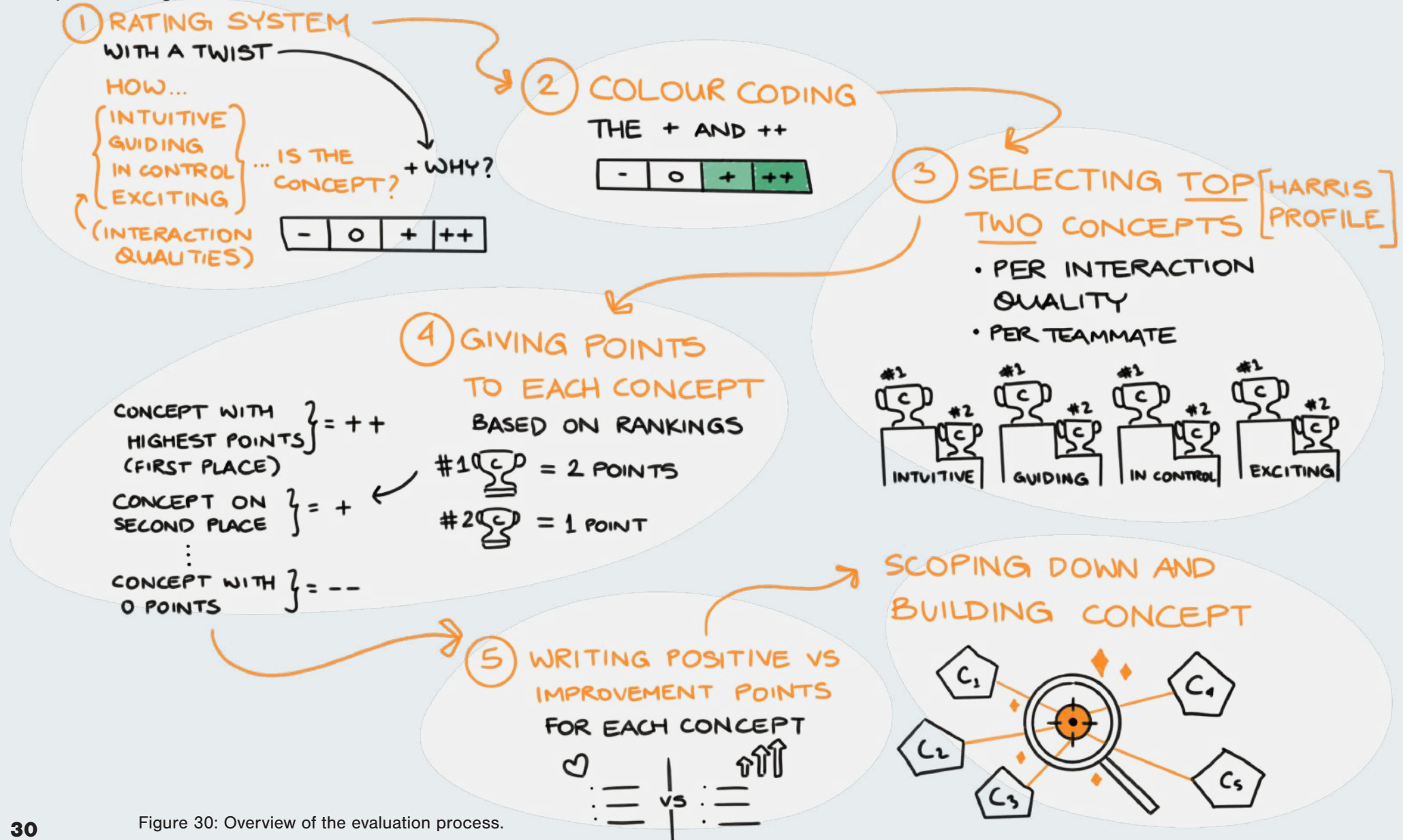


Figure 30: Overview of the evaluation process.

Harris profiles

As seen in the 'Evaluation Overview' (Figure 30), we started the process by rating each concept based on our interaction qualities (**Step 1**), while also supporting it with the reasons behind our ratings (they 'why'). It helped us reflect on **what aspects of each concept we thought were aligned with the list of requirements** by presenting them in a structured table (Appendix J). The order of importance of the requirements was considered for the evaluation process, which can be seen in Chapter 3.

After colour coding the ratings (**Step 2**), we were able to visually see **which concepts were the strongest in which requirements**. Using Harris Profile as the core method allowed us to individually select our top two concepts per interaction quality. This was done as a **starting point of converging** strong points of all the concepts to one main concept (**Step 3**). The steps 1 to 3 are further detailed out in Appendix J.

To continue the convergence process and rating system, we gave **points for the concepts** based on their previous rankings (**Step 4**). The concept which was ranked the highest earlier, was given 2 points, the second highest was given 1 point, and so on. This gave us a **clearer idea of which concepts did better** with respect to certain interaction qualities.

In order to ensure that we **do not skip out on important design points** considered by the other concepts, we documented the **positive vs improvement points (Step 5)**, which was taken into consideration for the final step - scoping and building our initial plan for the final concept.

Appendix J showcases the Harris Profile's complete matrix from every step of the evaluation process in more detail.



Concept 1 - Less buttons

	-	-	+	+
Intuitive				
In control				
Guiding				
Exciting				

Figure 31: The Harris Profile of concept 1: 'Less Buttons'.



Strong points

- The initial conscious choice of your cooking mode
- Change of icons as you swipe a long
- Distinct boiling icon
- Preheat animation with showing that it is in boost mode
- Showing measured and current temperature
- Visual cue for connecting the probe
- Settings icon
- The single boost button on the hob



Pain points

- The colors could have a bigger opacity difference
- The steps in temperature in PLUS mode should be bigger
- Clearer distinction of CCelsius and PLUS
- Missing the integration of the pre sets within the PLUS function



Concept 2 - Big family

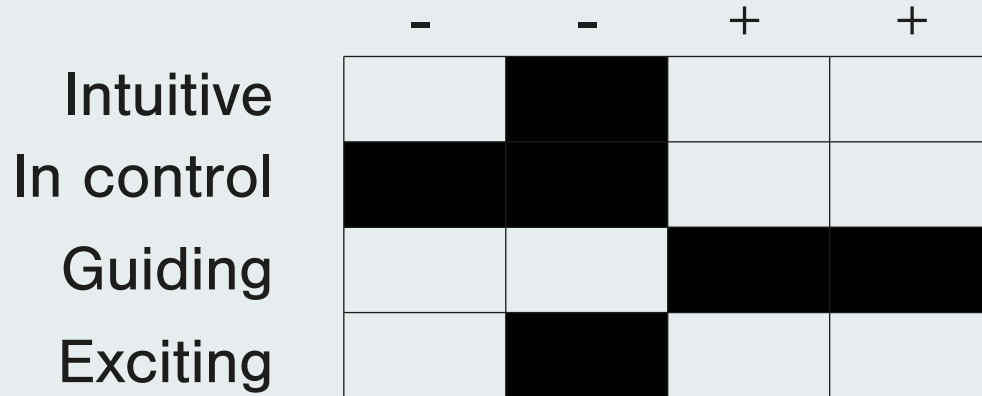


Figure 33: The Harris Profile of concept 2: 'Big Family'.

+ Strong points

- The guest function was a nice differentiation
- Nice that you could set the preset temperatures for yourself
- Nice that the app had the instruction videos
- Icon of back and Celsius were nice

- Pain points

- The lock on every hob could be confusing since people might want to press it (and this can be hot)
- Seeing so many functions at the same time might be an overload for the user
- The timers on top of the IRIS slider looked a little cramped
- It could be confusing to have the temperatures displayed next to the presets



Concept 3 - Voice control

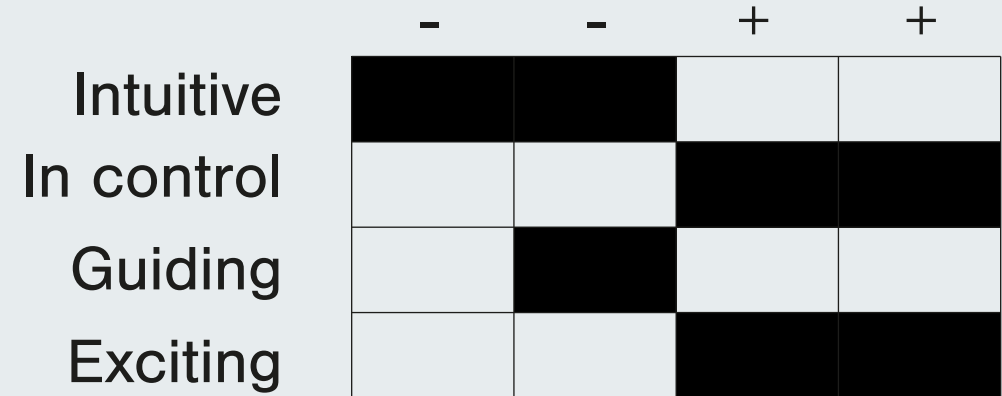


Figure 32: The Harris Profile of concept 3: 'Voice control'.

+ Strong points

- The hob was very in line with what you wanted it to do
- A clear cue that the probe was connected
- No text guidance was needed as it was all audio
- More aware of what the error was and how you could solve it
- All the actions you take are conscious
- Clear cue of the start of pre heat the waiting of the pre heat is made more comfortable
- Presets linked to actions

- Pain points

- The voice might get annoying over time
- Does not make clear you can not use the pan while in pre heat
- Would get chaotic if you have to take multiple actions at the same time
- Will it still work in a loud environment?
- In unlocking the chef you might want more control instead of delegating the task for the hob to do



Concept 4 - Touchscreen

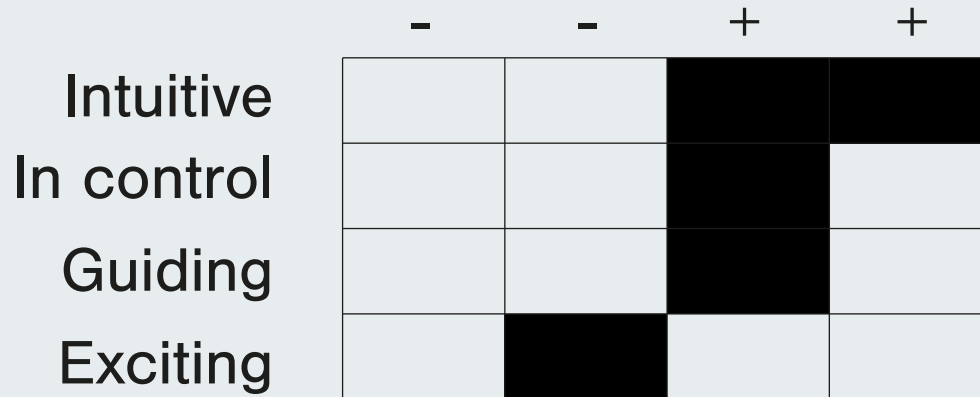


Figure 34: The Harris Profile of concept 4: 'Voice control'.

+ Strong points

- The conscious choice of which mode you will be cooking
- Clear guidance in when you can go back
- You started off from set temperatures in the PLUS cooking
- You were guided in the connection of the Celsius which made it clearer you could start cooking
- Better control in the plus function and showing in which cooking zone you are without being locked
- You only saw functions that actually could be used
- The timer visual in the inner circle improved the quick glance experience

- Pain points

- A lot of text on the interface
- The boost icon could be confused with the pre heat
- Touchscreen on the hob is expensive and hard to control
- Swiping movements can be difficult on a hob
- The size of the screen is hard to integrate on the smaller hob



Concept 5 - Learning the hob

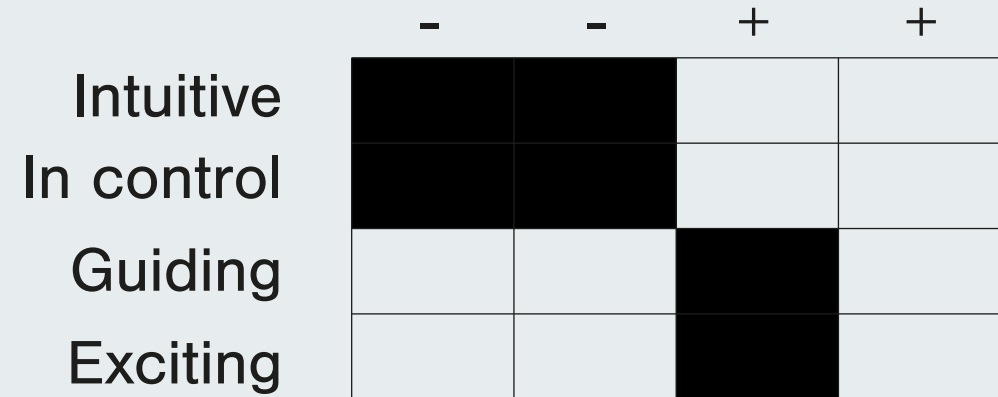


Figure 35: The Harris Profile of concept 5: 'Learning the hob'.

+ Strong points

- It divided the manual in sections, so you had an active choice of what part you want to learn more about
- Explaining the error in a visual way
- Using gamification to encourage user to try the ultra function(Plus/Celsius)
- Building up a Atag communication by recording using history and creating Atag CV which can be shared among users

- Pain points

- The gamefication may doesn't fits ATAGs brand identity
- It would be confusing to have it on the hob interface
- User might not really be interested in the record of use
- The data didn't give so much assistance.

Results of Harris profile

The entire evaluation process with the Harris Profiles clearly shows that **Concept 4 (Touch Screen)** had achieved the best results, see outcome in Figure 36. While all concepts did have important points we have taken into consideration towards the end of our convergence process, this concept had the strongest points in relation with our list of requirements. Consequently, Concept 4 (Touch Screen) served as a **base point to start the final design proposal**, on to which **other concepts' strong points would be integrated** (the exact points considered can be seen in 'Convergent Strategy' - Chapter 4).

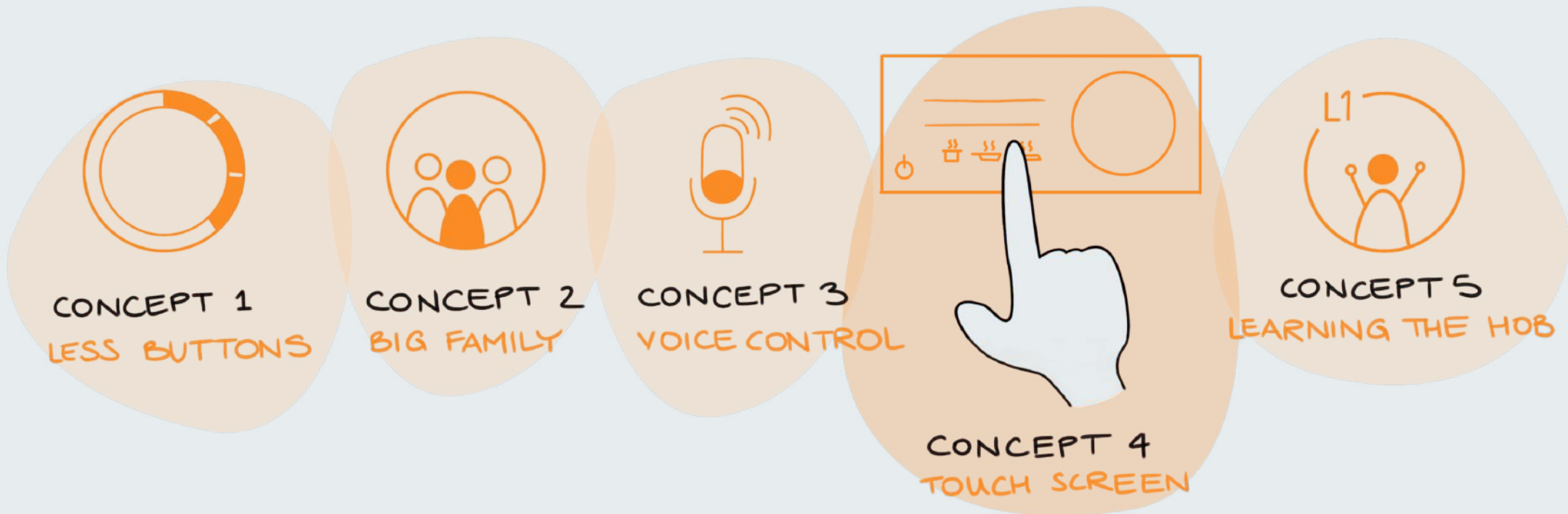
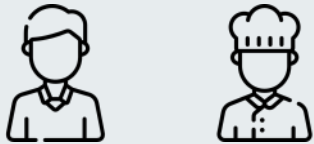


Figure 36: Visualisation of the outcomes of the harris profiles.

Redesign Scope

As we were summarizing our design goals and scope, we realized that there were too many focal points to consider. Hence, we asked ourselves a crucial question: **“What would we change in the user flow to have the most significant impact?”** This question helped us identify the core of our redesigned interface. By doing so, we decided to reanalyze the

user interaction flow and focus primarily on the PLUS mode, which is the most complex and feature-rich section of the interface (see Figure 37). By narrowing our scope, we aim to refine our design and achieve a significant enhancement in user interaction with minimal changes.



Focus

Clearly differentiate between simple cooking and chef-level cooking to meet the needs of both beginners and experienced chefs. With our redesign putting emphasis of the experiential chefs as described in Chapter 1.

MIN

Minimal Interface Changes

Commit to minimal modifications to the user interface but maximize changes to the interaction flow. The design should be intuitive and straightforward, reducing the learning curve and simplifying user interactions to enhance usability.

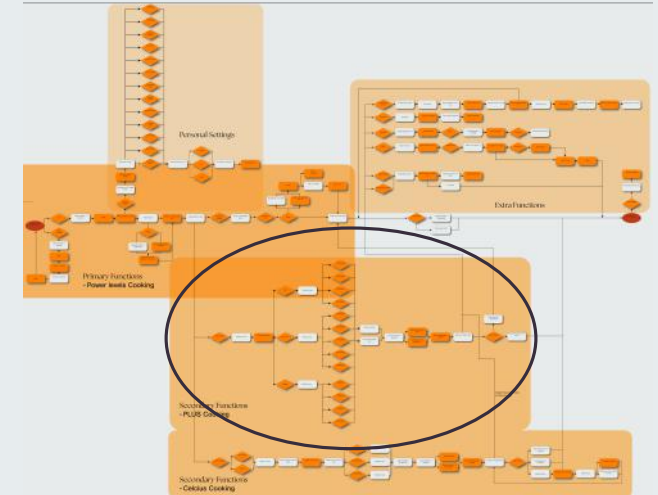


Figure 37: The PLUS user flow circled in the full user flow.



PLUS Mode

Focus on the PLUS mode in the final output, aiming to clearly articulate the intended adjustments in temperature and the interaction between various cooking modes.



Reframing Grill Extension

Grill will not be a separate preset. We incorporate the grill function into the bridge mode settings, using the same presets to ensure consistency. This approach ensures that the grill feature is better integrated, simplifying the user experience.



Integrate Celsius Function

To make the learning curve of the Celsius function in line with the Plus mode we propose aligning Celsius as a distinct feature on top of the PLUS mode, providing advanced temperature control for precise cooking tasks. But within the coherent story of the PLUS cooking

4.

Redesign Proposal

In the last chapter we converged the concepts to one. In this chapter we will elaborate on that design and frame it as the redesign proposal for the new Atag induction hob. For that we will explain how the strong points of the different concepts could be converged into this one design. Moreover, we will explain how this design interacts with the user and show how the interface has been designed.

Quick Chapter glance:

- Current interaction
- Convergent strategy
- The Redesign
- Improved interaction
- Conclusion
- Recommendations
- Next Steps

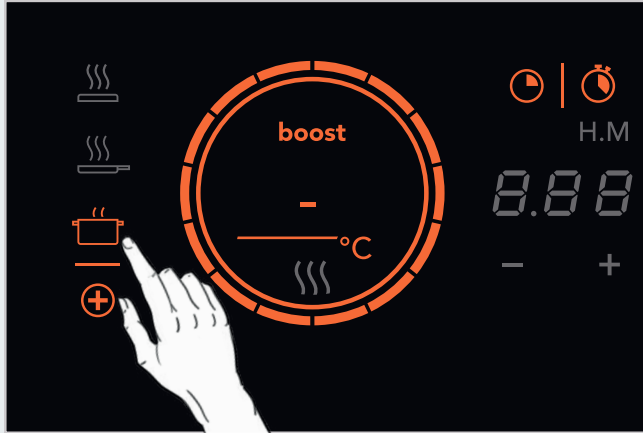


Current interaction

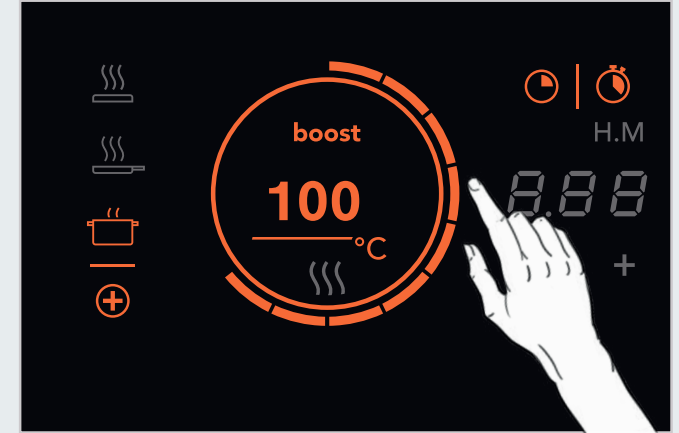
To show the improvement of the interaction, we first want to point out the current interaction, shown in Figure 38. In the next pages we explain the redesign and the new improved interaction.



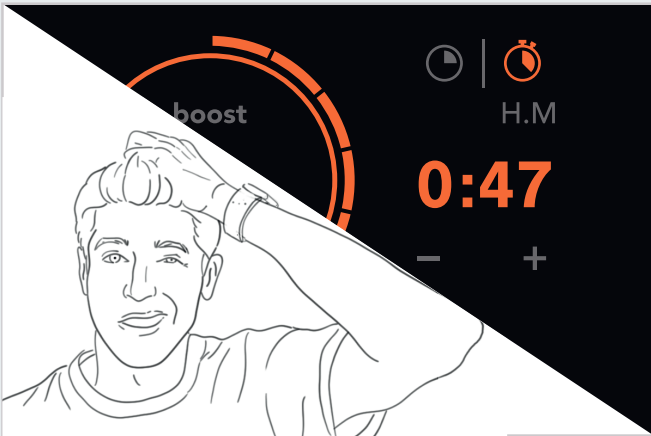
Richard just came home from a long day of work. He is a bit tired but still excited and ready to start cooking for him and his wife.



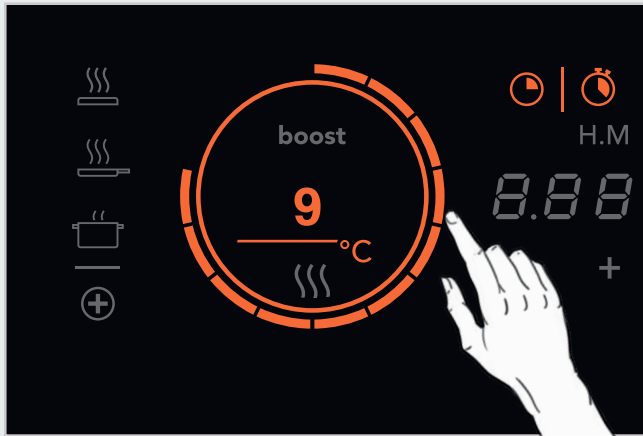
He is excited to give the plus cooking settings a try and starts up the settings. The chicken needs to be baked at 120 degrees Celsius.



When in the plus cooking he is a bit lost in the presets. He cannot find the 120 degrees and cannot change the temperature how he wanted to.



When not changing the settings for a bit to long, all of a sudden the hob starts and a countdown begins. Richard is stressed out, and starts baking his chicken immediatly.



Oh no the chicken is burning and he cannot change the temperature, after turning the iris slider he is suddenly in the power modes. O well, then just using these functions.



Unfortunately, the chicken was burned in the process. The dinner unfortunately resulted in only eating the side salad while feeling stressed and discontent.

Convergent strategy

With the convergent strategy we aimed to take all the strong points of every concept to concise that into one concept that fits our scope, interaction vision and design goal the best. So the design should be intuitive, let the user feel in control, guiding and exciting to use. In the list below, the strong points of the concepts that fits our new narrowed scope as stated in Chapter 3, on which we will later work to build the

final concept, are stated. In the recommendation, later on in this chapter, we elaborate further on the things that are outside the primary redesign focus, but would be valuable to take in account in future Atag redesign iterations.

Strong Points to integrate

- The user made a conscious choice of what mode he wanted to cook with upfront (1,4)
- There was a change of icons as you swiped along the temperatures in the PLUS function, making the user guided instead of locked with a range with that icon preset. (1,4)
- There was a distinct boiling icon, which was clearer product feedback. (1,4)
- The preheat animation communicated to the user that the hob was boosting. (1)
- A seperate settings icon. (1)
- A simplified display for guest cooking or for a quick meal. (2)
- A back button always present to prevent locking (except for when in pre-heat) (1,2,4)
- New improved Celsius icon. (1,2,5)
- All actions were made conscious. (1,2,3,4)
- The cue of the start of pre-heat was made very clear. (1,2,3,4)
- The presets within the PLUS cooking were linked to specific actions and communicated clear. (3,4)
- The presets offered a starting point for the temperature in stead of locking the user in a set of temperatures. (4)
- Better in control in the PLUS cooking since you could swipe along in the tempereatures with increments of 5. (4)
- The hob only displayed the function that actually could be used. (4)
- The quick glance visualisation of the timer. (4)

The numbers in brackets show of which concept they came.

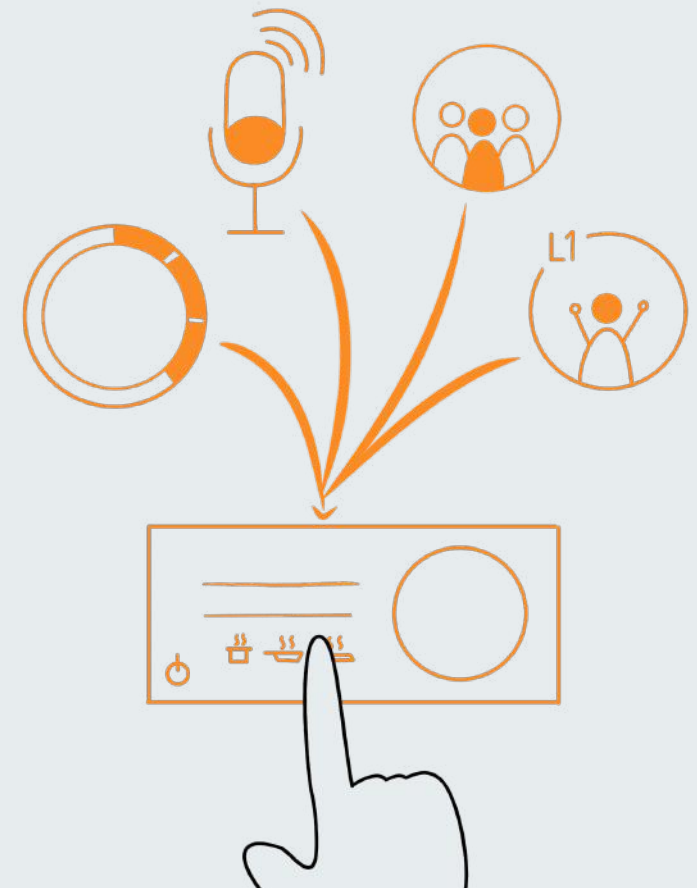
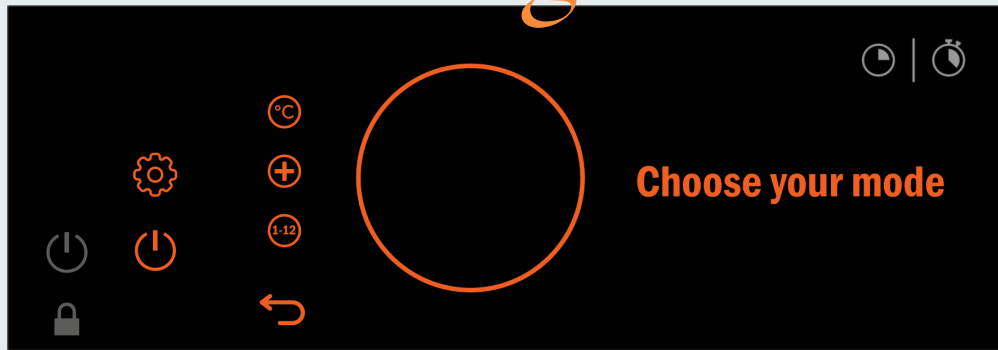


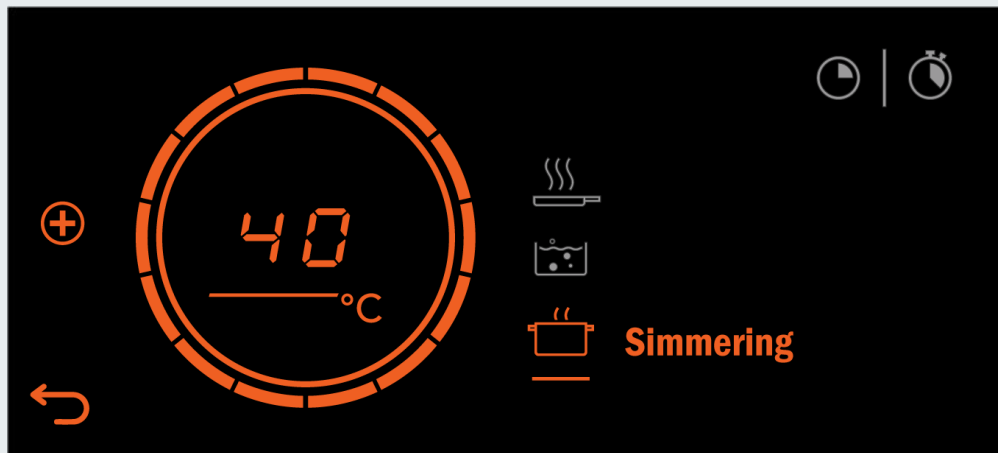
Figure 39: Visualisation of the convergent of the concept in Touchscreen.

The Redesign



Concious choice

When the hob is switched on, users are presented with a range of cooking modes to choose from, enabling them to quickly start cooking or provide a simpler display for guests. This feature empowers users to feel more in control of their cooking style. Additionally, the Celsius cooking icon has been refined for better visibility, and the settings are readily accessible once the hob is activated. To make the lock function of the hob more comprehensible, it has been separated from the on/off button. Instead of the ambiguous 'L' sign, an orange lock will indicate its activation (which is grey if not active), making it easier to understand.



Consistent presets

Once you've selected your mode, it will always be displayed on the left of the IRIS slider. The Celsius and plus cooking modes in the new concept tell a coherent story, as they use the same icons and basepoint temperatures. You can find more details about the preset temperatures on the next page. Additionally, the back button has been added to the display, making it easy to navigate through modes or adjust the temperature setting.

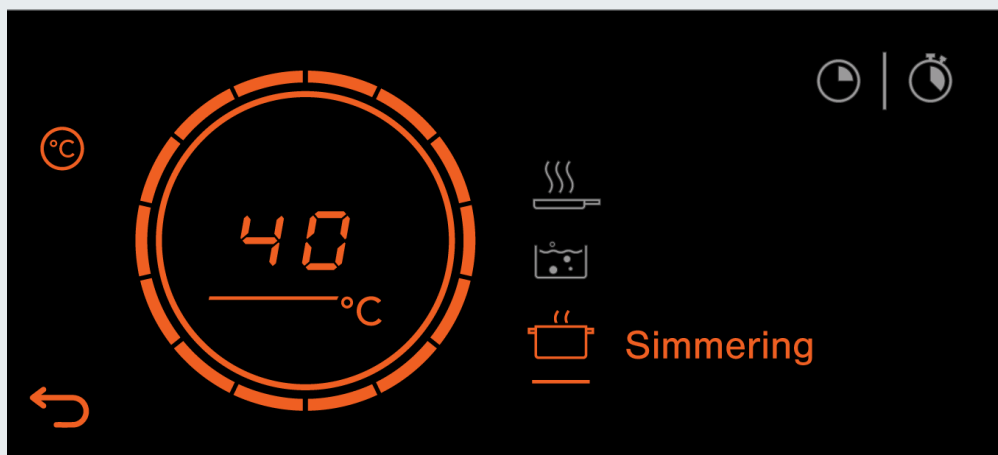


Figure 40: Overview of the changed icons and presets of interface

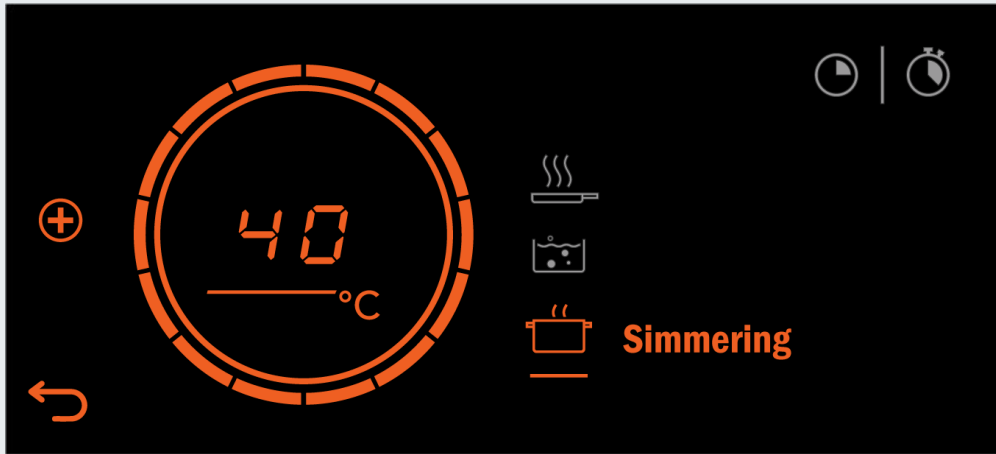


Figure 41: Overview of the presets of interface.

Presets

The display has been improved to make cooking more intuitive and easier to follow. It now provides users with information on various presets, including text next to each icon to help them understand which mode they're in. If you swipe to change the temperature in increments of 5 degrees and go past the start point of another icon, the icon will automatically change to match your intended setting.

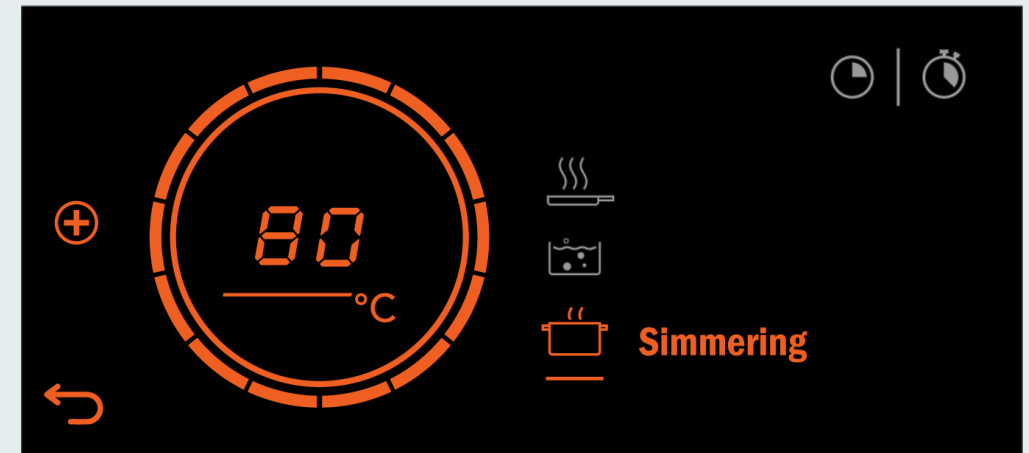
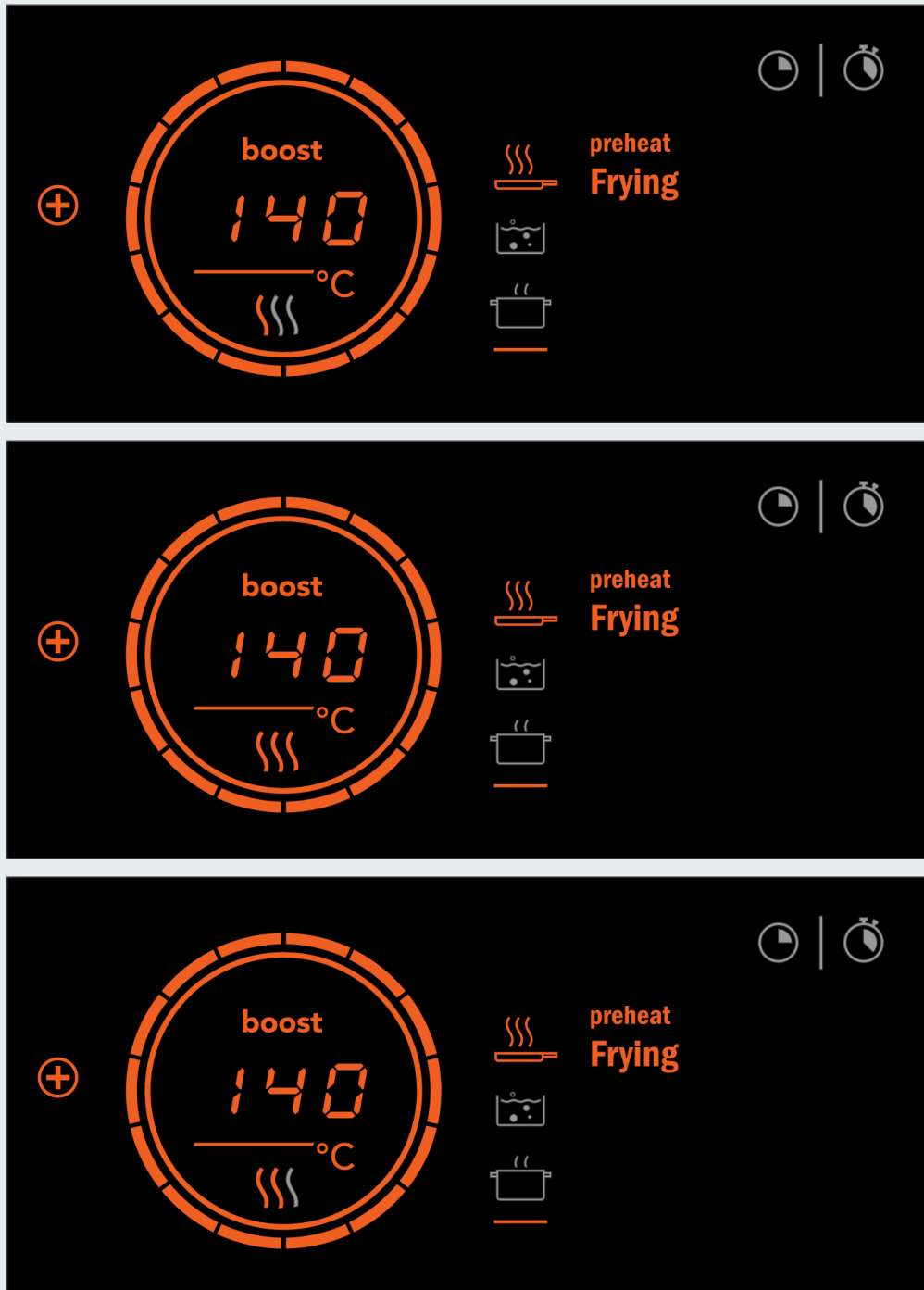


Figure 42: Example temperature outside the preset temperatures.

Setting temperature presets

Additionally, the way the temperature presets are configured has been improved. Instead of providing a fixed set of temperatures, the preset now serves as a starting point for temperature adjustment. For instance, when selecting the frying option, the temperature display automatically shows 140. From this point on, users can increase or decrease the temperature in increments of 5 degrees, giving them a greater sense of control and avoiding being limited to a predetermined range of temperatures.



Preheat

The preheat function has undergone significant changes to make it more user-friendly and intuitive. During phase 1, users felt most a lack of control during the preheat. Due to the fact that the hob is locked during the technical setup of and getting to the temperature of the preset. To address that the user is 'locked', the back button is no longer displayed to communicate this to the user. Additionally, instead of the countdown, which can be stressful, a filling heating sign is now displayed beneath the temperature to guide the user in the process. To indicate that the hob is in preheat mode, this is displayed above the mode they are currently in, To indicate that the hob is in preheat mode, this is displayed above the current mode. Also, to show that the pan is boosting (the reason why the user is locked for a short time), the hob displays the word "boost", keeping the user informed throughout the process.

Figure 43: Overview of the new improved preheat function.

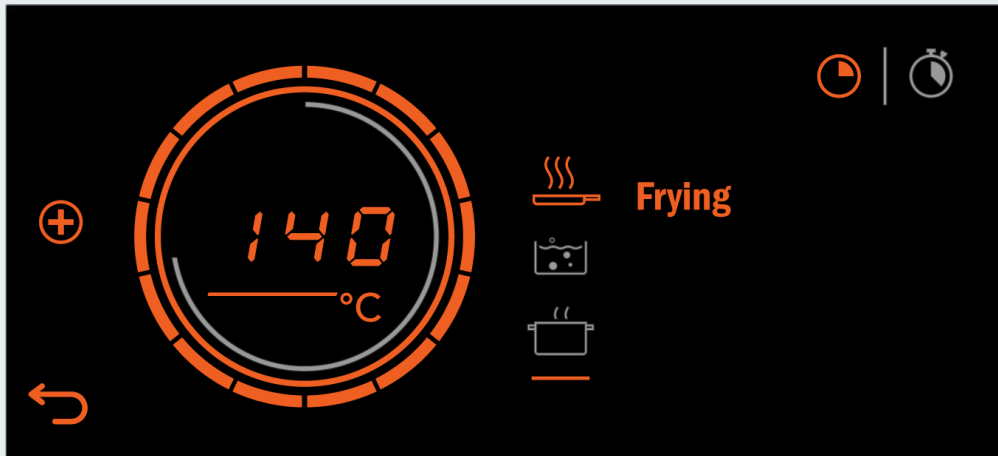


Figure 45: The visualisation of the timer function with only the time bar.

Timer function

Finally, in our effort to improve the cooking experience, we have enhanced the timer function to make it more intuitive and easy to use. The timer works the same way as before, but now a small grey bar is displayed within the IRIS slider to show the progress of time. To reduce cognitive load and avoid stress while cooking, the numbers representing the time disappear and only appear after the timer function is pressed. This means that the primary way to check the remaining time is through the grey bar. This feature allows for a quick glance at the timer's progress and helps users feel excited instead of pressured while cooking.

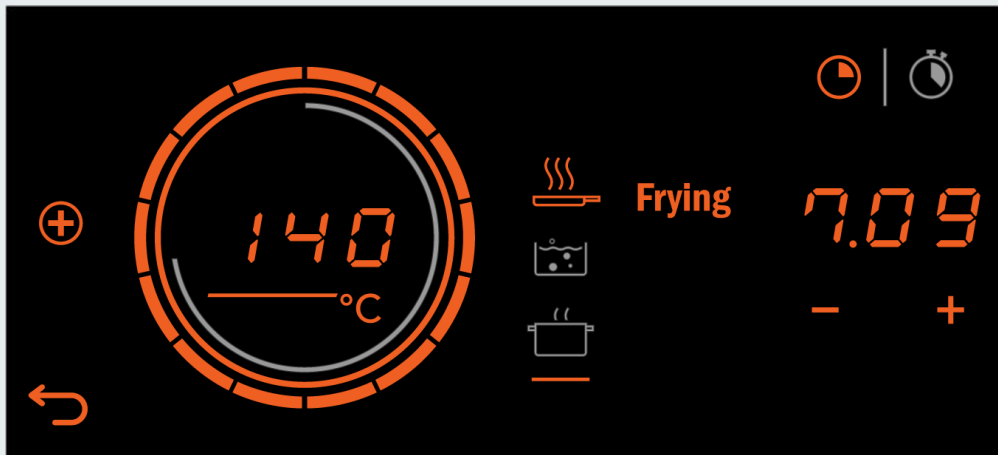


Figure 44: The visualisation of the timer function with numbers.

Improved interaction

Now the redesign is explained, it is important to understand what it changes to the interaction as well. To make this possible an improved interaction story board is shown in Figure 46.



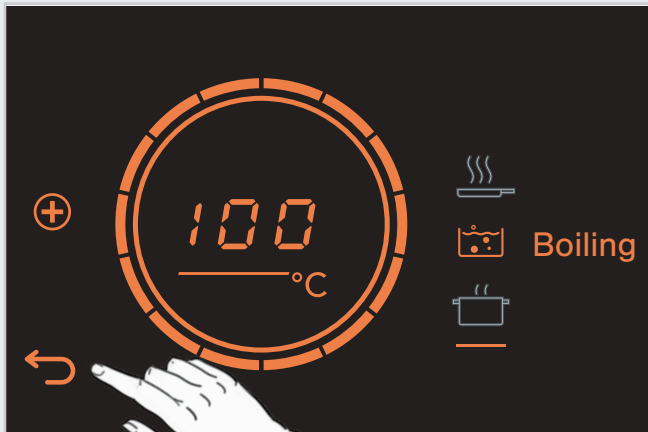
Richard just came home from a long day of work. He is a bit tired but still excited and ready to start cooking for him and his wife.



The hob asks in which mode he wants to cook. Thinking about it shortly, he decides to cook with the plus cooking settings.



The chicken has to be cooked at 140 degrees. Luckily Richard has the freedom to change the presets temperature. He now can set it to exactly 120 degrees.



Oh no, he made a small mistake for the temperature and has to change it. Luckily he can easily go back with the return button.



Now the temperature is set he can start cooking. He noticed that the hob is preheating. So he waits patiently until the bar is fully filled.



What a lovely dinner with his wife. The chicken to the salad is the perfect addition.

Figure 46: The improved user scenario.

Conclusion

The redesign incorporates the best elements of five interaction concepts. Our primary focus was to eliminate the pain points discovered in phase one. We examined the current flow of the product and limited the redesign to the Plus cooking feature. By keeping our interaction vision in mind, we addressed the feeling of being out of control during cooking and the lack of intuitiveness in the process. We achieved this by making the process quicker and more intuitive, while the hob guided the user, creating an exciting experience when using the new Atag induction hob. Figure 47 shows which pain points are addressed and Appendix K depict the new flow with improvements that makes the redesign stronger, with the aim of achieving our design goal: to provide users with confidence. The next page depicts the flow of the core part we have improved.

- ! : the pain points recognized in Phase 1
- ! : addressed pain points with the redesign.

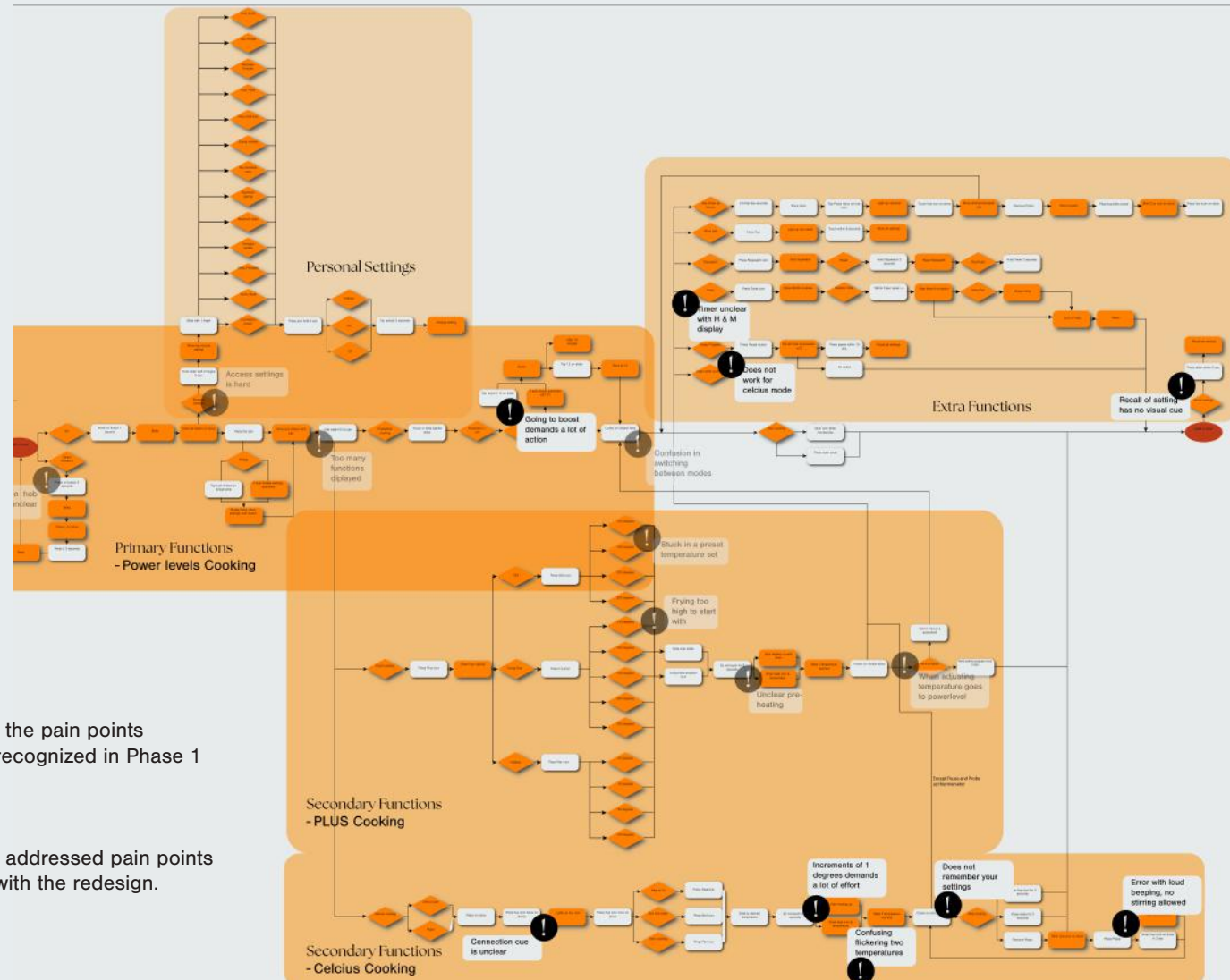


Figure 47: The addressed pain points with the concept, the ones with less transparency, are addressed.

New User Flow

The primary focus of the redesign was to enhance the user flow of the Plus Functions. Figure 48 demonstrates the new and improved user flow, which is noticeably simpler and more streamlined compared to the original one shown in Figure 47. See for the full new user flow comparison Appendix L.

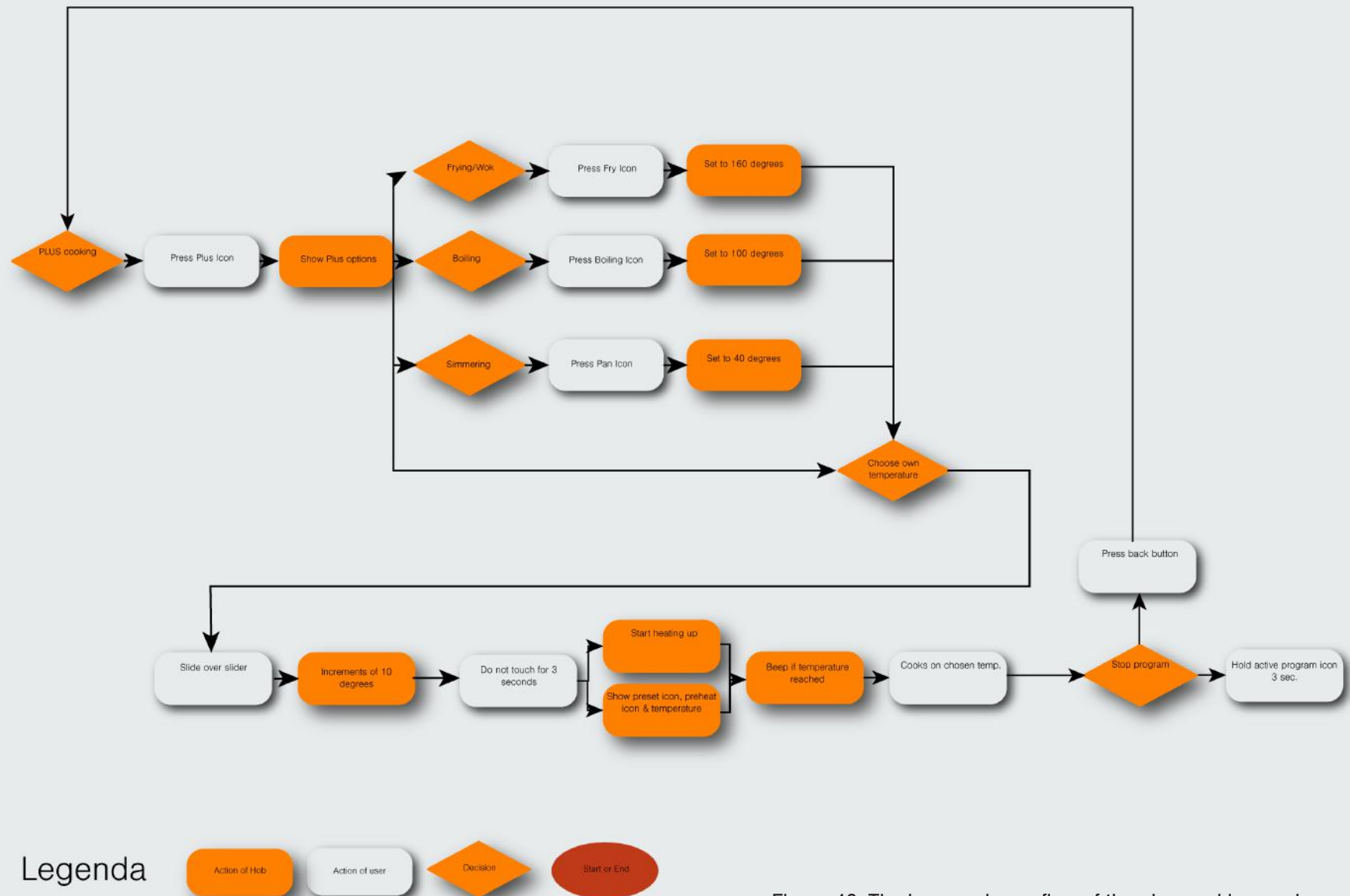


Figure 48: The improved user flow of the plus cooking mode.

Recommendations

We have focused our attention on merging concepts related to the scope of our design. However, during this process, we have identified several areas that could benefit from future enhancements. To this end, we have

compiled a list of recommendations for upcoming design iterations. Our aim is to provide guidance and inspiration for future improvements.

Probe: Show connection cue



Figure 49: Example of connection cue

Introduce auditory or visual feedback when the probe is connected to the hob. This feature aims to mitigate user confusion and reinforce their confidence in product operation. By providing clear, immediate feedback, users can be assured of successful connections, promoting a more intuitive and satisfying experience.

Error communication

Error

Water

No Pan

Figure 52: Example of showing errors.

In case of an error, display prompt messages or provide voice reminders on the screen to guide users in identifying the cause of the error without needing to consult the manual. This approach will make the error more intuitive and allow users to quickly rectify or repair it, preventing a decrease in user confidence and control.

Quick start guide

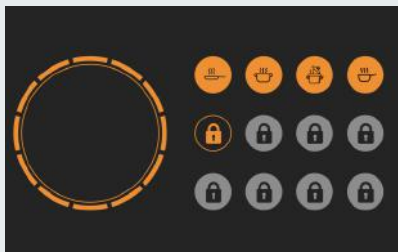


Figure 50: Example of the quick start guide.

Develop an additional quick start guide that focuses on real usage scenarios rather than the traditional feature-based layout of standard manuals. This guide should help new users quickly become acquainted with the hob's functionalities within their practical contexts, facilitating a smoother learning curve and more effective use of the product right from the start.

Boost button



Figure 51: Example of the integration of the boost button.

Given the confusion that the semi-automatic interaction of the Boost function can cause, consider implementing a dedicated touch switch for the Boost feature. This modification would allow users to engage the function more straightforwardly, enhancing their understanding and ease of use. By isolating this function with a distinct control, users can more confidently utilize the Boost feature without second-guessing their actions.

Next Steps

This report concludes Phase 2 which started with a recap of Phase 1, the start of brainstorming and conceptualization and the evaluation of concepts that led to forming an initial concept redesign.

The next steps take us towards Phase 3 where we detail out the formulated concept redesign through series of prototyping, testing and iteration. We hope to conclude with valuable and practical redesign recommendations through our final design. (Figure 53)

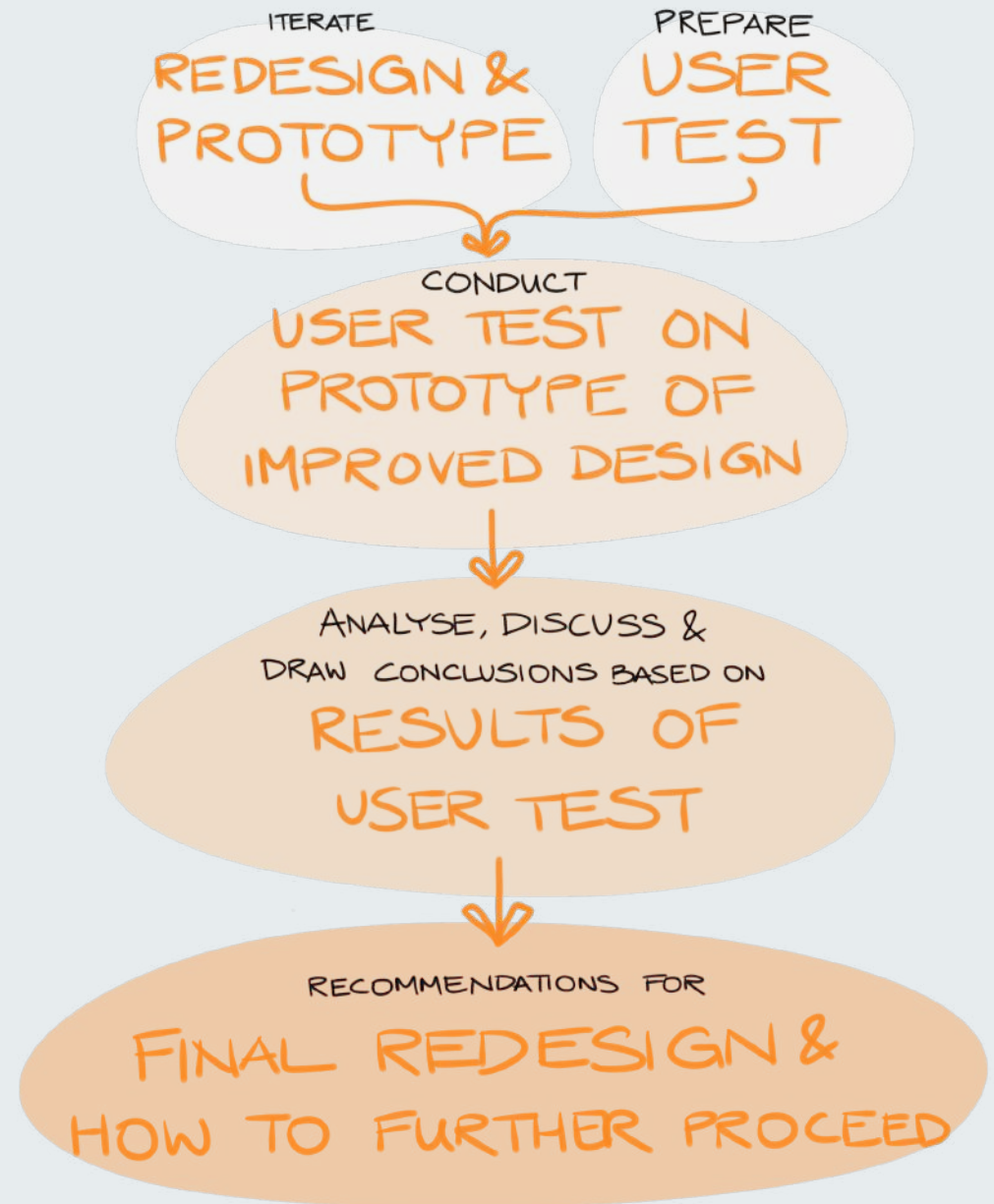


Figure 53: Visualisation of the stepping stones of the next page.

Appendix

A: Pain points user flow

B: How Might We's brainstorm

C: Figure Storming brainstorm

D: Eidetic Image brainstorm

E: What if brainstorm

F: Concept 2: Big Family

G: Concept 4: Touch screen

H: Concept 5: Learning the hob

I: Videos of Concepts

J: Harris profile

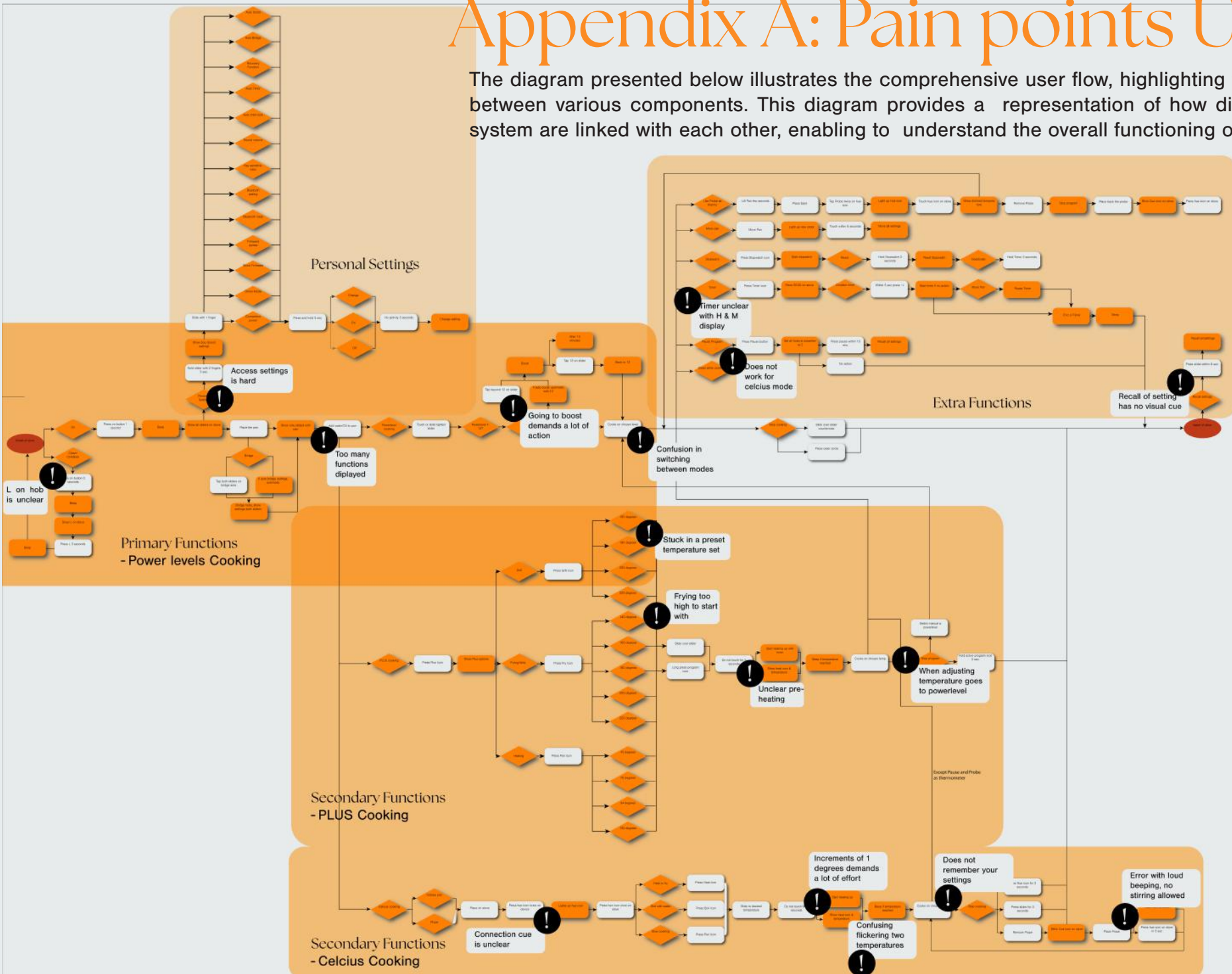
K: Improved user flow

L: PLUS function user flows



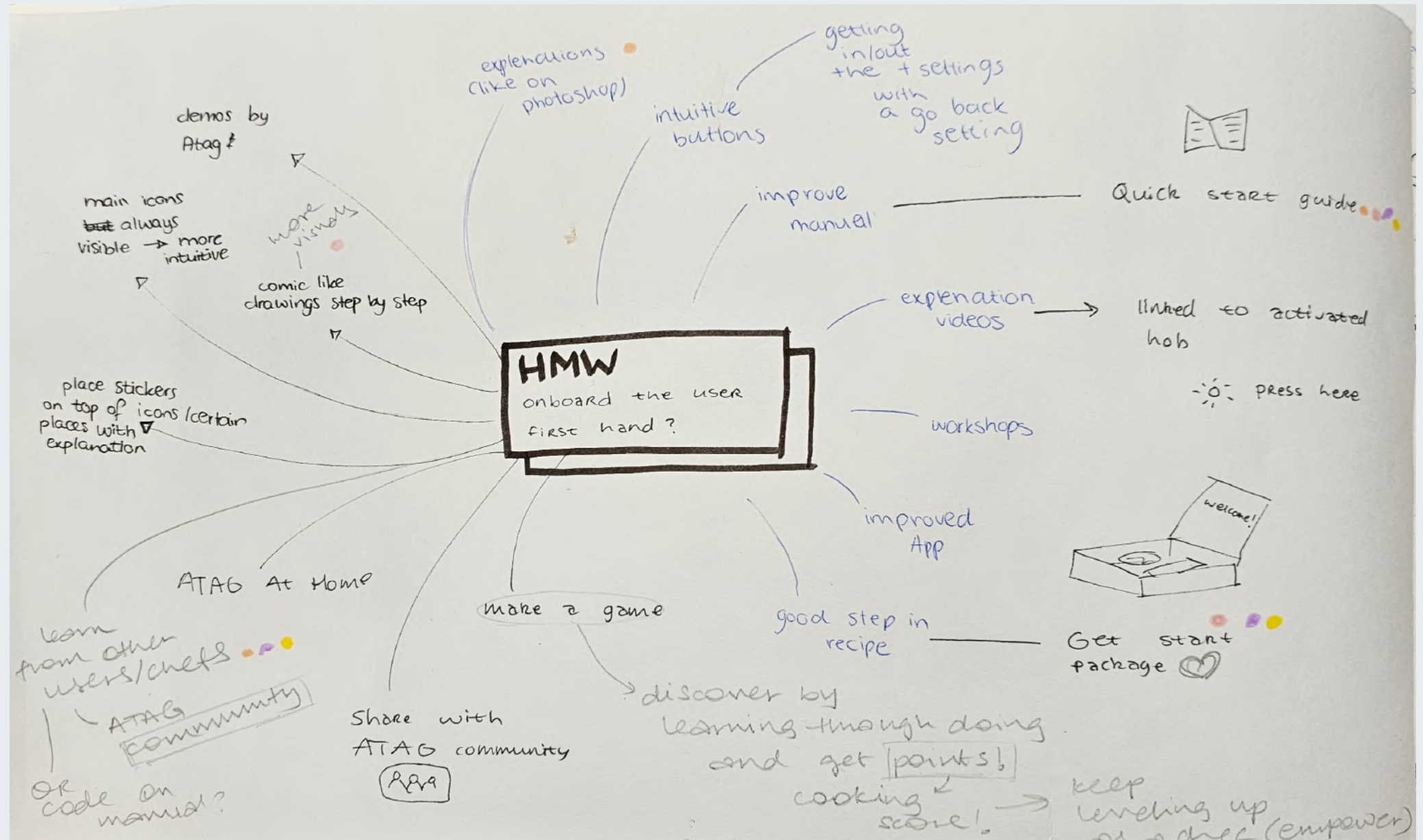
Appendix A: Pain points User flow

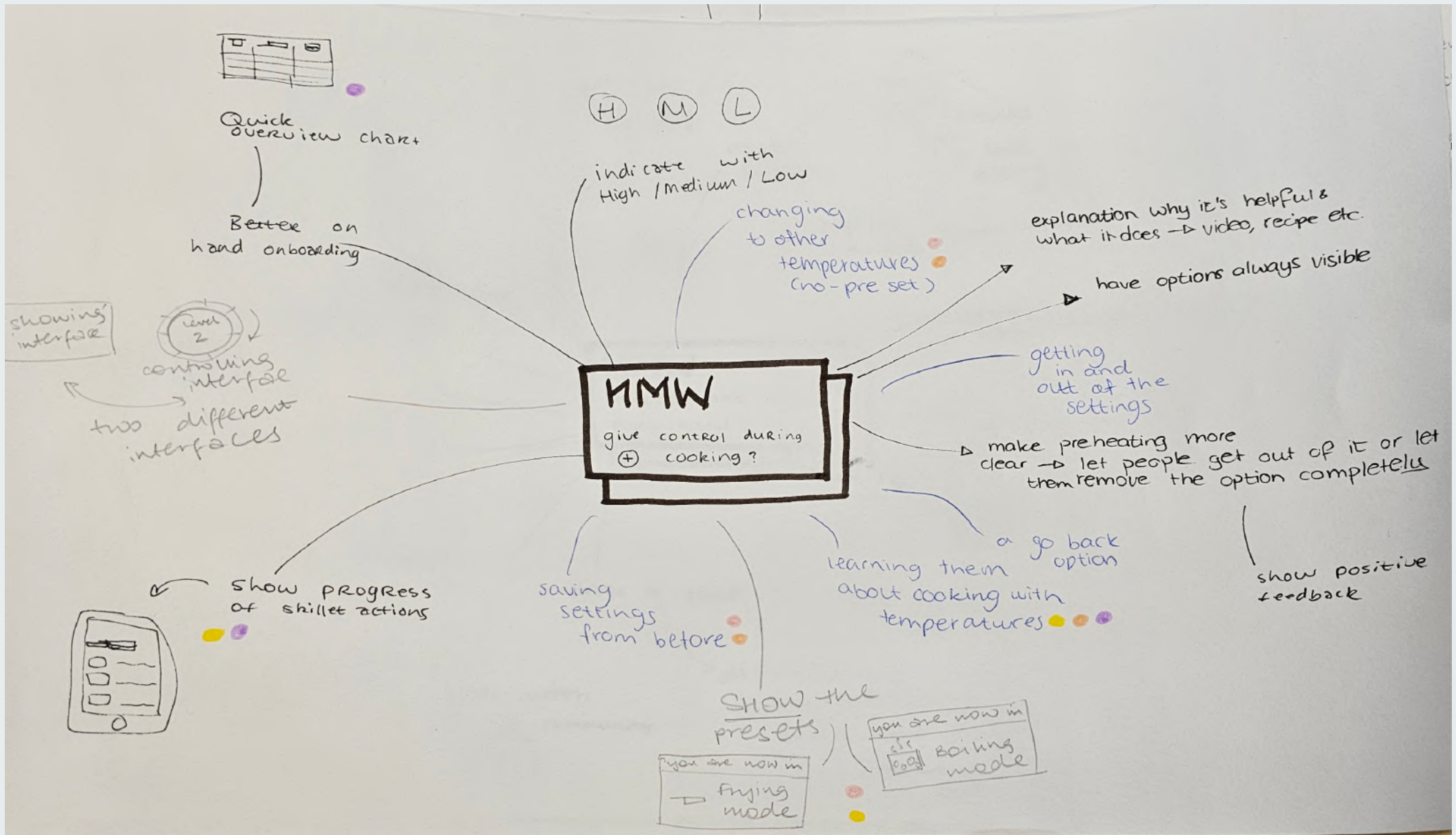
The diagram presented below illustrates the comprehensive user flow, highlighting the interconnection between various components. This diagram provides a representation of how different parts of the system are linked with each other, enabling to understand the overall functioning of the system.



Appendix B: How might we's Brainstorm

This appendix shows the complete overview of the how might we's brainstorm, as well as the concluding pages. The little dots next to some ideas were the most promising ones according to every teammate.





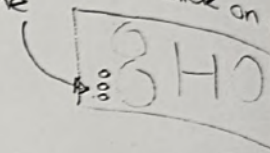
HMW

switch between cooking settings?

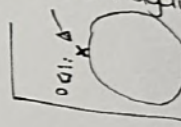


lock function to press when then you can switch this

always visible icons to click on when moving to other zone



a cross on the interface of settings → the different zones are visible then again

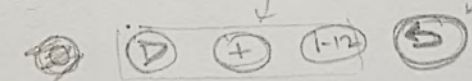


turn knob

press probe on



separate functions and allow to undo



hold the desired setting cooking

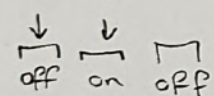


sliding bar

a back button

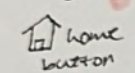
removing pan from hob

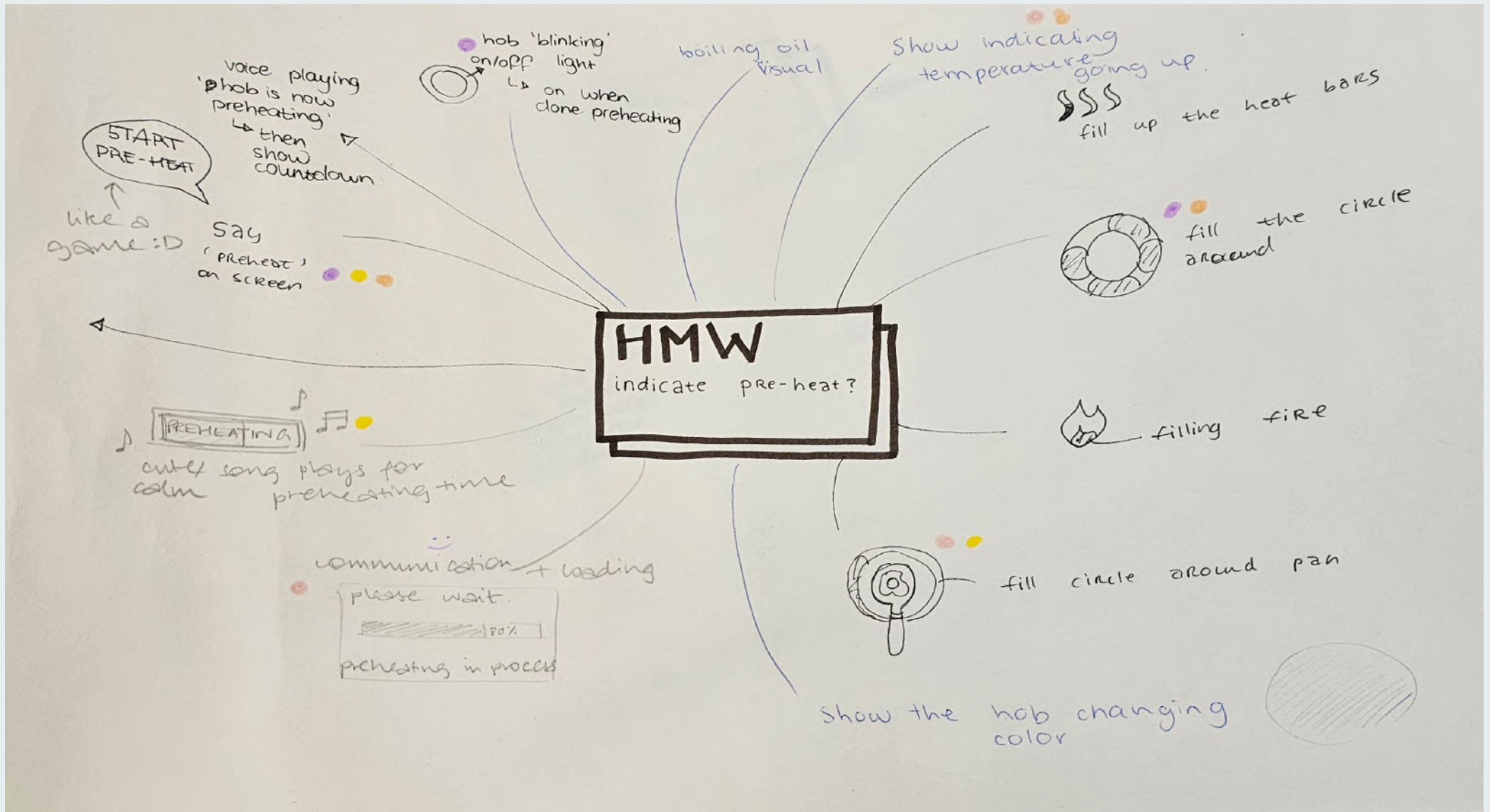
buttons that are clicking in & out

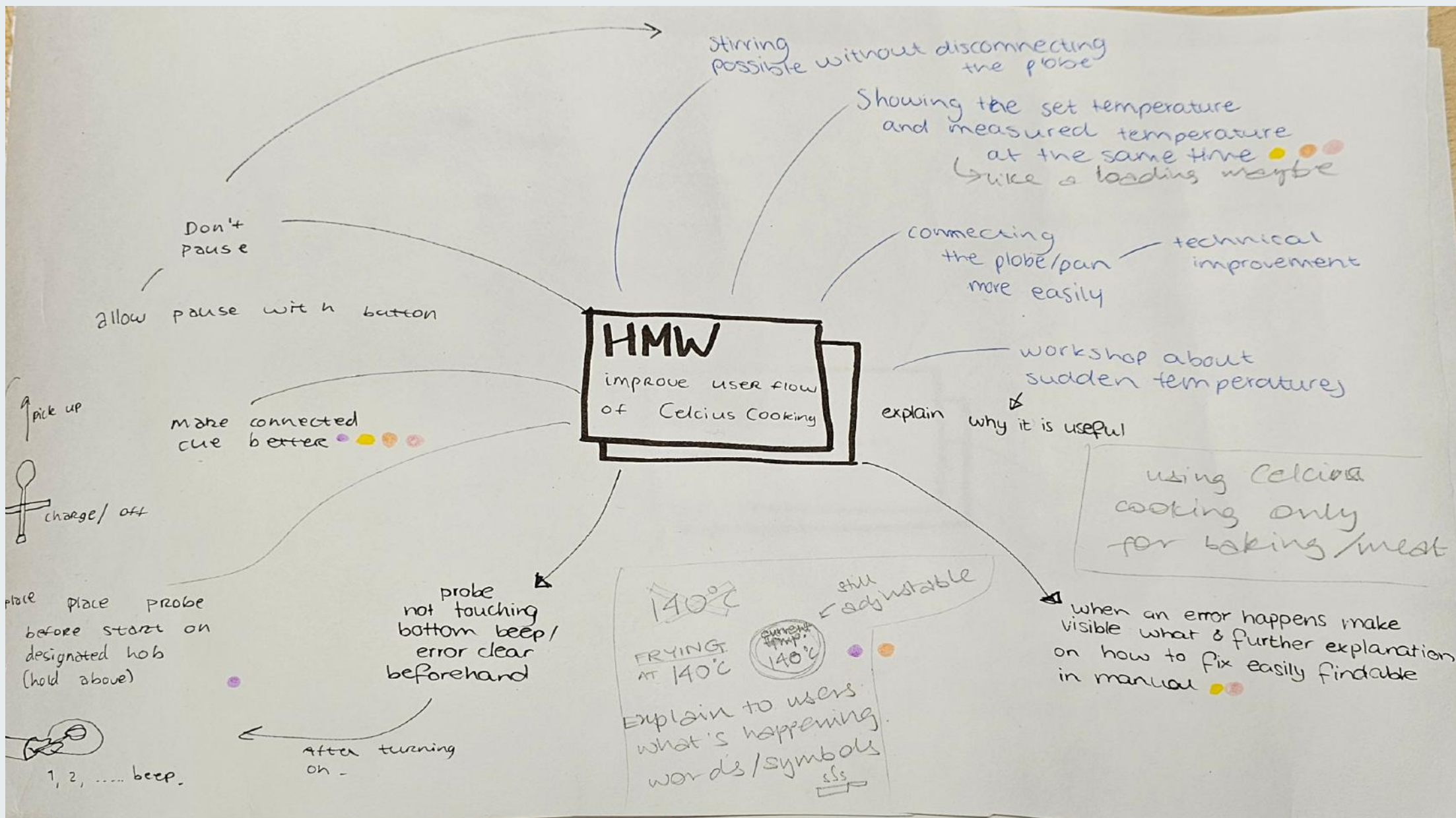


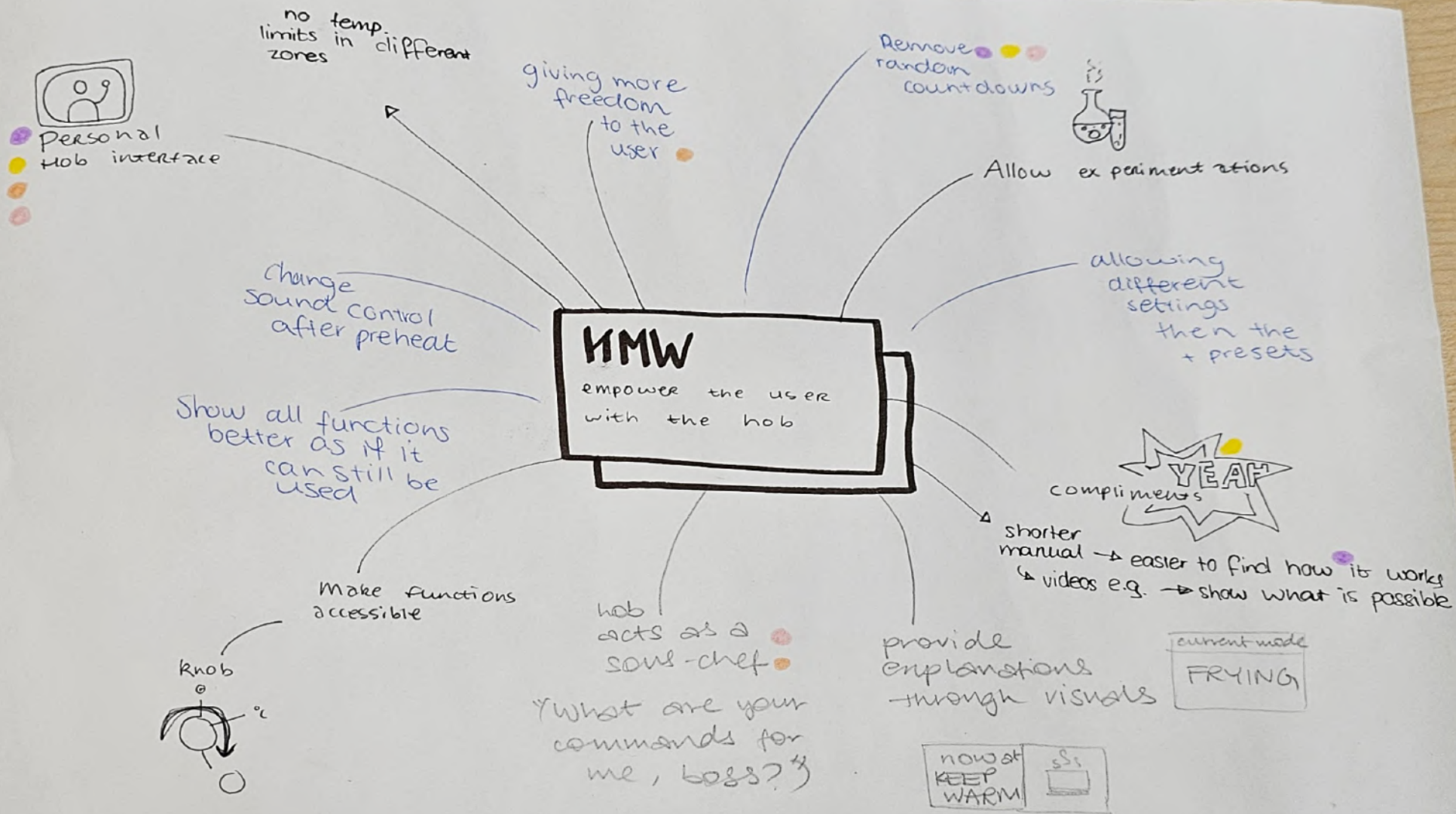
only locks when the pan is placed

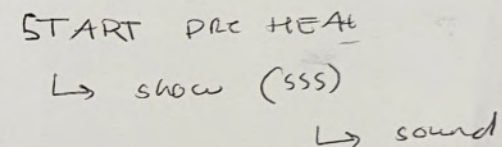
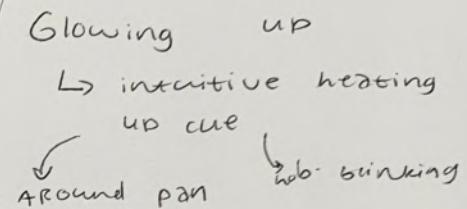
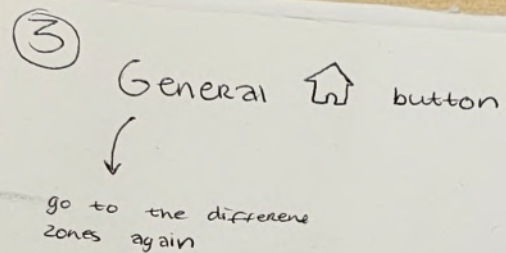
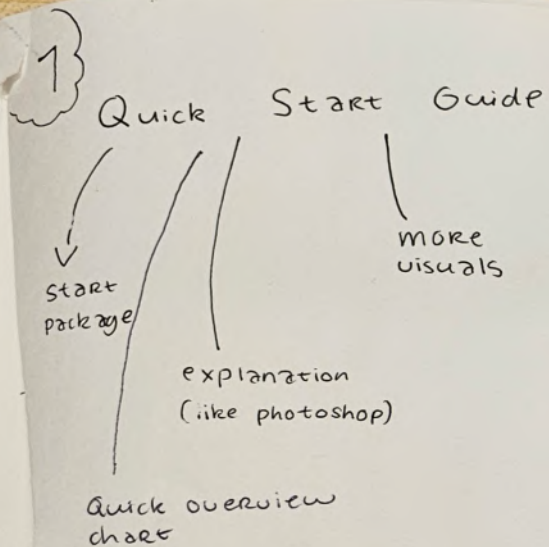
OR home button





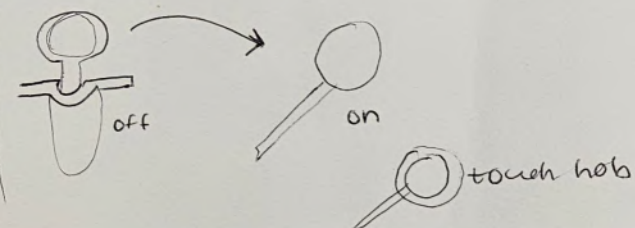
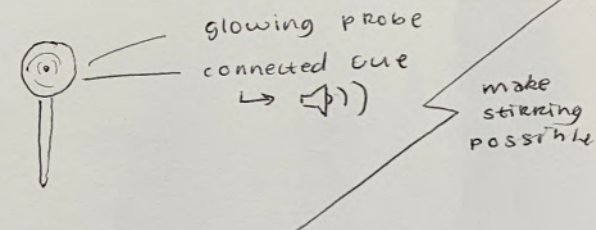
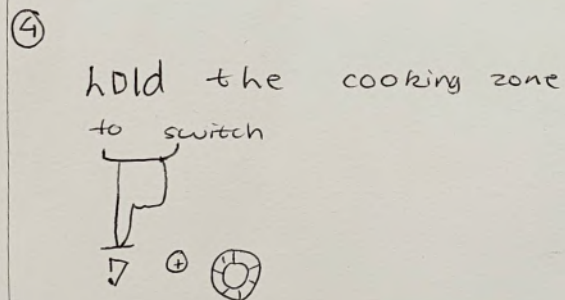
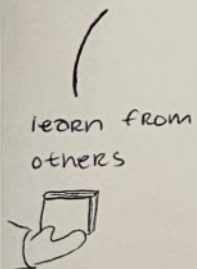






conclusion
page

ATAG community

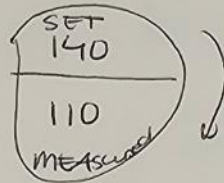


2

Show which
pre set range



no Lock in Presets



Show at the
same time



→ saving settings
PERSONAL
Hob interface

Remove countdowns
3, 2, 1.....

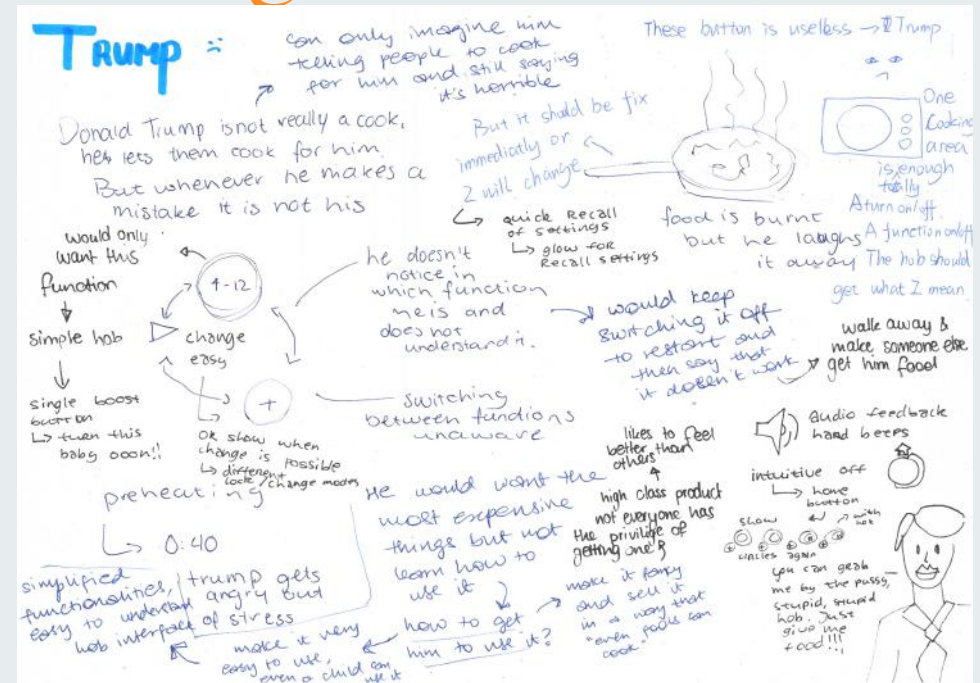
Hob acts as
sous-chef



Show
progress
of skillet
actions

Appendix C: Figure Storming Brainstorm

The following pages contain the results of the Figure Storming brainstorming session. Each brainstorming page is displayed below for each individual. Additionally, the conclusions have been separated and organized into personality points of the different figures that led to different functionality points.



Donald Trump

Personality points:

- needs to be simple made for a child
- wants most expensive thing
- errors are never his fault
- A fast and easy interaction
- nothing unexpected should happen

Functionality points:

- Very clear icons and product feedback
- An home button
- Quick recall settings, with clear guidance (glow for recall)
- A simple boost setting

Gordon Ramsey

- wants to use multiple hobs at once
 - ↳ quick switching modes, heat & pans
- no time for trying to connect bluetooth YES!
 - ↳ quick & little amount of steps
- knows how to cook → wants himself in control → no restrictions by hob or control
 - ↳ personalized hob
 - ↳ 'design' it himself to own preferences
- filming while cooking so may be change some sound effects
- no obstacles while cooking
 - ↳ personalisable
- animations on cooking hob
 - ↳ fire growing
- Easy to clean - Can't left my cooking area such a mess
 - ↳ (Stay smells good)
- looks back to what usages he had
 - ↳ learning curve
 - ↳ seeing learning curve and settings to look back for improvement
 - ↳ memory of his favorite settings → saves him to use again
- let's make it fun! and fast!

Additional Notes:

- begin. → simple recipe guidelines
- adv. → more pans advanced multi
- expert → ☆ dinner
- Quick & dirty experimental packages → allow some dirt on the hob before error
- how button is always available
- Also the rebus function? → can be see talk to the hob
- allow experimenting with temperatures, timing, pans, hobs.
- biobreak, frying activated
- boost function - high temperatures
- less fancy and more functional
- Touch V prompt - great
- Slide X too slow
- Slider design
- settings to save him to use again
- thrill

Personality points:

- Quick pace
- He wants all control
- No errors or obstacles
- Advanced settings for his chef mode

- Quick connection settings (Celcius settings)
- Personalised experimental interface
- Easy to clean in between cooking
- Memory of his settings
- Product feedback that communicates well (so doesn't interrupt his work flow)

Ronaldo

on phone 'plan' what hob needs to do → phone icon

have the hob cook for him → egg boil setting

place every thing wait → cooks food for you

being able to use multiple hobs at the same time to cook for the whole family

massive hob for massive house

getting things cooked for him by a chef → on can the hob be this?? 'I go, give me chicken & rice

chef should be able to give fine dining experience and go beyond his cooking skills with ease

different functions should work on different cooking zones at the same time

hob should be souschef and not control cooking

Lazy mode / learn mode

Just one click →

The ~~hob~~ UI button will change visible →

Semi → Auto Induction hob

if the cuisine going wrong (smells terrible, ~~burned~~ ^{burnt}) The hob can detect at the same time asap. and change the mode / do something.

Safe mode

healthy food → optimal food for a pro sporter → ch. base cooking on that → temp. cooking important

changeable temperature and pre sets - strict diet for sporting, maybe some presets on health

he is quick responding in the plate football, this has to be kinda

biggie buttons for big fingers

chef

Personality points:

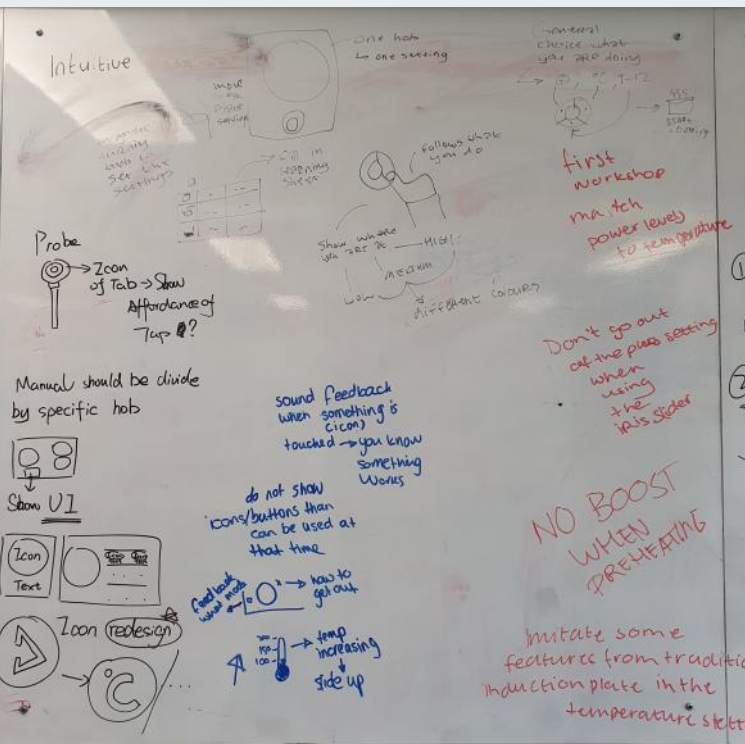
- Lazy mode
- Healthy cooking for the gains
- Time restricted
- A big size house

- Bigger buttons
- Good clear onboarding -> make it personable
- Food plan
- 'One button' Experience
- Personal presets
- A guiding product / he is your sous chef
- Changeable temperature settings within PLUS cooking

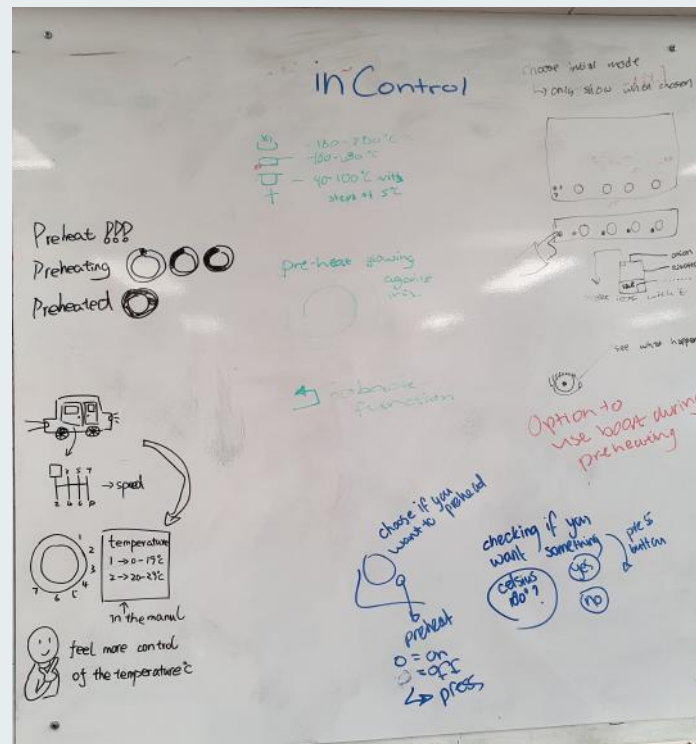
Appendix D: Eidetic Image Brainstorm

The images presented below illustrate the outcome of the Eidetic Image Brainstorm that revolved around the characteristics of our interaction vision qualities.

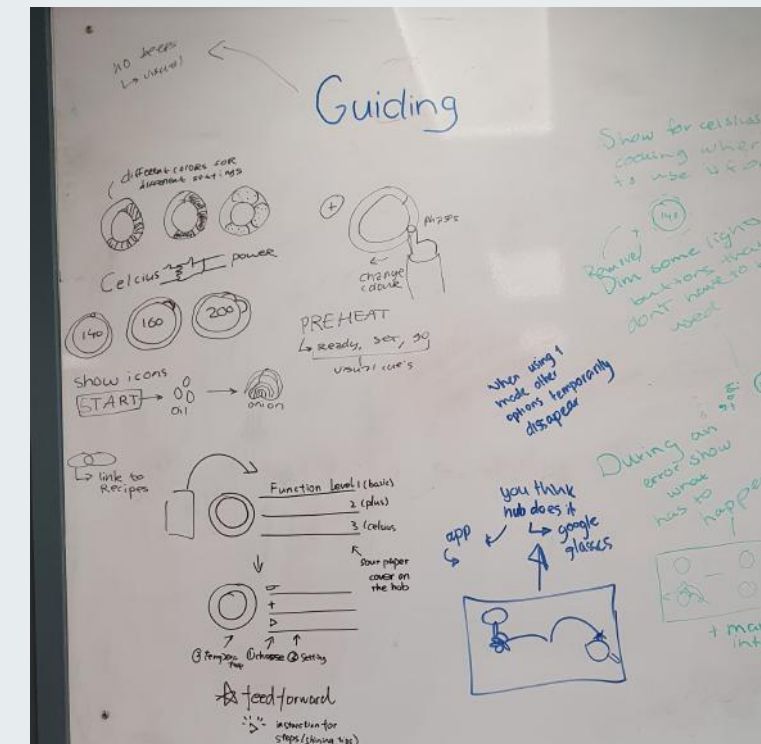
Intuitive



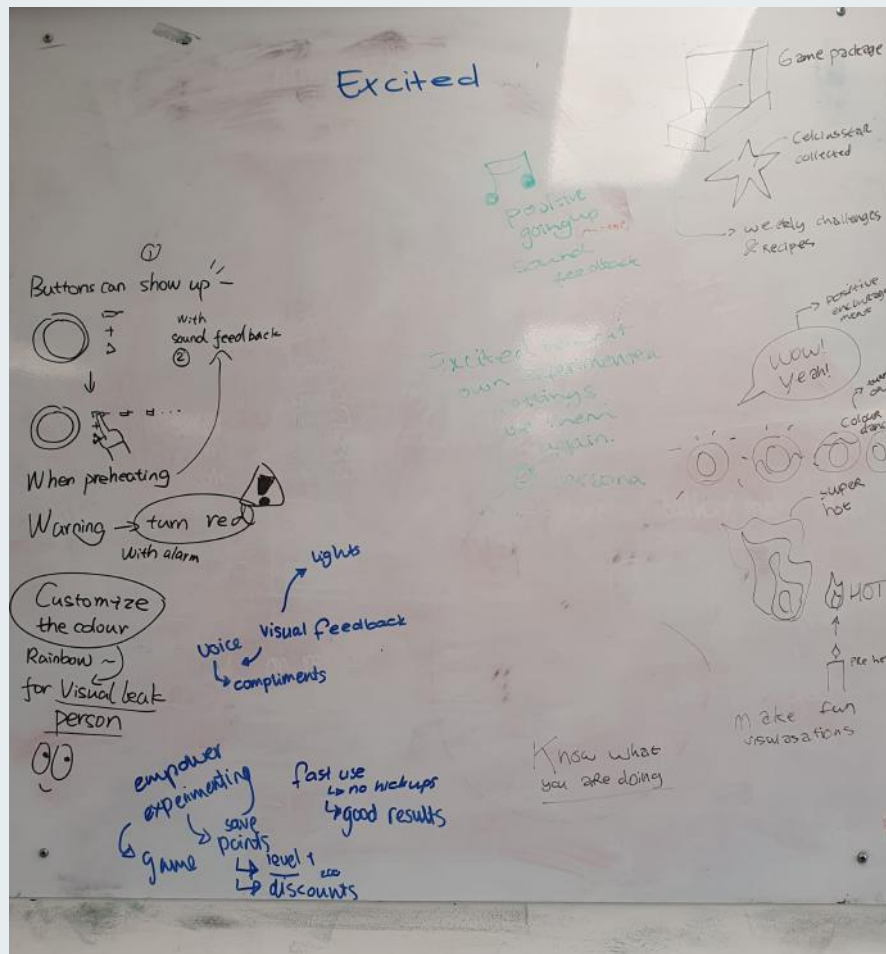
In Control



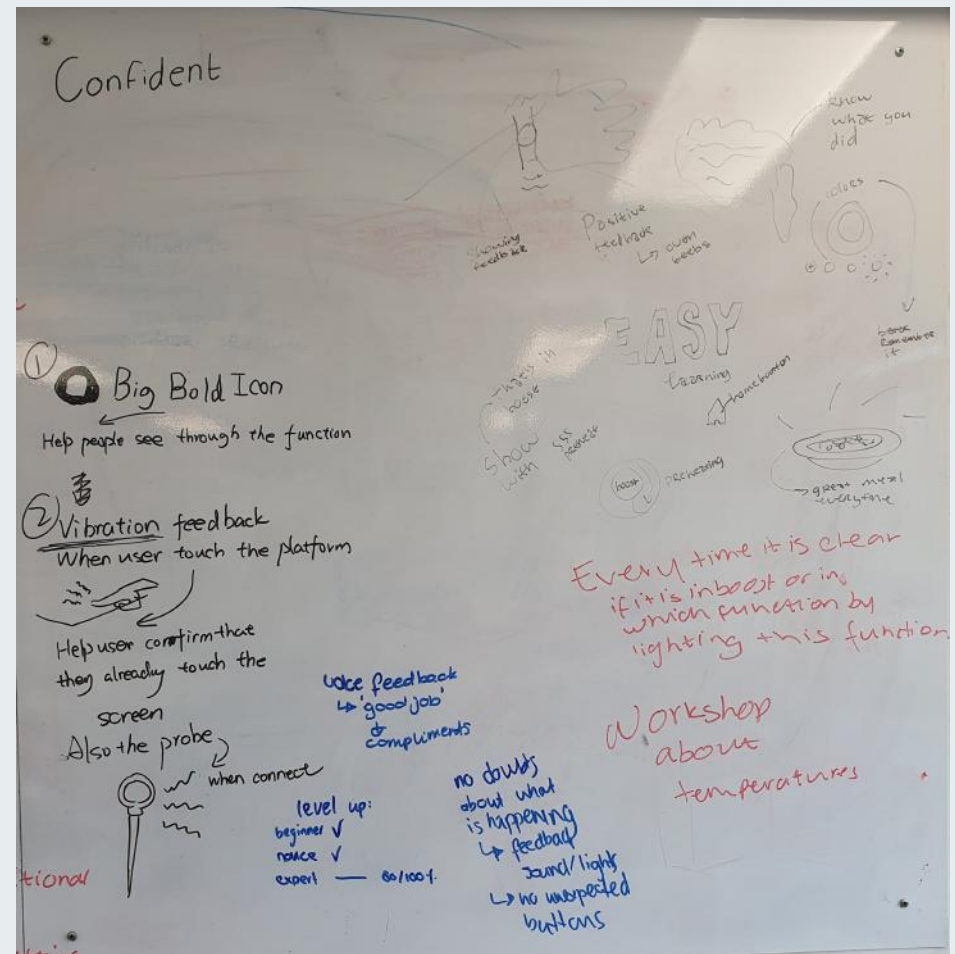
Guiding



Excited



Confident



Appendix E: What if Brainstorm

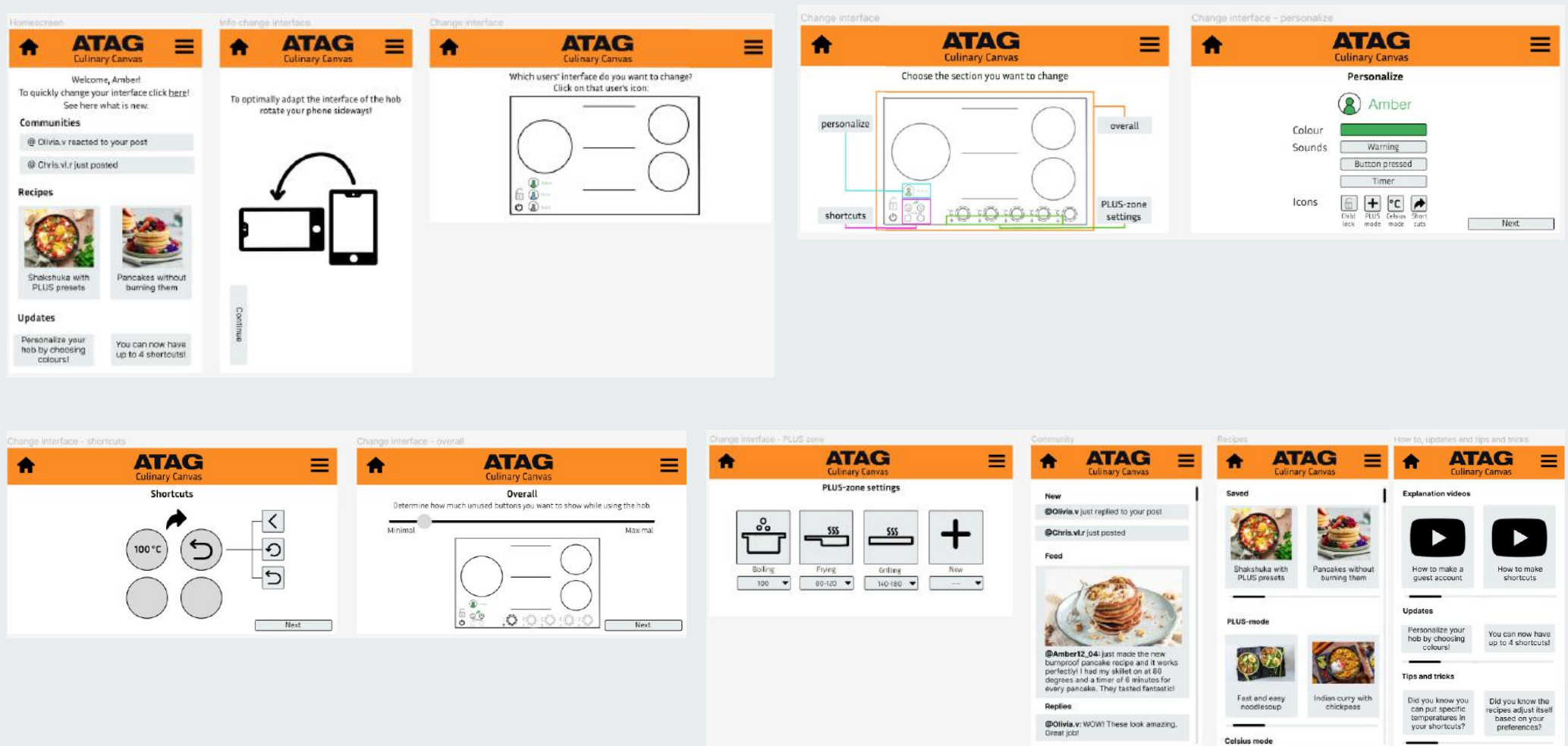
The images below depict the various “what if” questions that were generated during the what if brainstorming session. Later, we grouped these questions into the five main most interesting categories, as mentioned on the right side.

1. **What if** the hob had less buttons?
2. **What if** the hob had a persona's setting?
3. **What if** the hob was controlled with a touchscreen?
4. **What if** the hob was only voice controlled?
5. **What if** learning the hob was like a game?



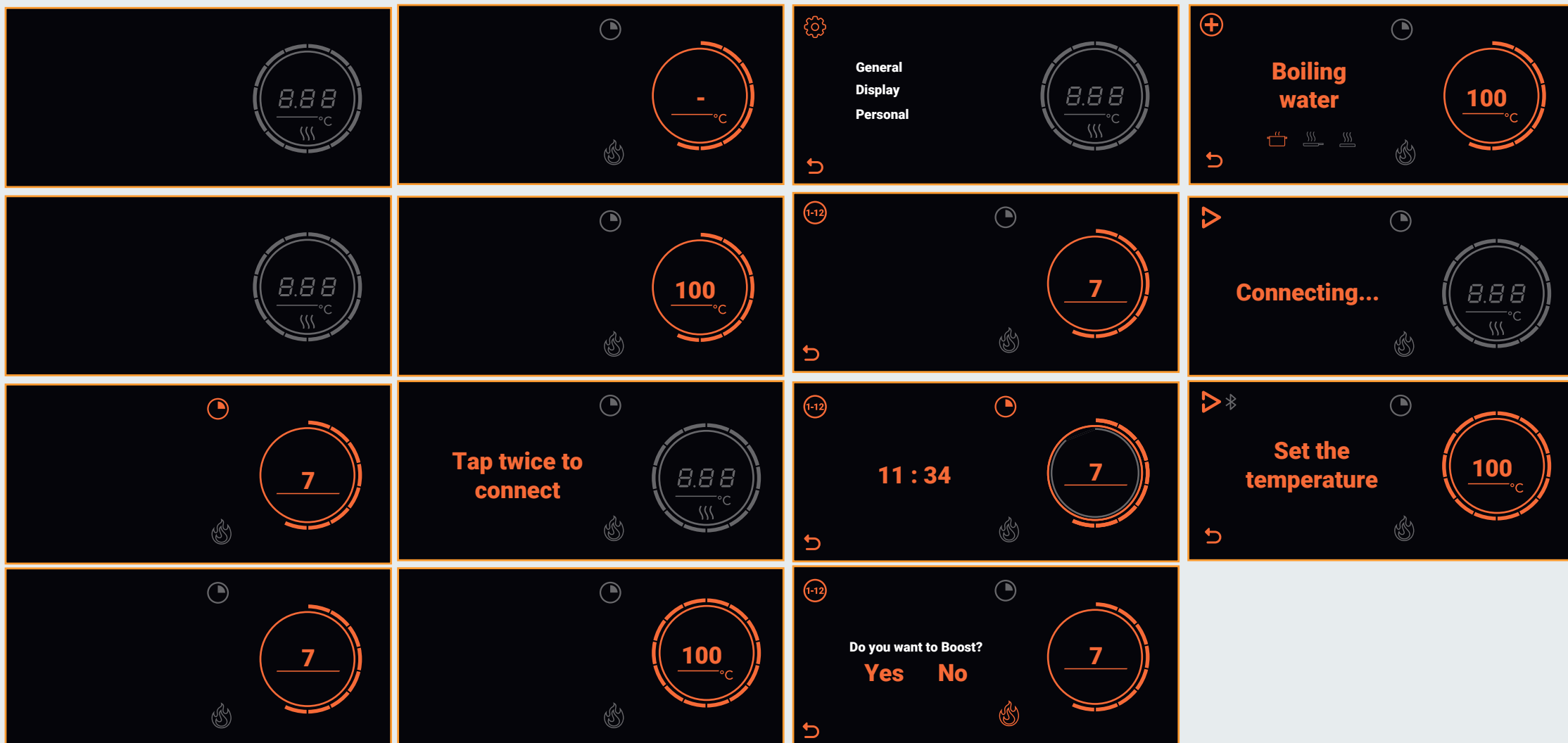
Appendix F: Concept 2 - Big family

The app interface can be seen in the visuals below:



Appendix G: Concept 2 - Touchscreen

For concept 4 this are all the interface screen and a video that shows how it works:



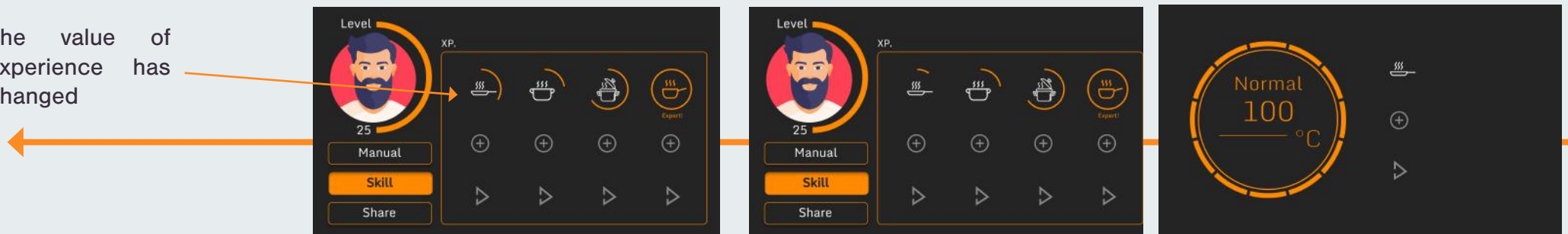
Appendix H: Concept 5 - Learning the hob

Below is the interaction flow of the button and the respond on the app when learning the hob with the digital guide.



When different function buttons are touched, an experience bar briefly appears around the button to display the changes. To avoid confusion with other functions, this experience bar will only appear for 2 seconds before the data is automatically recorded into the app. This ensures clarity and minimizes distractions while providing a quick visual feedback to the user.

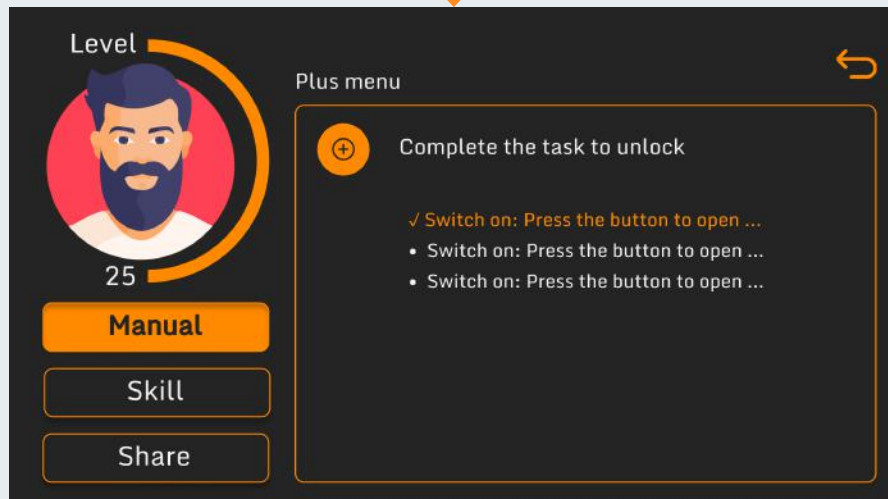
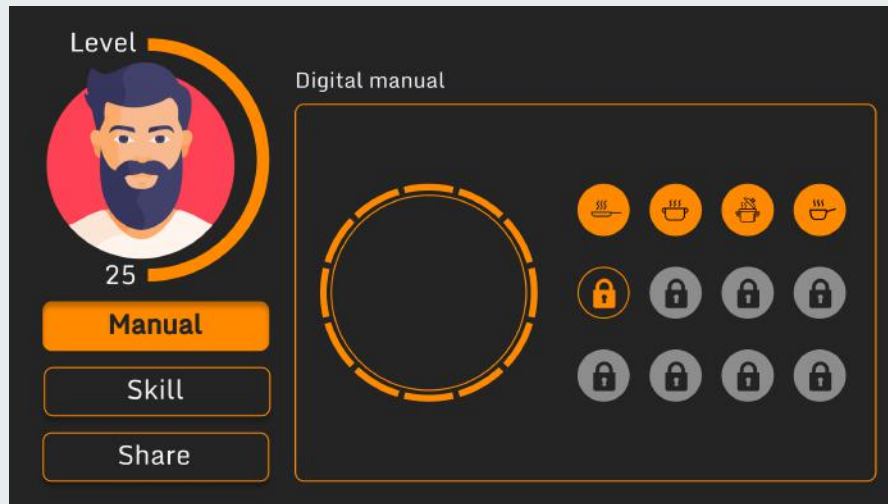
The value of experience has changed



Usage data for each button on the device will be meticulously recorded to provide users with detailed insights into their interactions. By employing a comparative method, the system will showcase the frequency and patterns of simpler versus more complex function usage. This strategic approach is designed to encourage users to explore and engage with the advanced features available on the device. By highlighting underutilized capabilities and offering guided suggestions for their application, users are not only motivated to experiment with these more sophisticated functions but are also supported in understanding how they can enhance their overall experience. This method leverages user curiosity and learning to foster a more enriching interaction with the product.

Below is the basic functions introduction of the digital manual.

Quick guide



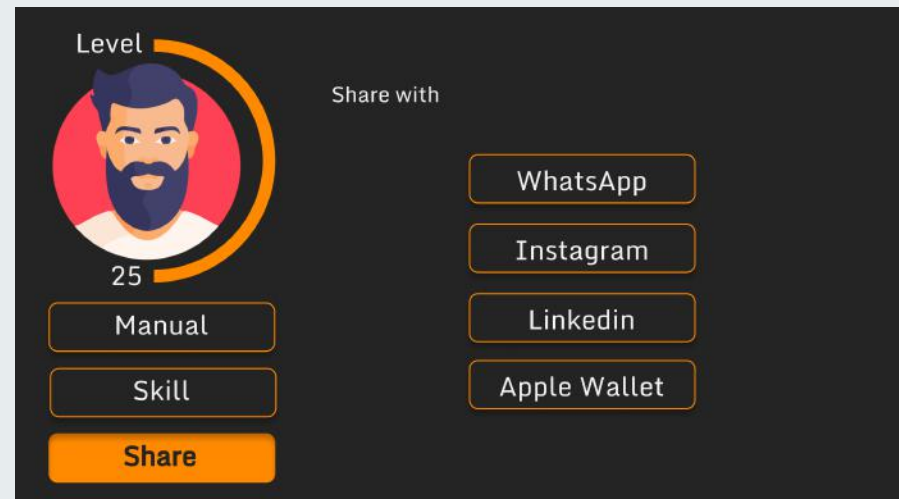
Quick guide is design for the first time use or the quick preview of the functions. It can detect the user interaction with the hob.

Skill



This is the page of display the skills recorded by daily use.

Share



On this page you can share your progress on several social media platforms.

Appendix I: Videos of concepts

Links to the video demonstrations of each of our concepts can be accessed below:



<https://youtu.be/Yo7hZcHTIb8>



<https://youtu.be/KbdNUJYoT4M>



<https://youtu.be/FfJjfpszkyk>



https://youtu.be/yc_3rb-ruvQ



<https://youtu.be/TpjBzogHnsg>

Appendix J: Harris profile

This appendix shows the harris profiles per participant and the comments that were given during this evaluation.

Participant 1

Intuitive	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen	Concept 5 Learning the Hob
Buttons lit up when on / showing functionality when activated	++ : it is clear what options can be chosen at this point. There is not more shown than necessary to not make it confusing. Why?	Why?	+ : there is no buttons to be used, but because it is voice controlled you feel like you can choose any option at any time Why?	++ : this way it is not too confusing Why?	o : did not cover this Why?
The degrees related to the presets were clear	++ : there are no real presets, you can determine your own temperatures. But, different icons with different temperatures are there to advise on what it can be used for. Why?	Why?	++ : it is clear that the hob sets a certain temperature that is associated with a function, but you can change it yourself Why?	++ : there is no real presets as you can determine your own temperatures. But you can click on the boiling etc. which is a good way to quickly get to more used temps. Why?	+ : i imagine this will be shown in the online learning thing Why?
Preset icons were clear	++ : all icons are very clear and not limiting Why?	Why?	+ : it is clear that you can change the temperatures in PLUS and Celsius cooking yourself Why?	++ Why?	++ : this will be explained in the online learning thing Why?
The Celsius cooking icons were clear	++ : it is clear that you need to connect with Bluetooth to use celsius cooking. With the suggested improvement of removing the PLUS setting and then placing the bluetooth icon next to it, i think it may be confusing if you need the bluetooth for celsius or can cook without it. Why?	Why?	o : no icons are used, but with voice control it feels more free Why?	++ : it is clear that you need to connect to Bluetooth and when it is connected. Also the way of changing the temperatures is nice Why?	+ : see above Why?
Certain buttons were known due to previous use (like power levels and boiling preset)	++ : yes it was clear to me which buttons are meant for what, because of previous knowledge. Why?	Why?	o : no buttons to be used, so you can think of what to say to make something happen Why?	++ : the placement and visualisation of the touchscreen make it clear how to use buttons and when Why?	++ : the online manual learning makes you know the symbols and what to use it for Why?
It was clear if celsius cooking probe was connected	o : it does not really include where it shows if the probe is still connected or not. But for initial connection it does clearly show. Why?	Why?	++ : the hob clearly tells you to wait or if it is connected or not Why?	++ : The text on the hob clearly shows what is happening with the probe and if it is connected or not Why?	o : did not cover this, but i imagine it will clearly teach you when it is connected or not Why?
The letters on the hob was clear	++ : i think there are not much letters present. Only icons, but this is clear Why?	Why?	o : same as before Why?	++ : the letters are removed which makes it more clear Why?	++ : the online learning manual shows exactly what each function does Why?
Stopwatch is understood	o : this concept did not include the stopwatch Why?	Why?	o : this concept did not include the stopwatch Why?	o : no stopwatch only timer Why?	++ : see above Why?
The ON/OFF button was immediately found	o : same as before Why?	Why?	- : it shows a on/off button, but you are supposed to use voice control. Might be confusing Why?	o : same as before Why?	++ : did not cover this, but i imagine you learn this in the 'game' Why?
Action of placing the pan was automatic	o : same as before Why?	Why?	+ : the voice says that you can place the pan, which makes it intuitive Why?	o : same as before Why?	o : did not cover this Why?

In Control	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen	Concept 5 Learning the Hob
PLUS cooking makes you feel in control	+ : you can determine own temperatures, but the suggestion of removing the PLUS is also good, since it does not really add anything. Why?	Why?	++ : you tell hob what to do and can still change if hob does something Why?	++ Why?	o : i feel it is still locking, but because you have a good understanding of a function before you unlock it you know what you are doing Why?
Pre heating countdown makes you feel in control	++ : the countdown makes complete sense and does not feel limiting Why?	Why?	+ : i would not say you can control the preheat countdown, but it is clearly told what is happening and what you should do Why?	+ you can't get out but clearly states what is happening Why?	+ : it still feels locking, but you know clearly what is happening due to learning thing Why?
Could switch with ease between cooking options	++ : the addition of home button is amazing Why?	Why?	++ : you only have to talk Why?	++ go back options is good addition Why?	+ : see above Why?
Bluetooth probe does connect automatically or with ease	o : did not test if probe connects, but showing that you need to connect to bluetooth and if it has initially connected is very clear. Why?	Why?	++ : connection of the probe happens with only one command Why?	o : did not include this Why?	o : same as before Why?
Can easily continue using the hob after error	o : there was no error visible, but i imagine you could press home to go back to original settings Why?	Why?	++ : clearly tells what the error is and what to do to stop it Why?	o : did not include Why?	+ : i think the hob can communicate errors better but because you have an online learning environment you have more knowledge on what is happening Why?
When it responds to you as you predict it to (timer)	o : did not include timer Why?	Why?	o : did not include timer Why?	+ clear Why?	++ : because you clearly know what is happening to unlock a function i think it will do what you predict it to do Why?
Manual is easy to read and find specific topics (but it is the old manual)	o : did not use manual Why?	Why?	o : did not use manual Why?	o : did not include this Why?	++ : the online manual seems very clear and to the point, not too busy or too long text Why?
Can pause the celsius cooking	o : i did not see a pause button in celsius cooking, but i doubt if you need it Why?	Why?	o : concept did not include pause mode, but i imagine that you can just use your voice and tell to pause it Why?	o : did not include this Why?	+ : this is same as before, but you have more knowledge Why?
Can be turned off even when some function is locked	++ : there is a home button so you can always go back to the start setup Why?	Why?	++ : by telling her what you want to happen it turns off Why?	++ the go back button is really great Why?	o : same as before Why?
You decide to turn the hob on	o : same as before Why?	Why?	++ : only when you say something, it happens Why?	o : same as before Why?	o : same as before Why?

Guiding	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen	Concept 5 Learning the Hob
Icons only show when you can use them	++ : see above Why?	Why?	o : did not include icons, but with voice control it is clear that you determine which mode you use Why?	++ Why?	o : this is still the same as before Why?
Preheat function is clear	++ not limiting or anything, is clear what is happening Why?	Why?	++ : it exactly tells you what is happening and when it is done Why?	++ Why?	+ : everything is more clear because of the app and learning way Why?
Celsius cooking icon appears when ready to use	+ : celsius icon is present on home menu and then when you press it you are on celsius mode. So you can always click it, but right now can't use it if you don't connect to bluetooth which is good Why?	Why?	o : the icon is always there, but once again you tell hob what to do and then get feedback Why?	+ Why?	o : same as before Why?
Has presets to cook with	++ for PLUS and Celsius mode you can increase by a certain amount, but there are no limits within these settings which is great. Why?	Why?	++ : the presets are there, but it is not limiting, you can still change them Why?	++ Why?	+ : same presets are before, but more knowledge on functions Why?
Icons are clear • Celsius cooking • Letters on screen	++ very clear, no confusion Why?	Why?	o : no icons are changed Why?	++ Why?	++ : due to the app i feel you have better understanding of all different icons and letters Why?
Icons appear after turning on	++ : not too much icons are shown when they can't be used to that makes it very clear and less crowded. Why?	Why?	o : did not include icons Why?	++ not too much shown, very clear Why?	+ : same as before, but you have a clearer understanding Why?

Exciting	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen	Concept 5 Learning the Hob
The hob allows you to take control in the PLUS cooking making you not locked in cooking	++ : no limits anymore which makes PLUS cooking way more enjoyable. Why?	Why?	++ : you tell hob what to do and can still change it. Why?	+ Why?	- : the hob still feels locked because that part has not changed but you know now more what to do Why?
Being able to cook a new dish on a new hob was exciting (using power levels - sense of familiarity gives comfort) +	o : the hob looks very clean and sleek, which is nice. Why?	Why?	++ : this is very new and innovative, i would enjoy this and feel empowered Why?	o : did not include this, but i can imagine that the sleek look of the hob makes you excited to cook Why?	++ : the entire functionality becomes more clear and when you really understand what the options are i feel it will be a very exciting product to use Why?
Presets (when understood) were exciting to use because they were new	++ : it is a good way to cook and nice with the visual icons to see suggestions of why you could use it. Why?	Why?	++ : it is a nice way to cook and feel that the hob does what you want it to do with only a voice command. Why?	+ Why?	+ : nice way to cook and clear what they are for Why?
Indication the countdown has started (preheat)	++ : The heating is way more clear and without countdown, it just shows if it is heating up or not and how warm it is. But does not limit you in placing your pan or make it more confusing. Why?	Why?	+ : it is clear what is happening, but when you have to wait that takes a bit long and you might start to wonder what is going on now Why?	++ it clearly shows what is happening with heating up Why?	+ : you know that there is a countdown function and how looks so this is less unexpected Why?
Cleaning did not give a lot of beeps	o : did not test this Why?	Why?	o : did not test this, but as said before it clearly tells you what is happening and why and what you need to do Why?	o : did not test this Why?	o : did not test this Why?
Suitable sound effects	o : did not test sounds Why?	Why?	++ : voice sounds very calm and the calming music in between feels relaxing Why?	o : did not test this Why?	o : did not test sounds Why?
The timer function that could be set up for every single hob	o : did not test this Why?	Why?	o : did not test this Why?	o : did not test this Why?	o : did not test this Why?
Celsius was found interesting when used for baking or specific cooking (like for meat)	+ : if PLUS mode is removed, i think Celsius mode will be used more often and therefore feel more familiar and like a good addition to the hob. Why?	Why?	+ : Celsius cooking is more clear and by saying what you want to do, the hob does that for a specific product Why?	+ Why?	+ : it is more clear what to use for Why?

Participant 1

	Less buttons	Voice control	Touch screen	Learning the hob	
Intuitive					
Buttons showing when activated					1st: touch screen 2nd: less buttons
The degrees related to the presets were clear					
Preset icons were clear					
The Celcius cooking icons were clear					
Certain buttons were known due to previous use					
It was clear if celcius cooking probe was connected					
The letters on the hob was clear					
Stopwatch is understood					
The ON/Off button was immediately found					
Action of placing the pan was automatic					
In control					
PLUS cooking makes you feel in control					1st: voice control 2nd: touch screen
Pre heating countdown makes you feel in control					
Could switch with ease between cooking options					
Bluetooth probe does connect automatically or with ease					
Can easily continue using the hob after error					
When it responds to you as you predict it to (timer)					
Manual is easy to read and find specific topics					
Can pause the celsius cooking					
Can be turned off even when some function is locked					
You decide to turn the hob on					
Guiding					
Icons only show when you can use them					1st: touch screen 2nd: less buttons
Preheat function is clear					
Celcius cooking icon appears when ready to use					
Has presets to cook with					
Icons are clear; Celcius cooking and the letters on screen					
Icons appear after turning on					
Exciting					
The hob allows you to take control in the PLUS cooking and not locked					1st: voice control 2nd: less buttons
Being able to cook a new dish on a new hob was exciting					
Preset (when understood) were exciting to use					
Indication when the countdown has started (preheat)					
Cleaning did not give a lot of sounds					
Suitable sound effect					
The timer function that could be set up for every single hob					
Celsius was found interesting when used					

Participant 2

Intuitive	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen	Concept 5 Learning the Hob
Buttons lit up when on / showing functionality when activated	++ It has improved, the categorization of functions was great too because it's easier to find (being around the Iris slider) Why?	+ yes here, everything is lit up on startup, after pan detection only one is highlighted which is important. Why?		++ fewer buttons here too The whole thing being touchscreen makes it look seamless. Why?	++ I loved that your 'expertise' level shows up briefly when you use the function and even teaches you how to use them in an accessible way this could make users explore the hob in a fun way Why?
The degrees related to the presets were clear	++ YES! the degrees changed corresponding to their function which was amazing. The user doesn't have to second guess it anymore. It was big and in the centre Why?	+ showing the degrees with the presets is great however it was a bit too small to read Why?		++ yes, very Why?	++ can be taught through this Why?
Preset icons were clear	++ Why?	++ quite clear, showing them on the hob as well as the app Why?		++ very clear with text stating what preset it was Why?	++ very clear Why?
The Celsius cooking icons were clear	++ Allowing the user to experiment with what level they want to cook according to their purpose of cooking is great. Gives the user knowledge on which level does what kind of cooking. Why?	+ clear! Why?		++ Yes they were clear as were the temperatures. I liked changing the temperature with intervals of 5 Why?	+ see above Why?
Certain buttons were known due to previous use (like power levels and boiling preset)	++ power levels icon could be improved (instead of concentric circles) but the placement of PLUS and Celsius was improved. Having the preset icons show as change the temperature is awesome. VERY intuitive. Why?	+ power level icons were same but the presets showed up with associated temperatures Why?		+ Icons seemed to be the same which may cause confusion but there were lesser at once so not as overwhelming Although, the preset say what it is, so great! Why?	+ same as above Why?
It was clear if celsius cooking probe was connected	o Why?	o not shown Why?		+ it was clear that the probe is connected. I feel like we could add one more step where it tells where and when to place the probe on the pan. The touch screen can be used to give instructions Why?	o did not test this Why?
The letters on the hob was clear	+ didn't see letters which I guess is a good thing? Why?	o seemed similar Why?		o No letters were seen which is good is it hours and minute or minutes and seconds? Why?	+ we could have a replacement or even a cheat sheet for learning the 'language' (letters and symbols) of the hob Why?
Stopwatch is understood	o Why?	o seemed to be same as before Why?		o I saw timer but not stopwatch Why?	+ timer/stopwatch could be gamified for sure! haha I imagine a turtle slowing walking on your app screen to a finish line Why?
The ON/Off button was immediately found	o same as before Why?	o same as before Why?		o Why?	o Why?
Action of placing the pan was automatic	o same as before Why?	o same as before Why?		o Why?	o not tested

In Control	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen	Concept 5 Learning the Hob
PLUS cooking makes you feel in control	+ It seems like this time I don't get locked and I also understand what preset im in Why?	- while the plus cooking icons were clear, it may become a lot of work to manage them though as it's on the app. The user may have to go back and forth a lot. Also, making the user select the degree for the preset and then have the preset (eg. boiling) not work may cause frustration. Why?		+ yes it does presets+temperature+text Why?	++ For sure, because you get an explanation but also a motivation to learn it and level up Why?
Pre heating countdown makes you feel in control	o Why?	o not shown Why?		+ no control but it makes you aware that you need to wait by showing the wiggly lines. Maybe clearly showing how much time you need to wait for preheating to finish, visually on the touch screen Why?	o not tested specifically but preheating could be like the timer thing mentioned above Why?
Could switch with ease between cooking options	+ need to explore the technical details as to if it is possible though Why?	+ seems like you can need to explore the technical details as to if it is possible though Why?		+ very easily I liked the modes being in the top left corner Why?	+ seems to be the same function but with fun explanations, so yes Why?
Bluetooth probe does connect automatically or with ease	+ bluetooth symbol is clever and easy to understand, pairing functionality was not tested Why?	o not shown Why?		+ it seems to connect easily. having clearer text specifying what to connect and when will make it clearer Why?	+ There could be a better way to connect through instructions on this manual. Why?
Can easily continue using the hob after error	o Why?	o not shown Why?		o not shown Why?	++ could get a soft alert and instructions as to what to do on the app like an obstacle during a quest Why?
When it responds to you as you predict it to (timer)	o Why?	o timer seems the same as before. the H.M. might still be confusing Why?		+ swiping gesture is nice and without letters. maybe the letters can make it clear another way Why?	+ could be fun to gamify it while informing the users of 'WHY' there is a timer. Like fun fact bubbles while they wait Why?
Manual is easy to read and find specific topics (but it is the old manual)	o Why?	+ there seem to be instructions on the app, making the manual/instructions/recipes all accessible and digital Why?		o no manual Why?	++ YES. Very awesome. Very simple. Very fun. I like how the manual doesn't feel like a manual but really an extension of the hob Why?
Can pause the celsius cooking	- I don't remember seeing a pause function. Might need it Why?	++ I liked how the lock shows up on all hobs clearly indicating that it can't be used. I saw this function as something that can be used for the pause function too Why?		+ no pause Why?	o not really tested or applicable here Why?
Can be turned off even when some function is locked	o same but yes Why?	o same but yes Why?		o Why?	o Why?
You decide to turn the hob on	o same but yes Why?	o same but yes Why?		o Why?	o not tested

Guiding	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen	Concept 5 Learning the Hob
Icons only show when you can use them	++ yes and i love how it's understandable instead of the user guessing which symbol means what. each icon is attached to a degreeWhy?	+ pan detection reduces the amount of icons seen on the hobWhy?		+ yesWhy?	++ Yes and also motivates you to unlock the ones you havent usedWhy?
Preheat function is clear	++ it is clear and understandable Shows you your selected temperature in the main highlighted part, while showing you the actual temperature in grey and small, above itWhy?	0 not shownWhy?		+ clear, needs visual loading sign to show how much total timeWhy?	++ Why?
Celsius cooking icon appears when ready to use	++ can only use after connecting to bluetooth which is super smartWhy?	++ it appears when in use while the plus cooking icons dim downWhy?		+ yes. not as many celsius functions were shownWhy?	++ YesWhy?
Has presets to cook with	++ preset design and use super efficient and understandableWhy?	+ adjustable presets on appWhy?		++ yesWhy?	++ Why?
Icons are clear • Celsius cooking • Letters on screen	+ didnt see letters on screen which is good but would like to know if there was an alternate design for it in this prototypeWhy?	+ icons are clear but showcase many options at once. need to consider cognitive load on userWhy?		+ didnt see letters or a replacement for it otherwise, clearWhy?	++ Very clearWhy?
Icons appear after turning on	++ simple and minimal amount of icons for clear understandingWhy?	+ they do, but seeing a lot of icons at once may be overwhelming to the userWhy?		+ simple and minimalWhy?	++ they do, similar but fun and VERY guidingWhy?

Exciting	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen	Concept 5 Learning the Hob
The hob allows you to take control in the PLUS cooking making you not locked in cooking	+ seems like itWhy?	0 doesnt seem to lock you inWhy?		0 Why?	+ sureWhy?
Being able to cook a new dish on a new hob was exciting (using power levels - sense of familiarity gives comfort)	0 it could be but the demo didnt include the act of cookingWhy?	0 it could beWhy?		0 Why?	++ YESWhy?
Presets (when understood) were exciting to use because they were new	+ while it is explained, it is still a new functionality for users and also helpfulWhy?	0 yes - because of the flexibility of control shown at first use no - because in the long term use it may be frustrating to keep changing it from the app? can we control the degrees from the hob?Why?		+ felt a bit similar but with more guidance for sureWhy?	++ new and you get to learn with easeWhy?
Indication the countdown has started (for preheat)	+ The countdown existed (with locking) because of Atag's way of heating it in boost mode (for preheat) before bringing it down to the desired level. In this prototype, there was no countdown, only an indication of how hot it was. We need to see if it is a suitable solution in the technical side.Why?	0 no timer was testedWhy?		+ preheat existed but not as excitingWhy?	0 preheat could be made exciting through gamificationWhy?
Cleaning did not give a lot of beeps	0 didnt test cleaningWhy?	0 didnt test cleaningWhy?		0 Why?	0 Why?
Suitable sound effects	0 didnt test soundWhy?	0 didnt test soundWhy?		0 Why?	0 Why?
The timer function that could be set up for every single hob	0 didnt test timerWhy?	0 timer seemed to be the sameWhy?		0 seems likeWhy?	0 sameWhy?
Celsius was found interesting when used for baking or specific cooking (like for meat)	+ i think the way of connecting bluetooth first ensures the user to use the probe, which means it can hint to use it for baking or something rather than the defaultWhy?	+ i can imagine using this for different recipes/ingredients specifically by learning from the appWhy?		0 not tested for specific cookingWhy?	+ i can imagine learning this for different kind of specific cooking like for different ingredients or even recipesWhy?

Participant 2

Participant 2		Less buttons	Big family	Touch screen	Learning the hob	
Intuitive						
Buttons showing when activated					1st: less buttons 2nd: touch screen	
The degrees related to the presets were clear						
Preset icons were clear					1st: less buttons 2nd: touch screen	
The Celcius cooking icons were clear						
Certain buttons were known due to previous use					1st: less buttons 2nd: touch screen	
It was clear if celcius cooking probe was connected						
The letters on the hob was clear					1st: less buttons 2nd: touch screen	
Stopwatch is understood						
The ON/Off button was immediately found					1st: less buttons 2nd: touch screen	
Action of placing the pan was automatic						
In control						
PLUS cooking makes you feel in control					1st: learning the hob 2nd: touch screen	
Pre heating countdown makes you feel in control						
Could switch with ease between cooking options					1st: learning the hob 2nd: touch screen	
Bluetooth probe does connect automatically or with ease						
Can easily continue using the hob after error					1st: learning the hob 2nd: touch screen	
When it responds to you as you predict it to (timer)						
Manual is easy to read and find specific topics					1st: learning the hob 2nd: touch screen	
Can pause the celsius cooking						
Can be turned off even when some function is locked					1st: learning the hob 2nd: touch screen	
You decide to turn the hob on						
Guiding						
Icons only show when you can use them					1st: learning the hob 2nd: less buttons	
Preheat function is clear						
Celcius cooking icon appears when ready to use					1st: learning the hob 2nd: less buttons	
Has presets to cook with						
Icons are clear; Celcius cooking and the letters on screen					1st: learning the hob 2nd: less buttons	
Icons appear after turning on						
Exciting						
The hob allows you to take control in the PLUS cooking and not locked					1st: learning the hob 2nd: less buttons	
Being able to cook a new dish on a new hob was exciting						
Presets (when understood) were exciting to use					1st: learning the hob 2nd: less buttons	
Indication when the countdown has started (preheat)						
Cleaning did not give a lot of sounds					1st: learning the hob 2nd: less buttons	
Suitable sound effect						
The timer function that could be set up for every single hob					1st: learning the hob 2nd: less buttons	
Celsius was found interesting when used						

Participant 3

Intuitive	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen	Concept 5 Learning the Hob
Buttons lit up when on / showing functionality when activated	+, you could still see the buttons, but through the settings you could adjust how bright you want it to be. When pressing the Lock shows on every hob what was a little confusing as one might want to press there and this could still be hot. Why?	O, the buttons remained the same, however you did not have to interact with it. Why?	++, YES, you could go back always, except during pre heat which made a lot of sense. The icons only were active when actually could be pressed. So it was very nice!	-, you saw when pressing the buttons at what level you was, but this maybe confusing and i would want to turn that off Why?
The degrees related to the presets were clear	++, as you could chose yourself what the temperatures would be this made more sense and therefore was easier to understand Why?	++, you said the action that you wanted to do and the hob itself turned it to the temperature related to the action which made it clear. Why?	+, the text framed what you could use the temperatures for, and you started off from the point you selected (so say boiling, you start from 100 degrees) what I really liked	O, it was clear because you knew but on the hob interface it didnt have a clear connection yet Why?
Preset icons were clear	++, you chose the icons for plus cooking so this made them visible and people knew what they meant. Why?	+, not changed, but it was more clear since you said the action of what you wanted to cook, so she setted up the temperature so the icons did not had to be that clear. Why?	+, the icons were clearer since the text provided what the icons could be used for.	O, not changed
The Celcius cooking icons were clear	++, very clear! Why?	O, not changed Why?	O (not included)	+, beetter icon Why?
Certain buttons were known due to previous use (like power levels and boiling preset)	+, you choosing the icons made it more clear, but there maybe can still be confusion if you see all the three options at th same time. Why?	O, no change in buttons Why?	+, only the hesan cue sign was a little bit confusing but since the text provided you feedback with the other vogue symbols it was clearer.	++, it was well explained so you knew what you were doing Why?
It was clear if celcius cooking probe was connected	O (not included) Why?	++, there was a clear audio cue of that the probe was connected. However, maybe the user also need to know what to do with the probe maybe? Why?	++, because you saw the life action of the connection of the probe, and this also stayed on the screen it was a visible and clear feedback whether you were connected.	O, not changed
The letters on the hob was clear	+, there were no letters on the hob, but maybe the one on the timer can still be confusing. But it was clearer having the temperature displayed next to the plus setting Why?	+, less letters were needed since the error was explained by the voice. Why?	+, it were not letters anymore but text, which made it much clearer! However, I am a visual person so would rather have very clear icons maybe other wise it can be an information overload? Reading a recipe and also reading your hob etc....	+, same as above Why? Why?
Stopwatch is understood	-, it looked a little cramped to have the icons next to all the actioin of the celcius and plus cooking. And the H.S. still is a little confusing maybe? Why?	O, not changed Why?	- for me personally, i would rather use the +/- buttons to set up the timer. As I already with my phone have a hard time setting up the timer with the swipping option. And I think with greasy fingers this would be even harder.	O, not changed Why?
The ON/Off button was immediately found	+, it was more clear as the lock button was not included in the on/off button any more. It was nice to have a guest account and to make fo them the primary settings very clear. Why?	++, it was easily turned on by saying it to be turned on Why?	++, YES! and I could imagine this with like a nice car sound animation, lov it.	O, not shown Why? Why? Why?
Action of placing the pan was automatic	O (not included)	+, the voice guides you in placing the pan in the activated hob. Why?	O (not included)	O, not shown Why?

In Control	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen	Concept 5 Learning the Hob
PLUS cooking makes you feel in control	+, Makes you feel more in control as you chose the temperature it started from so you have control of the hob. Why?	-, she set it up for you so you kind of don't have the control of what is happening after putting in the prompt. Why?	++, plus cooking is way more in control as the product feedback is way better. Furthermore, as you jump from beginning setting but still can adjust, you have free control of the temperatures Why?	+, more control because you are part of the game and you chose to unlock this plus setting. Why?
Pre heating countdown makes you feel in control	O did not change Why?	+, you do know you have to wait, but it does not give control as you just have to wait and not really know what it is doing Why?	++, no preheating but filling icon which was a nice abstraction. Why?	O, not changed Why?
Could switch with ease between cooking options	+, there was a back button but uncertain wether you are locked in the preheat to go bck Why?	O, not shown Why?	++, since there was a back button that brought you back to your previous step you could switch with more ease Why?	O, not shown Why?
Bluetooth probe does connect automatically or with ease	O not shown Why?	++, YES!! really easy. Why?	+, the guidance of the probe is clearer as you now know it is connecting. Why?	O, not shown, but you could add this as part of the game. Why?
Can easily continue using the hob after error	O not shown Why?	+, you were aware of what the error was and what you could do to solve it. Why?	O, not shown but could imagine it could be displayed what the error was and how it could be resolved. Why?	+, i cn imagine that the game can explain the error Why?
When it responds to you as you predict it to (timer)	+, the preheat might be still confusing as you lock, it is improved as you think of the icons and temperatures so it is easier to predict. Why?	+, not shown but i thinkn if she always understands what i am saying it will react as i expect and can be guiding in the process. Why?	+, I think it follows your action very well so that it does things as you predict. Why?	+, if you follow along in the app and this is a clear link, you can know how it reponds Why?
Manual is easy to read and find specific topics (but it is the old manual)	++, it seems like there are instructions in the app on how to set up the hob Why?	O, not shown, but with the prompt guidance you could understand the first inter. Why?	O, not shown Why?	++, th manu is divided in chapters and it is excited to unlock the different features on the hob. Why?
Can pause the celcius cooking	O not shown Why?	+, not shown, but i can imagine you can ask celsi to pause one zone with ease Why?	O, not shown Why?	O, not shown Why?
Can be turned off even when some function is locked	-, i think nothing changed here and you are still locked in the PLUS function Why?	+, not shown, but i think you can ask her to be turned off at any point. Why?	+, You can always go back which is helpful. And i think you can always turn it off also in preheating? Why?	O, not changed, but due to the gamefication you might know what is hapening Why?
You decide to turn the hob on	+, you chose your account so you decide to turn it on in your own account. Why?	++, you make a concious choice to turn the hob on by asking celsi to turn the hob on. Why?	+, everything was off before you turn it on and only the on button was present. Why?	O, not changed Why?

Guiding	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen	Concept 5 Learning the Hob
Icons only show when you can use them	+, this was the case, however they were still at one place and still remained a little visible which might confuse the user. Why? Why?	O, not shown Why?	++, only the things you can actually touch or set up are visible on the touchscreen. Why?	O, the game one disappeared which was good, but for the other interactions noting changed here Why?
Preheat function is clear	O, not shown Why? Why?	++, it is clear it is preheating and that you have to wait till you can start cooking.	+, it was clear, however it would be nice to show how it is preheating and you know should not use it Why?	O, not shown Why?
Celcius cooking icon appears when ready to use	-, the celcius was already there Why? Why?	+, i think the icons will be appearing along side what you are saying	++, especially with the addition of the bluetooth icon it would be very clear. Why?	O, not shown Why?
Has presets to cook with	++, since the user can chose the presets themselves and chose the temperature as well this helped knowing what you are doing Why? Why?	++, the presets are much more clear as they are linked to actions compared to solely the temperature.	++< I REALLY liked the presets how it set up here, I see a nice combination with mine ;). And also really liked the boost icon.	+! The presets are linked to your experience lvl but the presets still can be confusing Why?
Icons are clear • Celcius cooking • Letters on screen	+, the celcius C is much more clear now but a lot of icons are shown at the same time which might confuse the user. Why?	O, not changed	O, the icons still remained a little vague. Why?	+, The icons of cooking are clearer Why?
Icons appear after turning on	++, the icons after choosing your account appeared that were related to that account. Why?	O, not shown	++, after pand detection the menu shows Why?	O, not shown, but maybe the game icon on the hob can be confusing Why?

Exciting	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen	Concept 5 Learning the Hob
The hob allows you to take control in the PLUS cooking making you not locked in cooking	-, you do have more control with the choosing the temperature however you are i think still locked within. Why?	O, i think you do are locked Why?	++, you are locked, but you know why and it is clear that you are locked so it is not a burden now on Why?	+, it lets you take more control in the sense that you have a better understanding of what is going on Why?
Being able to cook a new dish on a new hob was exciting (using power levels - sense of familiarity gives comfort) •	++, since the hob feels more your own one could be excited every time to try the new shortcuts and stuff. Why?	+, the first time would be exciting but maybe after a while you could get annoyed Why?	+, the familiarity of the settings in the text that guides you was nice, but maybe in quick glance would be confusing if you have different hobs turned on Why?	++, it was exciting because it felt like a game. Why?
Preset (when understood) were exciting to use because they were new	+ yes because you could change the presets and share this as well with others. Why?	++, it was exciting as you said your action and the hob will set it up for you Why?	++, I think since you know what you are doing and why you are doing it and it helps you it helps making it exciting Why?	++, they were exciting because you can play up to become experts Why?
Indication the countdown has started (pre-heat)	O, not shown Why?	++, there was a clear cue the pre heat had started Why?	++, there is no countdown anymore but a nice visualisation Why?	O, not shown Why?
Cleaning did not give a lot of beeps	-, the lock was shown on every hob indicating that you can clean freely, however this can be dangerous when it is still hot. Why?	-, it did disrupt your user flow Why?	O, I can imagine that touchscreen can be more sensitive, however, since it is always in lock cleaning before use is easy Why?	O, not shown Why?
Suitable sound effects	O not shown Why?	++, the sounds were suitable and soft, however maybe can get irritating after a while Why?	O not shown Why?	O, not shown Why?
The timer function that could be set up for every single hob	-, I thought the timer function was a little bit cramped on the interface. Why?	O not shown Why?	+, it could be set up for every hob and I really liked the visualisation of the timer, BUT the slider is for my greasy fingers not that nice maybe Why?	O, not shown Why?
Celsius was found interesting when used for baking or specific cooking (like for meat)	O, not shown/ maybe only exciting when shown in the community app Why?	-, it would be interesting, but maybe less clear as with celcius you want to do our the tasks maybe yourself as your are entering 'chef mode' Why?	O remained the same Why?	O, not really changed Why?

Participant 3

	Big family	Voice control	Touch screen	Learning the hob		
Intuitive						
Buttons showing when activated					1st: touch screen 2nd: big family	
The degrees related to the presets were clear						
Preset icons were clear						
The Celcius cooking icons were clear						
Certain buttons were known due to previous use						
It was clear if celcius cooking probe was connected						
The letters on the hob was clear						
Stopwatch is understood						
The ON/Off button was immediately found					1st: touch screen 2nd: big family	
Action of placing the pan was automatic						
In control						
PLUS cooking makes you feel in control						
Pre heating countdown makes you feel in control						
Could switch with ease between cooking options						
Bluetooth probe does connect automatically or with ease						
Can easily continue using the hob after error						
When it responds to you as you predict it to (timer)					1st: touch screen 2nd: big family	
Manual is easy to read and find specific topics						
Can pause the celsius cooking						
Can be turned off even when some function is locked						
You decide to turn the hob on						
Guiding						
Icons only show when you can use them					1st: touch screen 2nd: big family voice control	
Preheat function is clear						
Celcius cooking icon appears when ready to use						
Has presets to cook with						
Icons are clear; Celcius cooking and the letters on screen						
Icons appear after turning on						
Exciting						
The hob allows you to take control in the PLUS cooking and not locked					1st: touch screen 2nd: voice control	
Being able to cook a new dish on a new hob was exciting						
Presets (when understood) were exciting to use						
Indication when the countdown has started (preheat)						
Cleaning did not give a lot of sounds						
Suitable sound effect						
The timer function that could be set up for every single hob						
Celsius was found interesting when used						

Participant 4

Intuitive	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 5 Learning the Hob
Buttons lit up when on / showing functionality when activated	++ I really like the first menu it makes it way easier and more obvious for people in which mode they are	++ I actually really like the idea that you can only use the buttons that are in use. This makes it very clear in which mode you are cooking.	++ Yes this is very true. This is pretty clear and easy.	0 This did not change for me
The degrees related to the presets were clear	++ I think it is a bit more clear how warm you are cooking in a simple way, maybe it can still need some more clear text/icons and presets	++ I really like the shortcuts, because this makes it quick to use and people are comfortable, I only think it can be a bit confusing, do you only have 3 temperatures to cook at? This would not give me the freedom to change the temperatures how I like it to change	++ I like that you can say what your cooking and it changes instantly to the right temperature	- I think it will be easy to learn in the game but maybe this was not clear to me how exactly. But I think the game is a good motivation to start learning
Preset icons were clear	++ did not change drastically but did improve already a bit, because the temperatures are connecting to the power level, this gives the user something to hold on to	++ I really like the idea of having shortcuts, it is clear to me how to use and I love the app with it. This make it way more easy to set, instead of how the settings are now. I do like the shortcuts, but can you use it per icon or not?	0 It did not change	++ I think this will be in the game, but unsure how this will prevent in the interface
The Celsius cooking icons were clear	++ I like the connecting icon, but can you still see it during using the probe is the icon shown in a smaller thing. I do really like the simplicity of only using the slider	- I don't think it is very clear to me, the Celsius cooking icons are maybe better with a C instead, but I do not see how the measured and set temperature are shown, does the temperature also only show on the left? And where is the slider for, I am still left with some confusion	0 not changed	0 Maybe not instantly but I think the learning process does make it easier to start learning instead of the manual
Certain buttons were known due to previous use (like power levels and boiling preset)	++ I think the icons did make it more clear where to use it for, but still a bit practice is needed	++ I really like the personal accounts and that you can alter it on the app. This makes also more sense to the app. The short cuts are nice, I am only confused if you put the short cuts on the side to which hob they are going to respond.	++ Yes I do understand how to use it. It is nice that you can just say boiling children and I already goes to a right temperature	++ You can see progress of how many times you used the buttons, I think this is nice, but maybe the placement, how is not quite visible to me. Can I only see it on the app or the progress bar on the hob itself, maybe this confuses me with other progress bars
It was clear if Celsius cooking probe was connected	++ very very clear, only maybe keep the icon in small somewhere	- It is not quite clear for me, I did not see any feedback or icons that could lit up when it is connected	0 It is nice to have some feedback with sound. Only speaking might be unclear some times, think of shouting kids or cooking with friend	0 Did not change a lot for me
The letters on the hob was clear	0 It did not change a lot	0 I think the letters did not change a lot, to me they were already clear	0 Not changed a lot	0 This did not change a lot for me
The ON/OFF button was immediately found	++ Yes I like that the pause button is deleted and it is clear with the settings as well. Maybe the settings button can be thought out better, like were and how the settings are gonna be presented	++ I like the on and off button, also the child lock is very clear. It is very nice that the pause and wifi icon are gone. This makes it very clear and intuitive to use. It would only confuse me if I can see the lock function on the hob, when this is not only in use for one hob. I think the icon on the hob would make me to press the icon, but when the hob is hot this can be quite dangerous.	- This is voice controlled, it is easy to use but I can use some malfunction on turning it on and off when you don't want it to. Maybe it is better to only start voice control when you want to instead of always.	0 This did not change a lot for me
Stopwatch is understood	0 did not change	0 I think the stopwatch is okay, but it did not change a lot	0 Easy to use but did not change a lot. I am still in doubt with the speaking	0 did not change a lot for me
Action of placing the pan was automatic	0 This did not change a lot	0 For me when all the int sliders are on it would be confusing if there is whether pan detection or I can put all the hobs on, because to me it would make me to use the int slider when I see it. I do really like the idea of using the app and adjusting this, I am only curious how this can be done in a simple way, maybe this is something that almost everyone like just to show the hob in use, and the app doesnot have to be overcomplicated	0 It is automatically done if you say on which zone you are. The only thing I am insecure about is when it is noisy in the room, maybe you do need some buttons in stead of only voice.	0 This did not change a lot for me

In Control	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 5 Learning the Hob
PLUS cooking makes you feel in control	- I do feel like so, but I think I still will be stuck in the preset levels and the preheating a bit.	I like the pre sets, I only do not think it is nice that I can only see what the shortcuts are, I would like to change the temperatures in plus cooking	- I think the hob is more in control, while they are setting the temperature for you, but it is easy to use	++ when learning yes but did not change intuitively
Pre heating countdown makes you feel in control	++ I like the preheating. It is nice, not to commanding but clear	I donnot think the preheating changed, this would stress me out because I see the time running down	++ I like the icon for the preheating, it is clear but not commanding, I dont know if I would like the music all the time. It can be quite distracting	0 did not change
Could switch with ease between cooking options	++ I think it is clear when you can switch and how to switch, it makes me a bit more patient to wait until I can go back	I am not sure how to go back to the functions. When I still see an icon in gray I would expect, I can maybe use it? But is there a going back option?	++ I think you can change always with voice	0 did not change
Bluetooth probe does connect automatically or with ease	++ It shows very clear when it is connecting and when it is connected. The big bluetooth logo helps a lot	I think it is still the same, so maybe not that clear	++ It connects automatically when saying connect probe. This is very nice. But does the hob know you are going to use the pan or the probe?	0 did not change, but maybe easier to learn
Can easily continue using the hob after error	0 did not change a lot	I am actually not sure I donnot think this is implemented and not improved.	++ I like how the hob tells you what to do with an error. It is clear and guiding	++ I think when learning in the game it will be fine, but did not change in the interface
When it responds to you as you predict it to (timer)	++ I think the feedback is very nice that you are getting back, some placements can I think be a bit better. It is predictable but not commanding	I do think I will understand it, it only looked quite small so maybe that will stress me if I have to set the timer.	++ Yes it does. It is also clear when there is an error it says what to do	0 when learning yes, but maybe not intuitively
Manual is easy to read and find specific topics (but it is the old manual)	0 did not change	I think with the app this is so much better. I really like the idea of the videos and the community.	Did not change	++ I think the game will make the manual way more easy. Maybe it is nice to show which parts of the manual are already learned
Can pause the Celsius cooking	0 I do not think this can be done	I donnot think I can do this, but I also do not see it as something I want to do so I see it as positive	I am not sure	0 did not change
Can be turned off even when some function is locked	++ I think this can be done, but the home button is also a nice one	I am actually not sure, I think you can because the icon is shown	0 I think this can be done that is nice	0 I am unsure, I think it did not change
You decide to turn the hob on	++ this is very clear and easy, not an overload of buttons	++ I think the turning on function is easy and the lock is very clear and understandable. It makes sense. Only maybe not the lock on the hobs	- I see some small functions happening here. Just like with your Siri. It comes on when you dont want it to	0 did not change

Guiding	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 5 Learning the Hob
Icons only show when you can use them	++ Yes they do, this is done in a clear way. The home button makes it clear how to get back	yes! I really like this option, it makes it very clear in which functions you are and what you can use	Did not change a lot	0 i am not sure
Preheat function is clear	++ It is very clear to me, but still not commanding. I am a bit confused about the boost and the preheating, it is used through each other?	I did not see the preheat, but i am not sure. It did not was clear to me.	the preheat function is very clear. i like the icon changing	0 did not change
Celsius cooking icon appears when ready to use	++ Yes it is not an overload and you can see in which icon you are	I am not sure, dont think this is implied	did not change a lot, do think it is easier now to connect to probe since it are less steps	0 i am not sure
Has presets to cook with	+ Yes it does, but maybe i feel a bit stuck in the plus cooking because it only has a few settings	Yes i like the short cuts, but i am unsure if i can still use different temperatures	I like that you can say what your cooking and it instantly goes to the right temperature	0 i think you will learn about it but did not change
Icons are clear • Celsius cooking • Letters on screen	+ They are pretty clear after learning them. Maybe it would be better with some text as well	I think i like the icons, but maybe there is a better one for the C, i feel like the C is sometimes used in different context for induction cooking	i dont think the icons changed a lot	0 did not change
Icons appear after turning on	++ Yes i like the first overview. Being conscious of what to use	I like that you can alter how many icons are visible	Did not change a lot	0 i think it does but did not change a lot

Exciting	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 5 Learning the Hob
The hob allows you to take control in the PLUS cooking making you not locked in cooking	++ it shows better when it is preheating and you have to wait, this would make me more patient, the home button makes it clear how to switch.	i am not sure how the locking goes and how to go back easily	I do not know if i feel in control since the hob changes the temperature automatically. I think i would feel as if the hob is more in control	+ I think i will be incontrol after learning with the game
Being able to cook a new dish on a new hob was exciting (using power levels - sense of familiarity gives comfort)	0 i dont think it will be more exciting only more clear.	I really like the sharing the recepies this is very nice and that it changes to your presets	I think i would be excited but maybe only the first time, after it i would like it to react to buttons as well, as i see this as having less malfunctions	++ I see the app helping with this in the form of a game, this could be exciting
Presets (when understood) were exciting to use because they were new	+ I think they are already a lot more clear and exciting. Still a bit stuck in the presets	Yes the short cuts are clear and exciting especially in combination with the app	yes also the talking is new but i do not see it as a long lasting idea. i think it would annoy me. I like how easy you can go to the presets	+ I think it is a new way to learn and this can be exciting.
Indication the countdown has started (pre heat)	++ I like that the countdowns are gone it is still clear.	This is actually not really clear to me, i did not see a lot of improvement.	I like the countdown of the timer and the pre heat better.	0 did not change a lot
Cleaning did not give a lot of beeps	0 did not change	I think this did not change it would confuse me	I think it will say if it has an error this is what i like	0 did not change a lot
Suitable sound effects	0 did not change	The sound effects did not change	Yes but maybe a bit to much for me, also with talking maybe	0 did not change a lot
The timer function that could be set up for every single hob	0 did not change	I think the timer did not change a lot and is not new or exciting to me	the timer did not change a lot for me	0 did not change a lot
Celsius was found interesting when used for baking or specific cooking (like for meat)	0 It is a lot more clear, more exciting i do not really know	I dont think it is exciting to me now, did not change a lot	the interface did not change a lot. connecting is a lot better to me	0 did not change a lot

Participant 4

Less buttons

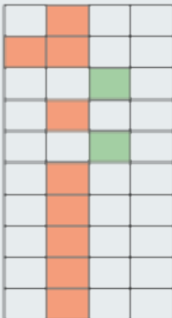
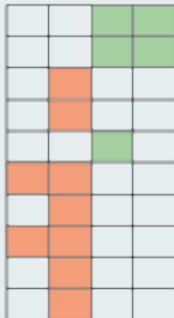
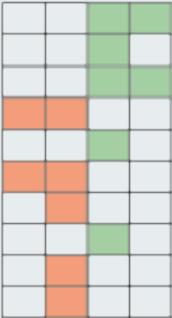
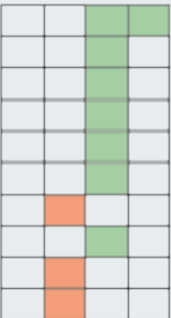
Big family

Voice control

Learning the hob

Intuitive

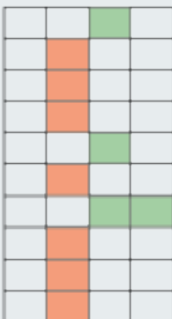
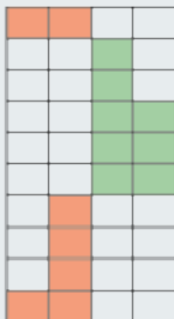
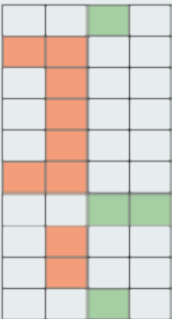
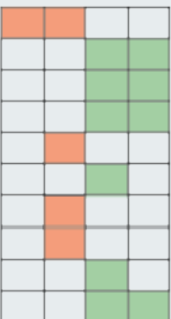
- Buttons showing when activated
- The degrees related to the presets were clear
- Preset icons were clear
- The Celcius cooking icons were clear
- Certain buttons were known due to previous use
- It was clear if celcius cooking probe was connected
- The letters on the hob was clear
- Stopwatch is understood
- The ON/Off button was immediately found
- Action of placing the pan was automatic



1st: less buttons
2nd: big family

In control

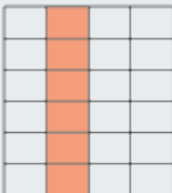
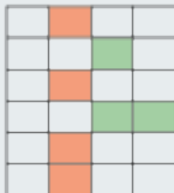
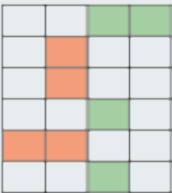
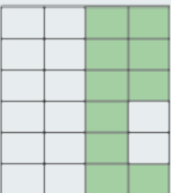
- PLUS cooking makes you feel in control
- Pre heating countdown makes you feel in control
- Could switch with ease between cooking options
- Bluetooth probe does connect automatically or with ease
- Can easily continue using the hob after error
- When it responds to you as you predict it to (timer)
- Manual is easy to read and find specific topics
- Can pause the celcius cooking
- Can be turned off even when some function is locked
- You decide to turn the hob on



1st: less buttons
2nd: voice control

Guiding

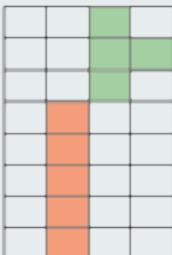
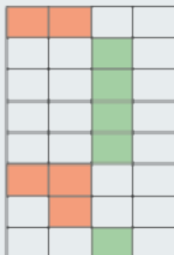
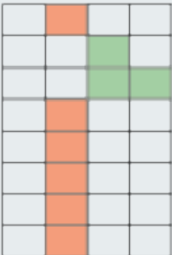
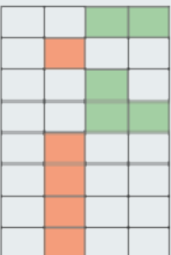
- Icons only show when you can use them
- Preheat function is clear
- Celcius cooking icon appears when ready to use
- Has presets to cook with
- Icons are clear; Celcius cooking and the letters on screen
- Icons appear after turning on



1st: less buttons
2nd: big family

Exciting

- The hob allows you to take control in the PLUS cooking and not locked
- Being able to cook a new dish on a new hob was exciting
- Preset (when understood) were exciting to use
- Indication when the countdown has started (preheat)
- Cleaning did not give a lot of sounds
- Suitable sound effect
- The timer function that could be set up for every single hob
- Celcius was found interesting when used



1st: less buttons
2nd: learning the hob

Participant 5

Intuitive	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen
Buttons lit up when on / showing functionality when activated	+ Yes	++ The function of customize make a great sense. You can adjust different appearance for different buttons.	++ Lights up with voice guide is more visible	++ Yes the buttons is clear on the display screen
The degrees related to the presets were clear	++ Yesssss and also with the ICONs I can see which cooking level I am using	O (Not included)	++ Yes with the voice guide I will get more information of the degrees	++ Yes the presets and the degree show on the same page
Preset icons were clear	O Not mentioned	+ Remove the preset function on the hob but you can adjust them on the app. Might need some instruction to use the app then people can know where is the preset function.	+ Even if the icons are not clear I can ask Celsi for help	O Not mentioned
The Celcius cooking icons were clear	++ Yes the buttons are clear	++ Yes the cooking style button will light up when turn on the hob.	O Not mentioned	++ Yes Text combine icon perform is better than icon only
Certain buttons were known due to previous use (like power levels and boiling preset)	O Not much about the preset function	++ Yes there will be a hint near by the function which indicate the history degree of choosing.	O Not included	+ Yes the previous use is shown well on the display screen and also active automatically
It was clear if celcius cooking probe was connected	+ Yes with the color change	O (Not included)	++ Yes!! I can know by Celsi without figure out the complex icons	++ Yes I like the text size!!!
The letters on the hob was clear	+ It would be nice if the layout is more clear and keep the same with different function(Plus and Celsius)	++ Yes the letter is readable	O Not included	++ Yes using short text is very intuitive and clear
The ON/OFF button was immediately found	++ Yes and can turn on for different cooking area directly	++ Yes, the on/off button is very clear at the first look. And also the lock function is visible.	++ Can be easily turn on/off by voice	O Keep the same
Stopwatch is understood	O Keep the same one	O Not very clear for me.	+ I suppose it will be understood	++ Yes adding the loop on the screen is helpful and the counting setting is very clear
Action of placing the pan was automatic	O Not mentioned	O (Not included)	+ Celsi can show me where should I place the pan	O Not included

In Control	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen
PLUS cooking makes you feel in control	++ Yes because I dont need to regonized every icons they just show up with different degrees	+ I can easily reach the PLUS cooking mode and set different degree for it	++ Yes, I can ask Celsi to set for me	+ Will be a little bit confuse when I facing a lot of icons
Pre heating countdown makes you feel in control	+ Yes I can see the simple icon and the color change	O (Not included)	++ Yes, with the changes of icons and the music I can understand the function and be clam down and simply wait for the pre heating	+ Yes I do feel the pre heating countdown is more visible and clear
Could switch with ease between cooking options	+ Yes The home button is easily to find	+ Yes It is clear that where the buttons are	+ Yes but I dont know if I can change options when Celsi is talking. If I switch too fast will I encounter a mess?	+ Yes I think is quick to find the home/back button in each functions
Bluetooth probe does connect automatically or with ease	O Not mentioned	O Not mentioned	O I cant tell	O Do not change
Can easily continue using the hob after error	O Not mentioned	O Not mentioned	++ Yes Celsi will remind me that there is a error and how can I deal with it. Clear instruction	O Not mentioned
When it responds to you as you predict it to (timer)	O Keep the same	O Not sure	+ Trust Celsi but maybe there will be some problem between some similar functions?	++ Not much unpredictable buttons
Manual is easy to read and find specific topics (but it is the old manual)	O Not mentioned	++ Yes with the app I can easily find how to change the preset and how to customize the appearance of the hob And I like the setting of guest, which contain less function and can be use without complex manual	+ I can ask Celsi instead of reading manual myself. However I might miss some functions without guiding.	O Do not change
Can pause the celsius cooking	O Not mentioned	+ The locking function is clear and will shown on every cooking area.	+ I suppose yes	O Keep the same
Can be turned off even when some function is locked	+ Yes the Icon is quite clear But the locked icon is as the same size as the ON/OFF button, I am not sure if these functions are in the same level or not	O Not sure if I can lock the functions	++ If we add the intelligence in the hob it wont be a problem	+ Yes the home/back button
You decide to turn the hob on	++ I know how to turn it on in the first use	+ I can easily turn on the hob.	++ Great experience!	++ Can be done without manual

Guiding	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen
Icons only show when you can use them	++ Yes	+ I think is a good idea to set them 50% visible when I am not using them. Cause the icons can still remind me that I can switch on different function.	+ Yes	+ Yes
Preheat function is clear	+ Yes the preheat icon is much more clear than before	O Not included	++ Yes with clear voice guiding	+ I like the loading page of the preheat function
Celsius cooking icon appears when ready to use	++ Yes it can be easily recognized	++ Yes it is clear	+ Yes	+ Yes and it keep appear when you are using it
Has presets to cook with	+ Yes but it will be hard to find how to adjust it on the screen	++ Yes there is a clear preset display	+ Yes I can ask for using the presets	O Not change a lot
Icons are clear • Celsius cooking • Letters on screen	+ Yes I like the idea of making them as a circle that I know the level of these functions	++ Yesyesyes	O No mentioned	++ Yes the Icons are clear in the whole experience
Icons appear after turning on	++ The Icons is visible	++ Yes and I can change the visible of the icons	++ Yes	+ Yes and the layout of the Icons are easy to understand

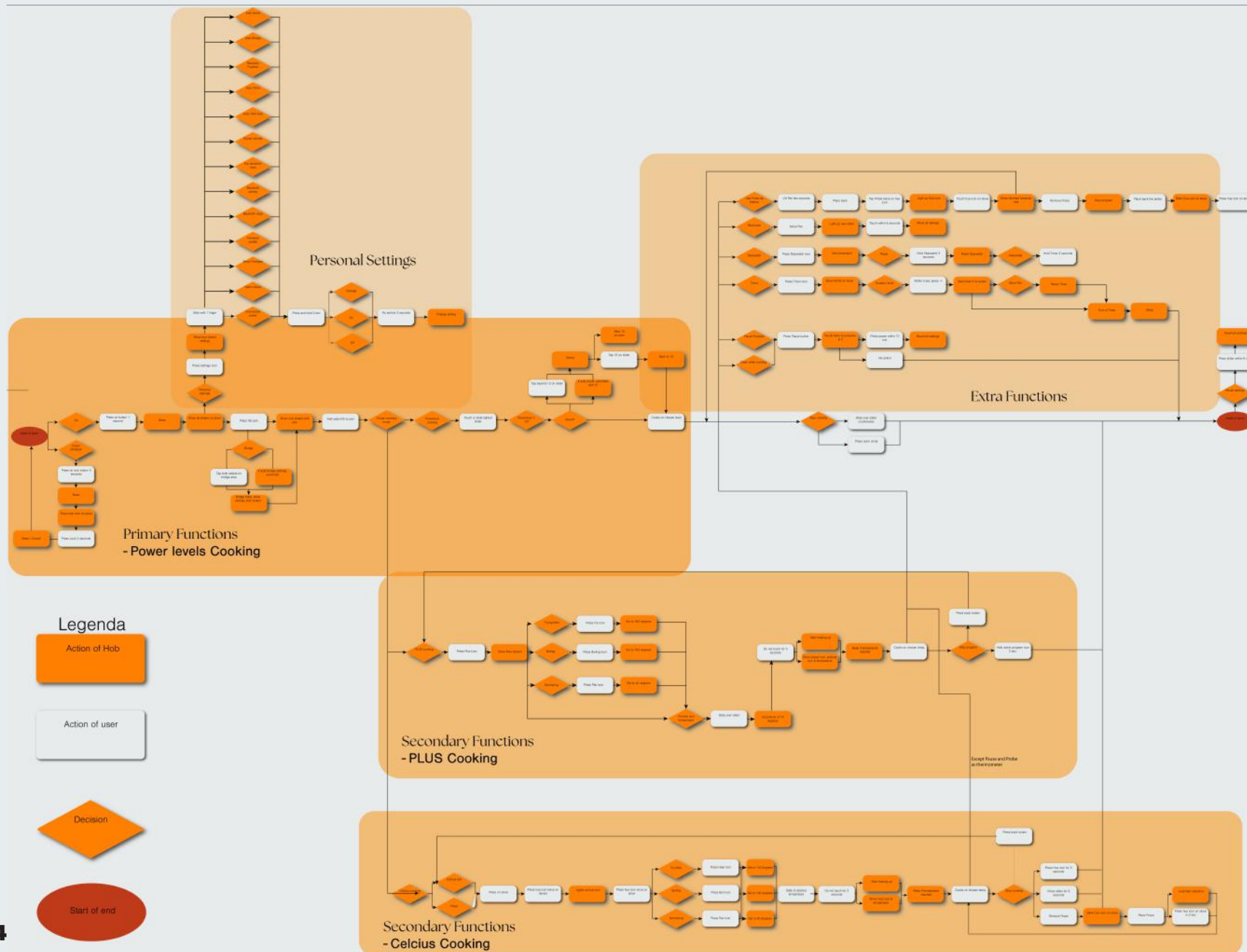
Exciting	Concept 1 - Less Buttons	Concept 2 - Big Family	Concept 3 - Voice Control	Concept 4 Touch Screen
The hob allows you to take control in the PLUS cooking making you not locked in cooking	+ Yes I like the home button which will show on each functions	+ Not very clear but I suppose that yes	O Not sure	
Being able to cook a new dish on a new hob was exciting (using power levels - sense of familiarity gives comfort)	++ Yes I will have a try on different cooking area, especially after done the preset.	++ Yes I am willing to set different color for my hob	++ Yes I can wait to using voice control again	+ Yes would be a good chance to try out the preset functions
Presets (when understood) were exciting to use because they were new	+ I feel like I cannot change a lot for the setting because of the less buttons but indeed the function is more easy than before.	++ Yes! Love to use the app to set presets because that will reduce the stress of making error on the hob	++ Yes is the first time I experience with the pre heating function which has a little explanation and the chill music	+ Yes and it is quite convenience to use
Indication the countdown has started	+ It is good to see the icon change	O Not mentioned	O Not mentioned	++ Yes the countdown slider is good
Cleaning did not give a lot of beeps	O Not mentioned	++ Yes cause I can easily know that if the cooking area was locked or not	+ Yes instead of beeps there is Celsi talking. Might feel a little bit annoying after using a long time?	O Not mentioned
Suitable sound effects	O Not mentioned	O Not mentioned	++ Yes the tone of the voice is not speedy and the setting of Celsi make me feel relax.	
The timer function that could be set up for every single hob	O Keep the same	O Not mentioned	O Not mentioned	++ Yes and the timer function is really cool that it is visible on the slider
Celsius was found interesting when used for baking or specific cooking (like for meat)	+ Not change a lot But atleast the icon can be found on the screen!!!	O Not included	++ There is a assistant for me to show me with specific cooking mode	O Not included

Participant 5

	Less buttons	Big family	Voice control	Touch screen		
Intuitive						
Buttons showing when activated					1st: touch screen 2nd: big family	
The degrees related to the presets were clear						
Preset icons were clear						
The Celcius cooking icons were clear						
Certain buttons were known due to previous use						
It was clear if celcius cooking probe was connected						
The letters on the hob was clear						
Stopwatch is understood						
The ON/Off button was immediately found					1st: voice control 2nd: less buttons	
Action of placing the pan was automatic						
In control						
PLUS cooking makes you feel in control						
Pre heating countdown makes you feel in control						
Could switch with ease between cooking options						
Bluetooth probe does connect automatically or with ease						
Can easily continue using the hob after error						
When it responds to you as you predict it to (timer)					1st: less buttons 2nd: big family	
Manual is easy to read and find specific topics						
Can pause the celsius cooking						
Can be turned off even when some function is locked						
You decide to turn the hob on						
Guiding						
Icons only show when you can use them					1st: less buttons 2nd: big family	
Preheat function is clear						
Celcius cooking icon appears when ready to use						
Has presets to cook with						
Icons are clear; Celcius cooking and the letters on screen						
Icons appear after turning on						
Exciting						
The hob allows you to take control in the PLUS cooking and not locked					1st: big family 2nd: voice control	
Being able to cook a new dish on a new hob was exciting						
Presets (when understood) were exciting to use						
Indication when the countdown has started (preheat)						
Cleaning did not give a lot of sounds						
Suitable sound effect						
The timer function that could be set up for every single hob						
Celsius was found interesting when used						

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Appendix K: Improved user flow



Appendix L: PLUS-functions user flows

The Figures X & X show the user flow of using the PLUS function. Figure X depicts the one without redesign, Figure X the one with the redesign.

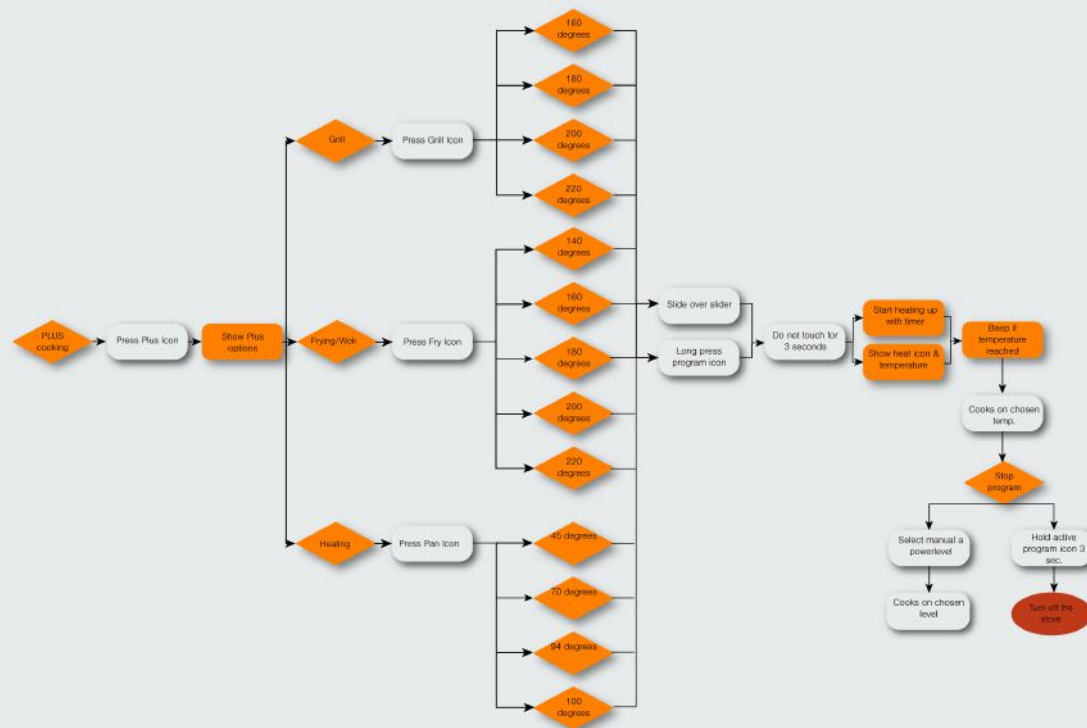


Figure 54: The old user flow of the plus function.

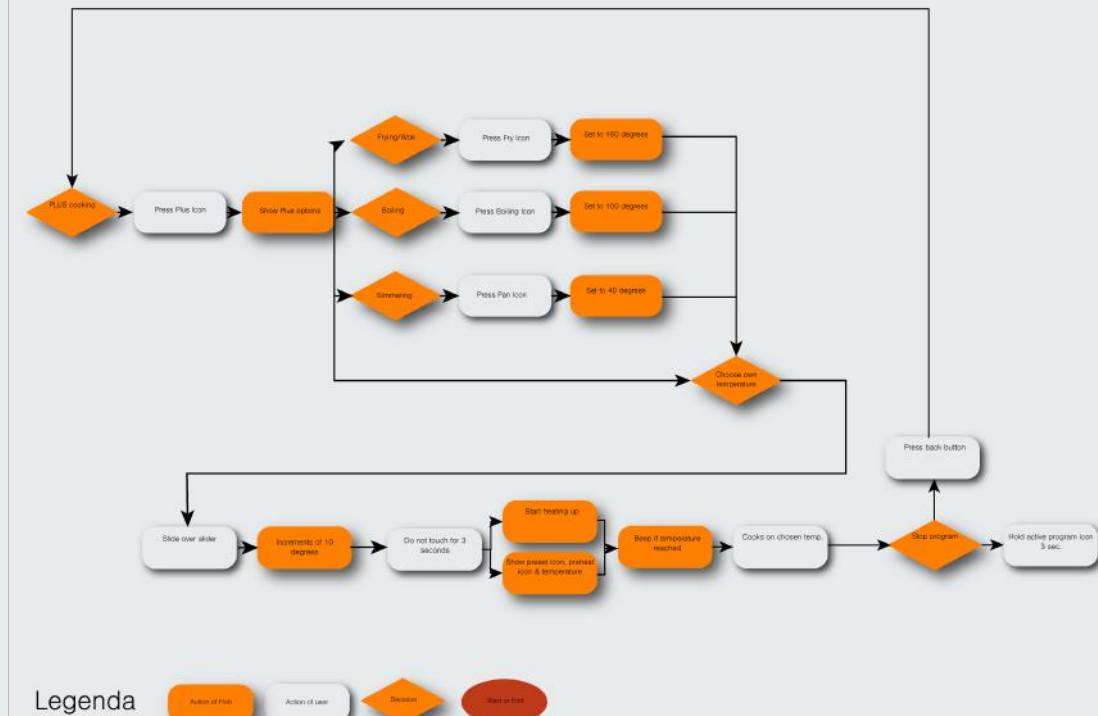


Figure 55: : The new user flow of the plus function.